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HARMONISED EUROPEAN STANDARD

IMT cellular networks;
Harmonised Standard for access to radio spectrum;
Part 26: Aerial User Equipment (UE)

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DEN/MSG-TFES-1505

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Foreword

This draft Harmonised European Standard (EN) has been produced by ETSI Technical Committee Mobile Standards Group (MSG), and is now submitted for the combined Public Enquiry and Vote phase of the ETSI Standardisation Request deliverable Approval Procedure (SRdAP).

For non-EU countries, the present document may be used for regulatory (Type Approval) purposes.

The present document has been prepared under the Commission's standardisation request C(2015) 5376 final [i.6] to provide one voluntary means of conforming to the essential requirements of Directive 2014/53/EU on the harmonisation of the laws of the Member States relating to the making available on the market of radio equipment and repealing Directive 1999/5/EC [i.1].

Once the present document is cited in the Official Journal of the European Union under that Directive, compliance with the normative clauses of the present document given in Tables A-1 and A-2 confers, within the limits of the scope of the present document, a presumption of conformity with the corresponding essential requirements of that Directive and associated EFTA regulations.

The present document is part 26 of a multi-part deliverable. Full details of the entire series can be found in part 1 [i.8].

Proposed national transposition dates	
Date of latest announcement of this EN (doa):	3 months after ETSI publication
Date of latest publication of new National Standard or endorsement of this EN (dop/e):	6 months after doa
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Modal verbs terminology

In the present document "**shall**", "**shall not**", "**should**", "**should not**", "**may**", "**need not**", "**will**", "**will not**", "**can**" and "**cannot**" are to be interpreted as described in clause 3.2 of the ETSI Drafting Rules (Verbal forms for the expression of provisions).

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1 Scope

1.0 General

The present document specifies technical characteristics and methods of measurements for the following equipment:

- Aerial User Equipment (UE) for Evolved Universal Terrestrial Radio Access (E-UTRA) and New Radio (NR).

NOTE: The relationship between the present document and essential requirements of Directive 2014/53/EU is given in annex A.

1.1 Operating bands

The aerial UE is capable of operating in all, or any part of the E-UTRA and NR frequency bands given in Table 1.1-1.

Table 1.1-1: E-UTRA and NR aerial UE operating bands

Band designation for operation as:		Uplink (UL) *operating band* aerial UE transmit F_{UL_low} to F_{UL_high}	Downlink (DL) *operating band* aerial UE receive F_{DL_low} to F_{DL_high}	Duplex Mode	Related EC/ECC decision
E-UTRA	NR				
1	n1	1 920 MHz to 1 980 MHz	2 110 MHz to 2 170 MHz	FDD	[i.15], [i.16] and [i.18]
3	n3	1 710 MHz to 1 785 MHz	1 805 MHz to 1 880 MHz	FDD	[i.13], [i.14] and [i.18]
7 (see note 3)	n7 (see note 3)	2 500 MHz to 2 570 MHz	2 620 MHz to 2 690 MHz	FDD	[i.17], [i.7] and [i.18]
8	n8	880 MHz to 915 MHz	925 MHz to 960 MHz	FDD	[i.13], [i.14] and [i.18]
20 (see note 3)	n20 (see note 3)	832 MHz to 862 MHz	791 MHz to 821 MHz	FDD	[i.4], [i.5] and [i.18]
28 (see notes 1 and 3)	n28 (see notes 1 and 3)	703 MHz to 748 MHz	758 MHz to 803 MHz	FDD	[i.9], [i.10] and [i.18]
38 (see note 3)	n38 (see note 3)	2 570 MHz to 2 620 MHz	2 570 MHz to 2 620 MHz	TDD	[i.17], [i.7] and [i.18]
N/A	n91 (see notes 2 and 3)	832 MHz to 862 MHz	1 427 MHz to 1 432 MHz	FDD	[i.4], [i.5] and [i.18]
N/A	n92 (see notes 2 and 3)	832 MHz to 862 MHz	1 432 MHz to 1 517 MHz	FDD	[i.4], [i.5] and [i.18]
N/A	n93 (see note 2)	880 MHz to 915 MHz	1 427 MHz to 1 432 MHz	FDD	[i.13], [i.14] and [i.18]
N/A	n94 (see note 2)	880 MHz to 915 MHz	1 432 MHz to 1 517 MHz	FDD	[i.13], [i.14] and [i.18]
N/A	n109 (see notes 2 and 3)	703 MHz to 733 MHz	1 432 MHz to 1 517 MHz	FDD	[i.11] and [i.18]

NOTE 1: In Europe, according to [i.9] and [i.10], radio equipment in band 28 is only allowed between 703 MHz and 736 MHz (F_{UL_low} = 703 MHz and F_{UL_high} = 736 MHz) for the transmitter and between 758 MHz and 791 MHz (F_{DL_low} = 758 MHz and F_{DL_high} = 791 MHz) for the receiver.
NOTE 2: Variable duplex operation does not enable dynamic variable duplex configuration by the network, and is used such that DL and UL frequency ranges are supported independently in any valid frequency range for the band.
NOTE 3: For those bands, mechanisms to ensure no-transmit zones for the aerial UEs are not considered in the current version of the present document.

NOTE: The relationship between the present document and essential requirements of article 3.2 of Directive 2014/53/EU [i.1] is given in annex A.

The present document covers requirements for E-UTRA and NR aerial UE from 3GPP™ Releases 18 defined in ETSI TS 136 101 [2] and ETSI TS 138 101-1 [3], respectively. This includes the requirements for E-UTRA and NR aerial UE operating bands from 3GPP™ Release 18 defined in ETSI TS 136 101 [2] and ETSI TS 138 101-1 [3].

The requirements for NR aerial UE in the present document apply to the combination of channel bandwidths, SCS and operating bands shown in Table 1.1-2. The channel bandwidths are specified for both the TX and RX path.

Table 1.1-2: Channel Bandwidths for each NR band

		NR band / SCS / UE Channel bandwidth									
NR Band	**SCS kHz**	**5 MHz**	**10 MHz (notes 1, 2)**	**15 MHz (note 2)**	**20 MHz (note 2)**	**25 MHz (note 2)**	**30 MHz**	**35 MHz**	**40 MHz**	**45 MHz**	**50 MHz**
n1	15	Yes	Yes	Yes	Yes	Yes Note 4	Yes Note 4		Yes Note 4		Yes Note 4
	30		Yes	Yes	Yes	Yes Note 4	Yes Note 4		Yes Note 4		Yes Note 4
	60		Yes	Yes	Yes	Yes Note 4	Yes Note 4		Yes Note 4		Yes Note 4
n3	15	Yes	Yes	Yes	Yes	Yes	Yes	Yes Note 4	Yes	Yes Note 4	Yes
	30		Yes	Yes	Yes	Yes	Yes	Yes Note 4	Yes	Yes Note 4	Yes
	60		Yes	Yes	Yes	Yes	Yes	Yes Note 4	Yes	Yes Note 4	Yes
n7	15	Yes	Yes	Yes	Yes	Yes	Yes		Yes		Yes
	30		Yes	Yes	Yes	Yes	Yes		Yes		Yes
	60		Yes	Yes	Yes	Yes	Yes		Yes		Yes
n8	15	Yes	Yes	Yes	Yes			Yes Notes 3 and 4			
	30		Yes	Yes	Yes			Yes Notes 3 and 4			
	60										
n20	15	Yes	Yes	Yes	Yes						
	30		Yes	Yes	Yes						
	60										
n28	15	Yes	Yes	Yes	Yes Note 5		Yes Note 5				
	30		Yes	Yes	Yes Note 5		Yes Note 5				
	60										
n38	15	Yes	Yes	Yes	Yes	Yes Note 4	Yes Note 4		Yes Note 4		
	30		Yes	Yes	Yes	Yes Note 4	Yes Note 4		Yes Note 4		
	60		Yes	Yes	Yes	Yes Note 4	Yes Note 4		Yes Note 4		
	60	Yes	Yes	Yes	Yes	Yes Note 4	Yes Note 4		Yes Note 4		Yes
n91	15	Yes	Yes Note 6								
	30										
	60										
n92	15	Yes	Yes	Yes	Yes						
	30										
	60										
n93	15	Yes	Yes Note 6								
	30										
	60										
n94	15	Yes	Yes	Yes	Yes						
	30										
	60										

				NR band / SCS / UE Channel bandwidth							
NR Band	**SCS kHz**	**5 MHz**	**10 MHz (notes 1, 2)**	**15 MHz (note 2)**	**20 MHz (note 2)**	**25 MHz (note 2)**	**30 MHz**	**35 MHz**	**40 MHz**	**45 MHz**	**50 MHz**
n109	15	Yes	Yes	Yes	Yes	Yes	Yes		Yes Note 3		Yes Note 3
	30	Yes	Yes	Yes	Yes	Yes	Yes		Yes Note 3		Yes Note 3
	60	Yes	Yes	Yes	Yes	Yes	Yes		Yes Note 3		Yes Note 3
NOTE 1: 90 % spectrum utilization may not be achieved for 30 kHz SCS. NOTE 2: 90 % spectrum utilization may not be achieved for 60 kHz SCS. NOTE 3: This aerial UE channel bandwidth applies only to downlink. NOTE 4: This aerial UE channel bandwidth is optional in this version of the present document. NOTE 5: For the 20 MHz bandwidth, the minimum requirements are specified for NR UL carrier frequencies confined to either 713 MHz to 723 MHz or 728 MHz to 738 MHz. NOTE 6: This aerial UE channel bandwidth is applicable only to uplink.											

2 References

2.1 Normative references

References are either specific (identified by date of publication and/or edition number or version number) or non-specific. For specific references, only the cited version applies. For non-specific references, the latest version of the referenced document (including any amendments) applies.

Referenced documents which are not found to be publicly available in the expected location might be found in the ETSI docbox.

NOTE: While any hyperlinks included in this clause were valid at the time of publication, ETSI cannot guarantee their long-term validity.

The following referenced documents are necessary for the application of the present document.

[1] ETSI TS 123 256 (V18.4.0) (04-2025): "5G; Support of Uncrewed Aerial Systems (UAS) connectivity, identification and tracking; Stage 2 (3GPP TS 23.256 version 18.4.0 Release 18)".

[2] ETSI TS 136 101 (V18.12.0) (02-2026): "LTE; Evolved Universal Terrestrial Radio Access (E-UTRA); User Equipment (UE) radio transmission and reception (3GPP TS 36.101 version 18.13.0 Release 18)".

[3] ETSI TS 138 101-1 (V18.14.0) (08-2026): "5G; NR; User Equipment (UE) radio transmission and reception; Part 1: Range 1 Standalone (3GPP TS 38.101-1 version 18.14.0 Release 18)".

[4] ETSI TS 138 521-1 (V18.7.0) (08-2025): "5G; NR; User Equipment (UE) conformance specification; Radio transmission and reception; Part 1: Range 1 Standalone (3GPP TS 38.521-1 version 18.7.0 Release 18)".

[5] ETSI TS 136 521-1 (V18.11.0) (03-2026): "LTE; Evolved Universal Terrestrial Radio Access (E-UTRA); User Equipment (UE) conformance specification; Radio transmission and reception; Part 1: Conformance testing (3GPP TS 36.521-1 version 18.11.0 Release 18)".

[6] ETSI TS 136 508 (V18.11.0) (03-2026): "LTE; Evolved Universal Terrestrial Radio Access (E-UTRA) and Evolved Packet Core (EPC); Common test environments for User Equipment (UE) conformance testing (3GPP TS 36.508 version 18.11.0 Release 18)".

[7] ETSI TS 138 508-1 (V18.7.0) (08-2025): "LTE; 5G; 5GS; User Equipment (UE) conformance specification; Part 1: Common test environment (3GPP TS 38.508-1 version 18.7.0 Release 18)".

[8] ETSI TS 136 523-1 (V18.10.1) (12-2025): "LTE; Evolved Universal Terrestrial Radio Access (E-UTRA) and Evolved Packet Core (EPC); User Equipment (UE) conformance specification; Part 1: Protocol conformance specification (3GPP TS 36.523-1 version 18.10.1 Release 18)".

[9] ETSI TS 138 523-1 (V18.3.0) (05-2025): "LTE; 5G; 5GS; User Equipment (UE) conformance specification; Part 1: Protocol (3GPP TS 38.523-1 version 18.3.0 Release 18)". .

[10] IEC 60068-2-1:2025 ED 7.0: "Environmental testing - Part 2-1: Tests - Test A: Cold".

[11] IEC 60068-2-2:2025 ED 6.0: "Environmental testing - Part 2-2: Tests - Test B: Dry heat".

2.2 Informative references

References are either specific (identified by date of publication and/or edition number or version number) or non-specific. For specific references, only the cited version applies. For non-specific references, the latest version of the referenced document (including any amendments) applies.

NOTE: While any hyperlinks included in this clause were valid at the time of publication, ETSI cannot guarantee their long-term validity.

The following referenced documents may be useful in implementing an ETSI deliverable or add to the reader's understanding, but are not required for conformance to the present document.

[i.1] Directive 2014/53/EU of the European Parliament and of the Council of 16 April 2014 on the harmonisation of the laws of the Member States relating to the making available on the market of radio equipment and repealing Directive 1999/5/EC.

[i.2] ETSI EG 203 336 (V1.2.1) (05-2020): "Guide for the selection of technical parameters for the production of Harmonised Standards covering article 3.1(b) and article 3.2 of Directive 2014/53/EU".

[i.3] Recommendation SM.329-13 (09/2024)): "Unwanted emissions in the spurious domain".

[i.4] Commission Decision 2010/267/EU of 6 May 2010 on harmonised technical conditions of use in the 790-862 MHz frequency band for terrestrial systems capable of providing electronic communications services in the European Union.

[i.5] ECC/DEC/(09)03: "Harmonised technical conditions for mobile/fixed communications networks (MFCN) operating in the band 791-862 MHz", Approved 30 October 2009, Amended 07 November 2025.

[i.6] Commission Implementing Decision C(2015) 5376 final of 4 August 2015 on a standardisation request to the European Committee for Electrotechnical Standardisation and to the European Telecommunications Standards Institute as regards radio equipment in support of Directive 2014/53/EU of the European Parliament and of the Council.

[i.7] ECC Decision 05(05): "Harmonised utilization of spectrum for Mobile/Fixed Communications Networks (MFCN) operating within the band 2500-2690 MHz", Approved 18 March 2005, Amended 05 July 2019.

[i.8] ETSI EN 301 908-1: "IMT cellular networks; Harmonised Standard for access to radio spectrum; Part 1: Introduction and common requirements".

[i.9] Commission Implementing Decision (EU) 2016/687 of 28 April 2016 on the harmonisation of the 694-790 MHz frequency band for terrestrial systems capable of providing wireless broadband electronic communications services and for flexible national use in the Union.

[i.10] ECC Decision (15)01: "Harmonised technical conditions for mobile/fixed communications networks (MFCN) in the band 694-790 MHz including a paired frequency arrangement (Frequency Division Duplex 2x30 MHz) and an optional unpaired frequency arrangement (Supplemental Downlink)", Approved 06 March 2015.

[i.11] ECC Decision 17(06): "The harmonised use of the frequency bands 1427-1452 MHz and 1492-1518 MHz for Mobile/Fixed Communications Networks Supplemental Downlink (MFCN SDL)", approved 17 November 2017, corrected 2 March 2018.

[i.12] ERC Recommendation 74-01: "Unwanted emissions in the spurious domain", Approved 1998, Amended 29 May 2019.

[i.13] Commission Implementing Decision (EU) 2022/173 of 7 February 2022 on the harmonisation of the 900 MHz and 1800 MHz frequency bands for terrestrial systems capable of providing electronic communications services in the Union and repealing Decision 2009/766/EC.

[i.14] ECC Decision (06)13: "Designation of the bands 880-915 MHz, 925-960 MHz, 1710-1785 MHz and 1805-1880 MHz for terrestrial UMTS, LTE, WiMAX and IoT cellular systems", Approved 01 December 2006, Amended 8 March 2019.

[i.15] Commission Implementing Decision (EU) 2020/667 of 6 May 2020 amending Decision 2012/688/EU as regards an update of relevant technical conditions applicable to the frequency bands 1 920-1 980 MHz and 2 110-2 170 MHz.

[i.16] ECC Decision (06)01: "The harmonised utilisation of the bands1920-1980 MHz and 2110-2170 MHz for mobile/fixed communications networks (MFCN) including terrestrial IMT systems", Approved 24 March 2006, Amended 8 March 2019.

[i.17] Commission Implementing Decision (EU) 2020/636 of 8 May 2020 amending Decision 2008/477/EC as regards an update of relevant technical conditions applicable to the 2 500-2 690 MHz frequency band.

[i.18] ECC Decision (22)07: "Harmonised technical conditions for the usage of aerial UE for communications based on LTE and 5G NR in the bands 703-733 MHz, 832-862 MHz, 880-915 MHz, 1710 - 1785 MHz, 1920-1980 MHz, 2500-2570 MHz and 2570-2620 MHz harmonised for MFCN". Approved 18 November 2022.

[i.19] ETSI TS 136 509 (V10.3.0) (09-2014): "LTE; Evolved Universal Terrestrial Radio Access (E-UTRA) and Evolved Packet Core (EPC); Special conformance testing functions for User Equipment (UE) (3GPP TS 36.509 version 10.3.0 Release 10)".

[i.20] ETSI TR 100 028 (all parts) (V1.4.1): "Electromagnetic compatibility and Radio spectrum Matters (ERM); Uncertainties in the measurement of mobile radio equipment characteristics".

3 Definition of terms, symbols and abbreviations

3.1 Terms

For the purposes of the present document, the following terms apply:

aerial User Equipment: aerial UE refers to User Equipment (UE) supporting Uncrewed Aircraft Systems (UAS) features and services

NOTE 1: An aerial UE is considered to have access to UAS services after it has performed a successful authentication and authorization with the UAS Service Supplier (USS) as described in ETSI TS 123 256 [1].

NOTE 2: An aerial UE is installed either on-board an Unmanned Aircraft (e.g. drones) or on-board manned aircraft (e.g. helicopter).

aerial UE transmission bandwidth configuration: set of resource blocks located within the UE channel bandwidth which may be used for transmitting or receiving by the aerial UE

channel bandwidth: RF bandwidth supporting a single NR RF carrier with the transmission bandwidth configured in the uplink or downlink of a cell

NOTE 1: The channel bandwidth is measured in MHz and is used as a reference for transmitter and receiver RF requirements.

NOTE 2: Channel bandwidth and the maximum transmission bandwidth configuration for one NR channel are described in Figure 3.1-1.

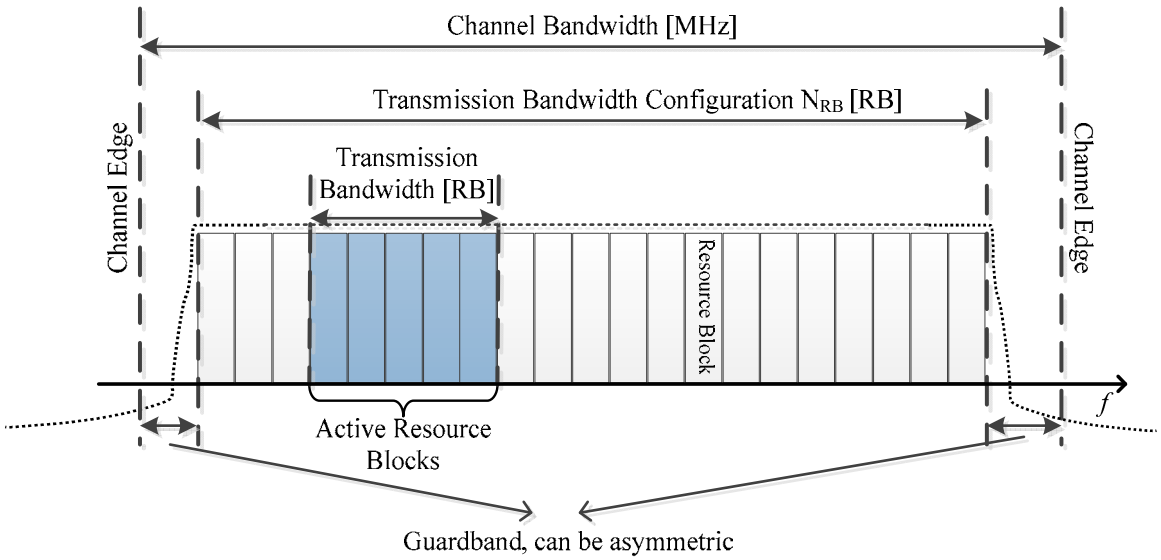


Figure 3.1-1: Definition of the channel bandwidth and the maximum transmission bandwidth configuration for one NR channel

channel edge: lowest and highest frequency of the carrier, separated by the channel bandwidth

mean power: power measured in the operating system bandwidth of the carrier

NOTE: The period of measurement is assumed to be at least one subframe (1 ms) unless otherwise stated.

network signalled value: signalling value sent from the BS to the aerial UE to indicate additional unwanted emission requirements to the UE

occupied bandwidth: width of a frequency band such that, below the lower and above the upper frequency limits, the mean powers emitted are each equal to a specified percentage β/2 of the total mean power of a given emission

operating band: frequency range (paired or unpaired) in which NR operates that is defined with a specific set of technical requirements

output power: mean power of one carrier of the aerial UE, delivered to a load with resistance equal to the nominal load impedance of the transmitter

reference bandwidth: bandwidth in which an emission level is specified

resource block: physical resource consisting of a number of symbols in the time domain and a number of consecutive subcarriers spanning 180 kHz in the frequency domain

sub-block: one contiguous allocated block of spectrum for transmission and reception by the same aerial UE

NOTE: There may be multiple instances of sub-blocks within an RF bandwidth.

sub-block bandwidth: bandwidth of one sub-block

transmission bandwidth: bandwidth of an instantaneous transmission from an aerial UE or BS, measured in Resource Block units

NOTE: See Figure 3.1-1.

3.2 Symbols

For the purposes of the present document, the following symbols apply:

Δf_{OOB}	Δ Frequency of out-of-band emission
$BW_{Channel}$	Channel bandwidth
BW_{GB}	Virtual guard band to facilitate transmitter (receiver) filtering above/below edge CCs
$BW_{Interferer}$	Channel Bandwidth of the interferer

BW_{UTRA}	Channel Bandwidth UTRA
F	Frequency
$F_{Interferer}$ (offset)	Frequency offset of the interferer
$F_{Interferer}$	Frequency of the interferer
$F_{Ioffset}$	Frequency offset of the interferer
F_C	Frequency of the carrier centre frequency
F_{DL_low}	The lowest frequency of the downlink operating band
F_{DL_high}	The highest frequency of the downlink operating band
F_{UL_low}	The lowest frequency of the uplink operating band
F_{UL_high}	The highest frequency of the uplink operating band
F_{edge_low}	The *lower edge* of aggregated channel bandwidth, expressed in MHz
F_{edge_high}	The *higher edge* of aggregated channel bandwidth, expressed in MHz
$F_{Interferer}$	Frequency of the interferer
F_{offset}	Frequency offset from F_{C_high} to the *higher edge* or F_{C_low} to the *lower edge*
$F_{offset,high}$	Frequency offset from $F_{C,high}$ to the upper *UE RF Bandwidth edge*, or from $F_{C,block,\ high}$ to the upper sub-block edge
$F_{offset,low}$	Frequency offset from $F_{C,low}$ to the lower *UE RF Bandwidth edge*, or from $F_{C,block,\ low}$ to the lower sub-block edge
F_{OOB}	The boundary between the NR out of band emission and spurious emission domains
	The power spectral density of the total input signal (power averaged over the useful part of the symbols within the transmission bandwidth configuration, divided by the total number of RE for this configuration and normalized to the subcarrier spacing) at the UE antenna connector, including the own-cell downlink signal or the power spectral density of the total input signal at the UE antenna connector (power averaged over the useful part of the symbols within a given bandwidth and normalized to the said bandwidth), including the own-cell downlink signal
	The total transmitted power spectral density of the own-cell downlink signal (power averaged over the useful part of the symbols within the transmission bandwidth configuration, divided by the total number of RE for this configuration and normalized to the subcarrier spacing) at the eNode B transmit antenna connector
	The total received power spectral density of the own-cell downlink signal (power averaged over the useful part of the symbols within the transmission bandwidth configuration, divided by the total number of RE for this configuration and normalized to the subcarrier spacing) at the UE antenna connector
	The received power spectral density of the total noise and interference for a certain RE (average power obtained within the RE and normalized to the subcarrier spacing) as measured at the UE antenna connector
L_{CRB}	Transmission bandwidth which represents the length of a contiguous resource block allocation expressed in units of resources blocks
	The power spectral density of a white noise source (average power per RE normalized to the subcarrier spacing), simulating interference from cells that are not defined in a test procedure, as measured at the UE antenna connector
	The power spectral density of a white noise source (average power per RE normalized to the subcarrier spacing), simulating interference in non-CRS symbols in ABS subframe from cells that are not defined in a test procedure, as measured at the UE antenna connector
	The power spectral density of a white noise source (average power per RE normalized to the subcarrier spacing), simulating interference in CRS symbols in ABS subframe from all cells that are not defined in a test procedure, as measured at the UE antenna connector
	The power spectral density of a white noise source (average power per RE normalized to the subcarrier spacing), simulating interference in non-ABS subframe from cells that are not defined in a test procedure, as measured at the UE antenna connector
$N_{Offs\text{-}DL}$	Offset used for calculating downlink EARFCN
$N_{Offs\text{-}UL}$	Offset used for calculating uplink EARFCN
N_{RB}	Transmission bandwidth configuration, expressed in units of resource blocks
N_{RB_agg}	Aggregated Transmission Bandwidth Configuration The number of the aggregated RBs within the fully allocated Aggregated Channel bandwidth
$N_{RB,c}$	The transmission bandwidth configuration of component carrier c, expressed in units of resource blocks
$N_{RB,largest\ BW}$	The largest transmission bandwidth configuration of the component carriers in the bandwidth combination, expressed in units of resource block
N_{tone}	Transmission bandwidth configuration for category NB1, expressed in units of tones
$N_{tone\ 3,75\ kHz}$	Transmission bandwidth configuration for category NB1 with 3,75 kHz sub-carrier spacing, expressed in units of tones

$N_{tone\ 15\ kHz}$	Transmission bandwidth configuration for category NB1 with 15 kHz sub-carrier spacing, expressed in units of tones
N_{UL}	Uplink EARFCN
N_{DL}	Downlink EARFCN
$N_{UL/DL}$	Uplink and Downlink EARFCN
NS_x	Network signalled value "x"
P	Number of cell-specific antenna ports
p	Antenna port number
P_{CMAX}	Configured maximum UE output power
P_{CMAX_L}	Lower bound for configured maximum UE output power
$P_{CMAX_L,c}$	Lower bound for configured maximum UE output power for serving cell *c*
$P_{CMAX_L_E\text{-}UTRA,c}$	Lower bound for configured maximum UE output power for serving cell c for E-UTRA
$P_{CMAX_L,f,c,NR}$	Lower bound for configured maximum UE output power for serving cell c for NR
$P_{Interferer}$	Modulated mean power of the interferer
P_{UMAX}	Maximum UE Power with possible power reduction due to modulation type, network signalling values and location near the edge of the band
R_{av}	Minimum average throughput per RB

3.3 Abbreviations

For the purposes of the present document, the following abbreviations apply:

ACLR	Adjacent Channel Leakage Ratio
ACS	Adjacent Channel Selectivity
A-MPR	Additional Maximum Power Reduction
BPSK	Binary Phase-Shift Keying
BS	Base Station
BW	BandWidth
C_RNTI	Cell Radio Network Temporary Identifier
CAA	Civil Aviation Authority
CC	Component Carrier
CH	CHannel
CP	Cyclic Prefix
CW	Continuous Wave
DCI	Downlink Control Information
DL	DownLink
DMRS	DeModulation Reference Signal
ECC	Electronic Communications Committee
EU	European Union
E-UTRA	Evolved Universal Terrestrial Radio Access
FDD	Frequency Division Duplex
HARQ	Hybrid Acknowledge ReQuest
IBB	In-Band Blocking
IE	Information Element
IMT	International Mobile Telecommunications
MAC	Medium Access Control
MBW	Measurement BandWidth
MPR	Maximum Power Reduction
MSG	Mobile Standards Group
NR ACLR	New Radio Adjacent Channel Leakage Ratio
NR	New Radio
NS	Network Signalling
OCNG	OFDMA Channel Noise Generator
OFDM	Orthogonal Frequency Division Multiplexing
OOB	Out Of Band
OP	OFDMA channel noise generator Pattern
PC3	Power Class 3
PDCCH	Physical Downlink Control CHannel
PDSCH	Physical Downlink Shared CHannel
PHS	Personal Handy-phone System
PUSCH	Physical Uplink Shared CHannel

QAM	Quadrature Amplitude Modulation
QPSK	Quadrature Phase Shift Keying
RB	Resource Block
REFSENS	REFerence SENSitivity power level
RF	Radio Frequency
RMC	Reference Measurement Channel
RoT	Rise over Thermal
RRC	Root Raised Cosine
RX	Receiver
SCS	SubCarrier Spacing
SS	System Simulator
TDD	Time Division Duplex
TE	Test Equipment
TH	Temperature High
TH/VH	High extreme Temperature/High extreme Voltage
TH/VL	High extreme Temperature/Low extreme Voltage
TL	Temperature Low
TL/VH	Low extreme Temperature/High extreme Voltage
TL/VL	Low extreme Temperature/Low extreme Voltage
TPC	Transmitter Power Control
TR	Technical Report
UAS	Uncrewed Aircraft Systems
UAS	Uncrewed Aerial Systems
UAV	Uncrewed Aerial Vehicle
UE	User Equipment
UL	UpLink
USS	UAS Service Supplier
UTRA	Universal Terrestrial Radio Access
UUAA	UAV USS Authentication and Authorization
VH	Higher extreme Voltage
VL	Lower extreme Voltage

4 Technical requirements specifications

4.1 Technical requirements specifications for E-UTRA

4.1.1 Environmental profile

The technical requirements of the present document apply under the environmental profile for operation of the equipment, which shall be in accordance with its intended use. The equipment shall comply with all the technical requirements of the present document at all times when operating within the boundary limits of the operational environmental profile defined by its intended use.

4.1.2 Conformance requirements

4.1.2.0 General

The requirements in the present document are based on the assumption that the operating band (see Table 1.1-1) is shared between systems of the IMT family (for band n3 and n8 also GSM) or systems having compatible characteristics.

4.1.2.1 Introduction

To meet the essential requirement under article 3.2 of Directive 2014/53/EU [i.1] and requirements given in ECC Decision (22)07 [i.18] for IMT aerial UE, a set of essential parameters in addition to those in ETSI EN 301 908-1 [i.8] have been identified. Table 4.1.2.1-1 provides a cross reference between these essential parameters and the corresponding technical requirements for equipment within the scope of the present document.

Table 4.1.2.1-1: Cross references for E-UTRA

Essential parameter	Corresponding technical requirements		Corresponding test suite
Transmitter spectrum mask	4.1.2.4	Transmitter Spectrum emissions mask	5.1.3.3
Transmitter unwanted emissions in the out-of-band domain	4.1.2.5	Transmitter adjacent channel leakage power ratio	5.1.3.4
Transmitter unwanted emissions in the spurious domain	4.1.2.6	Transmitter spurious emissions	5.1.3.5
Transmitter power limits	4.1.2.2	Transmitter maximum output power	5.1.3.1
Transmitter Power Control (TPC)	4.1.2.3	Transmitter minimum output power	5.1.3.2
Transmitter power accuracy	4.1.2.2	Transmitter maximum output power	5.1.3.1
Receiver unwanted emissions in the spurious domain	4.1.2.12	Receiver spurious emissions	5.1.3.11
Receiver blocking	4.1.2.9	Receiver Blocking characteristics	5.1.3.8
Receiver desensitization			
Receiver spurious response rejection	4.1.2.10	Receiver spurious response	5.1.3.9
Receiver radio-frequency intermodulation	4.1.2.11	Receiver Intermodulation characteristics	5.1.3.10
Receiver adjacent signal selectivity	4.1.2.8	Receiver Adjacent Channel Selectivity (ACS)	5.1.3.7
Receiver sensitivity	4.1.2.7	Receiver Reference Sensitivity Level	5.1.3.6
Aerial UE identifier	4.1.2.13	Aerial UE identifier	5.1.3.12
Aerial UE Authorization and Authentication	4.1.2.14	Aerial UE Authorization and Authentication	5.1.3.13

The transmitter and receiver characteristics are specified at the antenna connector(s) of the aerial UE.

4.1.2.2 Transmitter Maximum Output Power

4.1.2.2.1 Definition

The following aerial UE Power Classes define the maximum output power for any transmission bandwidth within the channel bandwidth unless otherwise stated. The period of measurement shall be at least one sub-frame (1 ms).

4.1.2.2.2 Limits

The aerial UE maximum output power shall be within the shown value in Table 4.1.2.2.2-1. These requirements do not take into account the maximum power reductions allowed to the aerial UE subject to certain transmission conditions specified in ETSI TS 136 101 [2], clauses 6.2.3 and 6.2.4.

Table 4.1.2.2.2-1: Maximum output power requirement

E-UTRA Band	Power Class 3	
	Lower Limit (dBm)	Upper Limit (dBm)
1	20,3	25,7
3	20,3 (note)	25,7
7	20,3 (note)	25,7
8	20,3 (note)	25,7
20	20,3 (note)	25,0
28	19,8	25,0
38	20,3	25,7

NOTE: For transmission bandwidths (ETSI TS 136 521-1 [5], clause 5) confined within F_{UL_low} and F_{UL_low} + 4 MHz or F_{UL_high} - 4 MHz and F_{UL_high}, the maximum output power requirement applies by reducing the lower limit by 1,5 dB.

4.1.2.2.3 Conformance

The conformance tests for this requirement are defined in clause 5.1.3.1.

4.1.2.3 Transmitter Minimum Output Power

4.1.2.3.1 Definition

The minimum controlled output power of the aerial UE is defined as the broadband transmit power of the aerial UE, i.e. the power in the channel bandwidth for all transmit bandwidth configurations (resource blocks), when the power is set to a minimum value. The minimum output power is defined as the mean power in at least one sub-frame (1 ms).

4.1.2.3.2 Limits

The minimum output power measured shall not exceed the values specified in Table 4.1.2.3.2-1.

Table 4.1.2.3.2-1: Minimum output power

	Channel bandwidth/minimum output power/measurement bandwidth					
	1,4 MHz	**3,0 MHz**	**5 MHz**	**10 MHz**	**15 MHz**	**20 MHz**
Minimum output power	For carrier frequency f ≤ 3,0 GHz: ≤ -39 dBm					
Measurement bandwidth	1,08 MHz	2,7 MHz	4,5 MHz	9,0 MHz	13,5 MHz	18 MHz

4.1.2.3.3 Conformance

The conformance tests for this requirement are defined in clause 5.1.3.2.

4.1.2.4 Transmitter Spectrum Emission Mask

4.1.2.4.1 Definition

The Out of band emissions are unwanted emissions immediately outside the assigned channel bandwidth resulting from the modulation process and non-linearity in the transmitter but excluding spurious emissions. This out of band emission limit is specified in terms of a spectrum emission mask and an adjacent channel leakage power ratio. The spectrum emission mask of the aerial UE applies to frequencies (Δf_{OOB}) starting from the ± edge of the assigned E-UTRA channel bandwidth.

4.1.2.4.2 Limits

The power of any aerial UE emission shall fulfil requirements in Table 4.1.2.4.2-1.

Table 4.1.2.4.2-1: General E-UTRA spectrum emission mask, E-UTRA bands ≤ 3 GHz

Spectrum emission limit (dBm)/ Channel bandwidth							
Δf_{OOB} (MHz)	**1,4 MHz**	**3,0 MHz**	**5 MHz**	**10 MHz**	**15 MHz**	**20 MHz**	**Measurement bandwidth**
0 to 1	-8,5	-11,5	-13,5	-16,5	-18,5	-19,5	30 kHz
1 to 2,5	-8,5	-8,5	-8,5	-8,5	-8,5	-8,5	1 MHz
2,5 to 2,8	-23,5	-8,5	-8,5	-8,5	-8,5	-8,5	1 MHz
2,8 to 5		-8,5	-8,5	-8,5	-8,5	-8,5	1 MHz
5 to 6		-23,5	-11,5	-11,5	-11,5	-11,5	1 MHz
6 to 10			-23,5	-11,5	-11,5	-11,5	1 MHz
10 to 15				-23,5	-11,5	-11,5	1 MHz
15 to 20					-23,5	-11,5	1 MHz
20 to 25						-23,5	1 MHz

NOTE 1: The first and last measurement position with a 30 kHz filter is at Δf_{OOB} equals to 0,015 MHz and 0,985 MHz.
NOTE 2: The first and last measurement position with a 1 MHz filter for 1 MHz to 2,5 MHz offset range is at Δf_{OOB} equals to 1,5 MHz and 2,0 MHz. Similarly for other Δf_{OOB} ranges.
NOTE 3: The measurements shall be performed above the upper edge of the channel and below the lower edge of the channel.
NOTE 4: For the 2,5 MHz to 2,8 MHz offset range with 1,4 MHz channel bandwidth, the measurement position is at Δf_{OOB} equals to 3 MHz.

4.1.2.4.3 Conformance

The conformance tests for this requirement are defined in clause 5.1.3.3.

4.1.2.5 Transmitter Adjacent Channel Leakage Power Ratio

4.1.2.5.1 Definition

Adjacent Channel Leakage Power Ratio (ACLR) is the ratio of the filtered mean power centred on the assigned channel frequency to the filtered mean power centred on an adjacent channel frequency.

E-UTRA Adjacent Channel Leakage Power Ratio (E-UTRA ACLR) is the ratio of the filtered mean power centred on the assigned channel frequency to the filtered mean power centred on an adjacent channel frequency at nominal channel spacing. The assigned E-UTRA channel power and adjacent E-UTRA channel power are measured with rectangular filters with measurement bandwidths specified in Table 6.6.2.3.3.1-1 in ETSI TS 136 521-1 [5].

UTRA Adjacent Channel Leakage Power Ratio is specified for both the first UTRA adjacent channel (UTRA ACLR1) and the 2nd UTRA adjacent channel (UTRA ACLR2). The UTRA channel power is measured with a RRC bandwidth filter with roll-off factor $\alpha = 0,22$. The assigned E-UTRA channel power is measured with a rectangular filter with measurement bandwidth specified in Table 4.1.2.5.2-2.

4.1.2.5.2 Limits

If the measured adjacent channel power is greater than -50 dBm then the measured E-UTRA ACLR shall be higher than the limits in Table 4.1.2.5.2-1.

Table 4.1.2.5.2-1: E-UTRA aerial UE ACLR

	Channel bandwidth/E-UTRA ACLR1/measurement bandwidth					
	1,4 MHz	**3,0 MHz**	**5 MHz**	**10 MHz**	**15 MHz**	**20 MHz**
E-UTRA ACLR1	29,2 dB	29,2 dB	29,2 dB	29,2 dB	29,2 dB	29,2 dB
E-UTRA channel Measurement bandwidth	1,08 MHz	2,7 MHz	4,5 MHz	9,0 MHz	13,5 MHz	18 MHz
Aerial UE channel	+1,4 MHz or -1,4 MHz	+3 MHz or -3 MHz	+5 MHz or -5 MHz	+10 MHz or -10 MHz	+15 MHz or -15 MHz	+20 MHz or -20 MHz

If the measured UTRA channel power is greater than -50 dBm then the measured UTRA ACLR1, UTRA ACLR2 shall be higher than the limits in Table 4.1.2.5.2-2.

Table 4.1.2.5.2-2: UTRA aerial UE ACLR

	Channel bandwidth/UTRA ACLR1/2/measurement bandwidth					
	1,4 MHz	**3,0 MHz**	**5 MHz**	**10 MHz**	**15 MHz**	**20 MHz**
UTRA ACLR1	32,2 dB	32,2 dB	32,2 dB	32,2 dB	32,2 dB	32,2 dB
Adjacent channel centre frequency offset (in MHz)	0,7 + $BW_{UTRA}/2$ / -0,7 - $BW_{UTRA}/2$	1,5 + $BW_{UTRA}/2$ / -1,5 - $BW_{UTRA}/2$	2,5 + $BW_{UTRA}/2$ / -2,5 - $BW_{UTRA}/2$	5 + $BW_{UTRA}/2$ / -5 - $BW_{UTRA}/2$	7,5 + $BW_{UTRA}/2$ / -7,5 - $BW_{UTRA}/2$	10 + $BW_{UTRA}/2$ / -10 - $BW_{UTRA}/2$
UTRA ACLR2	-	-	35,2 dB	35,2 dB	35,2 dB	35,2 dB
Adjacent channel centre frequency offset (in MHz)	-	-	2,5 + 3 × $BW_{UTRA}/2$ / -2,5 - 3 × $BW_{UTRA}/2$	5 + 3 × $BW_{UTRA}/2$ / -5 - 3 × $BW_{UTRA}/2$	7,5 + 3 × $BW_{UTRA}/2$ / -7,5 - 3 × $BW_{UTRA}/2$	10 + 3 × $BW_{UTRA}/2$ / -10 - 3 × $BW_{UTRA}/2$
E-UTRA channel Measurement bandwidth	1,08 MHz	2,7 MHz	4,5 MHz	9,0 MHz	13,5 MHz	18 MHz
UTRA 5 MHz channel Measurement bandwidth (see note 1)	3,84 MHz	3,84 MHz	3,84 MHz	3,84 MHz	3,84 MHz	3,84 MHz
UTRA 1,6 MHz channel measurement bandwidth (see note 2)	1,28 MHz	1,28 MHz	1,28 MHz	1,28 MHz	1,28 MHz	1,28 MHz
NOTE 1: Shall apply for E-UTRA FDD co-existence with UTRA FDD in paired spectrum. NOTE 2: Shall apply for E-UTRA TDD co-existence with UTRA TDD in unpaired spectrum. NOTE 3: BW_{UTRA} for UTRA FDD shall be 5 MHz and for UTRA TDD shall be 1,6 MHz.						

4.1.2.5.3 Conformance

The conformance tests for this requirement are defined in clause 5.2.3.4.

4.1.2.6 Transmitter Spurious Emissions

4.1.2.6.1 Definition

Spurious emissions are emissions which are caused by unwanted transmitter effects such as harmonics emission, parasitic emissions, intermodulation products and frequency conversion products, but exclude out-of-band emissions. The spurious emission limits are specified in terms of general requirements in line with Recommendation ITU-R SM.329-12 [i.3] and E-UTRA operating band requirement to address aerial UE co-existence.

4.1.2.6.2 Limits

The spurious emission limits in Table 4.1.2.6.2-2 apply for the frequency ranges that are more than Δf_{OOB} (MHz) from the edge of the channel bandwidth shown in Table 4.1.2.6.2-1.

The measured average power of spurious emission for general requirements shall not exceed the described values in Table 4.1.2.6.2-2.

The measured average power of spurious emission for E-UTRA operating band specific requirements to protected bands shall not exceed the described values in Tables 4.1.2.6.2-3 to 4.1.2.6.2-6.

Table 4.1.2.6.2-1: Δf_{OOB} boundary between E-UTRA channel and spurious emission domain

Channel bandwidth	1,4 MHz	3,0 MHz	5 MHz	10 MHz	15 MHz	20 MHz
Δf_{OOB} (MHz)	2,8	6	10	15	20	25

Table 4.1.2.6.2-2: General spurious emissions limits

Frequency range	Maximum level	Measurement bandwidth
9 kHz ≤ f < 150 kHz	-36 dBm	1 kHz
150 kHz ≤ f < 30 MHz	-36 dBm	10 kHz
30 MHz ≤ f < 1 000 MHz	-36 dBm	100 kHz
1 GHz ≤ f < 12,75 GHz	-30 dBm	1 MHz
12,75 GHz ≤ f < 5th harmonic of the upper frequency edge of the UL operating band in GHz	-30 dBm	1 MHz

The additional requirements in Table 4.1.2.6.2-3 apply for the frequency ranges that are more than Δf_{OOB} (MHz) from the edge of the channel bandwidth shown in Table 4.1.2.6.2-1.

Table 4.1.2.6.2-3: Spurious emission band aerial UE co-existence limits (network signalled value "NS_01")

E-UTRA Band	Spurious emission						
	Protected band	Frequency range (MHz)			Maximum Level (dBm)	MBW (MHz)	Comment
1	E-UTRA Band 1, 3, 7, 8, 20, 22, 28, 31, 32, 38, 40, 41,42, 43, 65, 67, 68, 72, 87, 88	F_{DL_low}	-	F_{DL_high}	-50	1	
	E-UTRA Band 34	F_{DL_low}	-	F_{DL_high}	-50	1	Note 3
	Frequency range	1 895	-	1 915	-15,5	5	Notes 3, 8
	Frequency range	1 915	-	1 920	+1,6	5	Notes 3, 8
3	E-UTRA Band 1, 7, 8, 20, 28, 31, 32, 33, 34, 38, 40, 41, 43, 65, 67, 68, 72, 87, 88	F_{DL_low}	-	F_{DL_high}	-50	1	
	E-UTRA Band 3	F_{DL_low}	-	F_{DL_high}	-50	1	Note 3
	E-UTRA Band 22, 42	F_{DL_low}	-	F_{DL_high}	-50	1	Note 2
7	E-UTRA Band 1, 3, 7, 8, 20, 22, 28, 31, 32, 33, 34, 40, 42, 43, 65, 67, 68, 72, 87, 88	F_{DL_low}	-	F_{DL_high}	-50	1	
	Frequency range	2 570	-	2 575	+1,6	5	Notes 3, 4
	Frequency range	2 575	-	2 595	-15,5	5	Notes 3, 4
	Frequency range	2 595		2 620	-40	1	Notes 3, 4
8	E-UTRA Band 1, 20, 28, 31, 32, 33, 34, 38, 40, 65, 67, 68, 72, 87, 88	F_{DL_low}	-	F_{DL_high}	-50	1	
	E-UTRA Band 3, 41	F_{DL_low}	-	F_{DL_high}	-50	1	Note 2
	E-UTRA Band 7	F_{DL_low}	-	F_{DL_high}	-50	1	Note 2
	E-UTRA Band 8	F_{DL_low}	-	F_{DL_high}	-50	1	Note 3
	E-UTRA Band 22, 42, 43	F_{DL_low}	-	F_{DL_high}	-50	1	Note 2
20	E-UTRA Band 1, 3, 7, 8, 22, 31, 32, 33, 34, 40, 43, 65, 67, 68, 72, 87, 88	F_{DL_low}	-	F_{DL_high}	-50	1	
	E-UTRA Band 20	F_{DL_low}	-	F_{DL_high}	-50	1	Note 3
	E-UTRA Band 38, 42	F_{DL_low}	-	F_{DL_high}	-50	1	Note 2
	Frequency range	758	-	788	-50	1	
28	E-UTRA Band 3, 7, 8, 20, 31, 34, 38, 41, 72, 87, 88	F_{DL_low}	-	F_{DL_high}	-50	1	
	E-UTRA Band 1, 22, 32, 42, 43, 65	F_{DL_low}	-	F_{DL_high}	-50	1	Note 2
	E-UTRA Band 1	F_{DL_low}	-	F_{DL_high}	-50	1	Notes 6, 9

E-UTRA Band	Spurious emission						
	Protected band	**Frequency range (MHz)**			**Maximum Level (dBm)**	**MBW (MHz)**	**Comment**
	Frequency range	758	-	773	-32	1	Note 3
	Frequency range	773	-	803	-50	1	
	Frequency range	470	-	694	-42	8	Notes 3, 7
38	E-UTRA Band 1, 3, 8, 20, 22, 28, 31, 32, 33, 34, 40, 42, 43, 65, 67, 68, 72, 87, 88	F_{DL_low}	-	F_{DL_high}	-50	1	
	Frequency range	2 620	-	2 645	-15,5	5	Notes 3, 5
	Frequency range	2 645	-	2 690	-40	1	Notes 3, 5

NOTE 1: F_{DL_low} and F_{DL_high} refer to each frequency range of the protected E-UTRA band.
NOTE 2: As exceptions, measurements with a level up to the applicable requirements defined in Table 4.1.2.6.2-3 are permitted for each assigned E-UTRA carrier used in the measurement due to 2nd, 3rd or 4th harmonic spurious emissions. Due to spreading of the harmonic emission, the exception shall be allowed for the first 1 MHz frequency range immediately outside the harmonic emission on both sides of the harmonic emission. This results in an overall exception interval centred at the harmonic emission of (2 MHz + N x L_{CRB} x 180 kHz), where N is 2, 3, 4 for the 2nd, 3rd or 4th harmonic respectively. The exception is allowed if the measurement bandwidth (MBW) totally or partially overlaps the overall exception interval.
NOTE 3: These requirements shall also apply for the frequency ranges that are less than Δf_{OOB} (MHz) in Table 4.1.2.6.2-1 from the edge of the channel bandwidth.
NOTE 4: This requirement shall be applicable for any channel bandwidths within the range 2 500 MHz to 2 570 MHz with the following restriction: for carriers of 15 MHz bandwidth when carrier centre frequency is within the range 2 560,5 MHz to 2 562,5 MHz and for carriers of 20 MHz bandwidth when carrier centre frequency is within the range 2 552 MHz to 2 560 MHz the requirement is applicable only for an uplink transmission bandwidth less than or equal to 54 RB.
NOTE 5: This requirement shall be applicable for any channel bandwidths within the range 2 570 MHz to 2 615 MHz with the following restriction: for carriers of 15 MHz bandwidth when carrier centre frequency is within the range 2 605,5 MHz to 2 607,5 MHz and for carriers of 20 MHz bandwidth when carrier centre frequency is within the range 2 597 MHz to 2 605 MHz the requirement is applicable only for an uplink transmission bandwidth less than or equal to 54 RB.
For carriers with channel bandwidth overlapping the frequency range 2 615 MHz to 2 620 MHz the requirement shall apply with the maximum output power configured to +19 dBm.
NOTE 6: As exceptions, measurements with a level up to the applicable requirement of -36 dBm/MHz is permitted for each assigned E-UTRA carrier used in the measurement due to 3rd harmonic spurious emissions. An exception is allowed if there is at least one individual RB within the transmission bandwidth (see Figure 5.4.2-1 in ETSI TS 136 521-1 [5]) for which the 3rd harmonic totally or partially overlaps the Measurement Bandwidth (MBW).
NOTE 7: This requirement shall be applicable in the case of a 10 MHz E-UTRA carrier confined within 703 MHz and 733 MHz, otherwise the requirement of -25 dBm with a measurement bandwidth of 8 MHz applies.
NOTE 8: This requirement shall be applicable for any channel bandwidths within the range 1 920 MHz to 1 980 MHz with the following restriction: for carriers of 15 MHz bandwidth when carrier centre frequency is within the range 1 927,5 MHz to 1 929,5 MHz and for carriers of 20 MHz bandwidth when carrier centre frequency is within the range 1 930 MHz to 1 938 MHz the requirement is applicable only for an uplink transmission bandwidth less than or equal to 54 RB.
NOTE 9: Applicable when the assigned E-UTRA carrier is confined within 718 MHz and 748 MHz and when the channel bandwidth used is 5 MHz or 10 MHz.

Table 4.1.2.6.2-4: Additional spurious emissions limits (network signalled value "NS_01")

E-UTRA band	Protected Frequency range	Maximum Level (dBm)	MBW (MHz)
20	470 MHz ≤ f ≤ 790 MHz	-65	8 MHz
NOTE: The conformance shall be assessed using the measurement position placed at the following centre frequencies: 474 MHz, 586 MHz, 690 MHz, 754 MHz, 770 MHz and 786 MHz.			

Table 4.1.2.6.2-5: Additional spurious emissions limits (network signalled value "NS_44")

Frequency range (MHz)	Channel bandwidth (MHz) / Spectrum emission limit (dBm)	Measurement bandwidth	Note
	1.4, 3, 5, 10, 15, 20 MHz		
2 620 ≤ f < 2 645	-15,5	5 MHz	Note
2 645 ≤ f ≤ 2 690	-40	1 MHz	
2 690 ≤ f ≤ 2 900	-50	1 MHz	
NOTE: For these adjacent bands, the emission limit could imply risk of harmful interference to UE(s) operating in the protected operating band.			

Table 4.1.2.6.2-6: Additional spurious emissions limits (network signalled value "NS_70")

Frequency range (MHz)	Channel bandwidth (MHz) / Spectrum emission limit (dBm)	Measurement bandwidth
	5, 10, 15, 20 MHz	
1 675 ≤ f ≤ 1 710	-40	1 MHz

4.1.2.6.3 Conformance

The conformance tests for this requirement are defined in clause 5.1.3.5.

4.1.2.7 Receiver Reference Sensitivity Level

4.1.2.7.0 General

The receiver characteristics are specified at the antenna connector(s) of the aerial UE.

4.1.2.7.1 Definition

Reference sensitivity measures the aerial UE's ability to receive data with a given average throughput for a specified reference measurement channel, under conditions of low signal level, ideal propagation and no added noise.

4.1.2.7.2 Limits

The throughput shall be ≥ 95 % of the maximum throughput of the reference measurement channels as specified in ETSI TS 136 521-1 [5], clauses A.2.2, A.2.3 and A.3.2 (with one sided dynamic OCNG Pattern OP.1 FDD/TDD for the DL-signal as described in ETSI TS 136 521-1 [5], clauses A.5.1.1 and A.5.2.1) with parameters specified in Tables 7.3.3-1 and 7.3.3-2 in ETSI TS 136 521-1 [5].

Table 4.1.2.7.2-1: Reference sensitivity QPSK $P_{REFSENS}$

Channel bandwidth							
E-UTRA Band	1,4 MHz (dBm)	3 MHz (dBm)	5 MHz (dBm)	10 MHz (dBm)	15 MHz (dBm)	20 MHz (dBm)	Duplex Mode
1	-	-	-99,3	-96,3	-94,5	-93,3	FDD
3	-101,0	-98,0	-96,3	-93,3	-91,5	-90,3	FDD
7	-	-	-97,3	-94,3	-92,5	-91,3	FDD
8	-101,5	-98,5	-96,3	-93,3	-	-	FDD
20	-	-	-96,3	-93,3	-90,5	-89,3	FDD
28	-	-99,5	-97,8	-94,8	-93,0	-90,3	FDD
38	-	-	-99,3	-96,3	-94,5	-93,3	TDD
NOTE 1: The transmitter shall be set to maximum output power level (ETSI TS 136 521-1 [5], Table 7.3.5-2).							
NOTE 2: The reference measurement channel is specified in ETSI TS 136 521-1 [5], clause A.3.2 with one sided dynamic OCNG Pattern OP.1 FDD/TDD as described in ETSI TS 136 521-1 [5], clauses A.5.1.1 and A.5.2.1.							
NOTE 3: The signal power is specified per port.							

The reference receiver sensitivity (REFSENS) requirement specified in Table 4.1.2.7.2-1 shall be met for an uplink transmission bandwidth less than or equal to that specified in ETSI TS 136 521-1 [5], Table 7.3.5-2.

4.1.2.7.3 Conformance

The conformance tests for this requirement are defined in clause 5.1.3.6.

4.1.2.8 Receiver Adjacent Channel Selectivity (ACS)

4.1.2.8.1 Definition

Adjacent Channel Selectivity (ACS) is a measure of a receiver's ability to receive an E-UTRA signal at its assigned channel frequency in the presence of an adjacent channel signal at a given frequency offset from the centre frequency of the assigned channel. ACS is the ratio of the receive filter attenuation on the assigned channel frequency to the receive filter attenuation on the adjacent channel(s).

4.1.2.8.2 Limits

The throughput R_{av} shall be $\geq$ 95 % of the maximum throughput of the reference measurement channels as specified in ETSI TS 136 521-1 [5] under the conditions specified in Table 4.1.2.8.2-2 and also under the conditions specified in Table 4.1.2.8.2-3.

Table 4.1.2.8.2-1: Adjacent channel selectivity

		Channel bandwidth					
Rx Parameter	**Units**	**1,4 MHz**	**3 MHz**	**5 MHz**	**10 MHz**	**15 MHz**	**20 MHz**
ACS	dB	33,0	33,0	33,0	33,0	30	27

Table 4.1.2.8.2-2: Test parameters for Adjacent channel selectivity, Case 1

Rx Parameter	**Units**	**Channel bandwidth**					
		1,4 MHz	**3 MHz**	**5 MHz**	**10 MHz**	**15 MHz**	**20 MHz**
Power in Transmission Bandwidth Configuration	dBm	REFSENS + 14 dB					
$P_{Interferer}$	dBm	REFSENS +45,5 dB	REFSENS +45,5 dB	REFSENS +45,5 dB	REFSENS +45,5 dB	REFSENS +42,5 dB	REFSENS +39,5 dB
$BW_{Interferer}$	MHz	1,4	3	5	5	5	5
$F_{Interferer}$ (offset)	MHz	1,4025	3,0075	5,0025	7,5075	10,0125	12,5025
NOTE 1: The transmitter shall be set to 4 dB below P_{CMAX_L} or $P_{CMAX_L_CA}$ as defined in clause 6.2.5 in ETSI TS 136 101 [2].							
NOTE 2: The interferer shall consist of the Reference measurement channel specified in clause A.3.2 of ETSI TS 136 521-1 [5] with set-up according to clause C.3.1 of ETSI TS 136 521-1 [5].							
NOTE 3: REFSENS as defined in clause 7.3.3 in ETSI TS 136 521-1 [5].							
NOTE 4: For DL category M1 UE, the reference sensitivity for category M1 in ETSI TS 136 521-1 [5], Table 7.3EA-1 and 7.3EA-2 should be used as REFSENS for the power in Transmission Bandwidth Configuration and $P_{Interferer}$.							
NOTE 5: For DL category M1 UE, the parameters for the applicable channel bandwidth apply.							

Table 4.1.2.8.2-3: Test parameters for Adjacent channel selectivity, Case 2

Rx Parameter	Units	Channel bandwidth					
		1,4 MHz	3 MHz	5 MHz	10 MHz	15 MHz	20 MHz
Power in Transmission Bandwidth Configuration	dBm	-56,5	-56,5	-56,5	-56,5	-53,5	-50,5
$P_{Interferer}$	dBm	-25					
$BW_{Interferer}$	MHz	1,4	3	5	5	5	5
$F_{Interferer}$ (offset)	MHz	1,4025	3,0075	5,0025	7,5075	10,0125	12,5025
NOTE 1: The transmitter shall be set to 24 dB below P_{CMAX_L} or $P_{CMAX_L_CA}$ as defined in clause 6.2.5 in ETSI TS 136 101 [2].							
NOTE 2: The interferer shall consist of the Reference measurement channel specified in clause A.3.2 of ETSI TS 136 521-1 [5] with set-up according to clause C.3.1 of ETSI TS 136 521-1 [5].							

4.1.2.8.3 Conformance

The conformance tests for this requirement are defined in clause 5.1.3.7.

4.1.2.9 Receiver Blocking Characteristics

4.1.2.9.1 Definition

The blocking characteristic is a measure of the receiver's ability to receive a wanted signal at its assigned channel frequency in the presence of an unwanted interferer on frequencies other than those of the spurious response or the adjacent channels, without this unwanted input signal causing a degradation of the performance of the receiver beyond a specified limit.

4.1.2.9.2 Limits

With parameters specified in Tables 4.1.2.9.2-1 and 4.1.2.9.2-2, the throughput shall be ≥ 95 % of the maximum throughput of the reference measurement channels as specified in clauses A.2.2, A.2.3 and A.3.2 (with one sided dynamic OCNG Pattern OP.1 FDD/TDD for the DL-signal as described in clauses A.5.1.1 and A.5.2.1) in ETSI TS 136 521-1 [5].

With parameters specified in Tables 4.1.2.9.2-3 and 4.1.2.9.2-4, the throughput shall be ≥ 95 % of the maximum throughput of the reference measurement channels as specified in clauses A.2.2, A.2.3 and A.3.2 (with one sided dynamic OCNG Pattern OP.1 FDD/TDD for the DL-signal as described in clauses A.5.1.1 and A.5.2.1) in ETSI TS 136 521-1 [5], except for the spurious response frequencies.

For Table 4.1.2.9.2-4 in frequency range 1, 2 and 3, up to $\max(24,\ 6\cdot\lceil N_{RB}/6\rceil)$ exceptions are allowed for spurious response frequencies in each assigned frequency channel when measured using a 1 MHz step size, where N_{RB} is the number of resource blocks in the downlink transmission bandwidth configuration. For these exceptions the requirements of clause 4.1.2.10 Spurious response are applicable.

With parameters specified in Table 4.1.2.9.2-5, the throughput shall be ≥ 95 % of the maximum throughput of the reference measurement channels as specified in clauses A.2.2, A.2.3 and A.3.2 (with one sided dynamic OCNG Pattern OP.1 FDD/TDD for the DL-signal as described in clauses A.5.1.1 and A.5.2.1) in ETSI TS 136 521-1 [5].

Table 4.1.2.9.2-1: In-band blocking parameters

Rx Parameter	Units	Channel bandwidth					
		1,4 MHz	3 MHz	5 MHz	10 MHz	15 MHz	20 MHz
Power in Transmission Bandwidth Configuration	dBm	REFSENS + channel bandwidth specific value below					
		6	6	6	6	7	9
$BW_{Interferer}$	MHz	1,4	3	5	5	5	5
$F_{Ioffset, case\ 1}$	MHz	2,1125	4,5075	7,5125	7,5025	7,5075	7,5125
$F_{Ioffset, case\ 2}$	MHz	3,5075	7,5075	12,5075	12,5125	12,5025	12,5075

NOTE 1: The transmitter shall be set to 4 dB below P_{CMAX_L} at the minimum uplink configuration specified in ETSI TS 136 101 [2] (Table 7.3.1-2 with P_{CMAX_L} as defined in clause 6.2.5).
NOTE 2: The interferer shall consist of the Reference measurement channel specified in clause A.3.2 of ETSI TS 136 521-1 [5] with a set-up according to clause C.3.1 of ETSI TS 136 521-1 [5].
NOTE 3: REFSENS as defined in clause 7.3.3 in ETSI TS 136 521-1 [5].
NOTE 4: For DL category M1 UE, the reference sensitivity for category M1 in ETSI TS 136 521-1 [5], Tables 7.3EA-1 and 7.3EA-2 should be used as REFSENS for the power in Transmission Bandwidth Configuration.
NOTE 5: For DL category M1 UE, the parameters for the applicable channel bandwidth apply.

Table 4.1.2.9.2-2: In-band blocking

E-UTRA band	Parameter	Units	Case 1	Case 2
	$P_{Interferer}$	dBm	-56	-44
	$F_{Interferer}$ (Offset)	MHz	= -BW/2 - $F_{Ioffset, case\ 1}$ and = +BW/2 + $F_{Ioffset, case\ 1}$	≤ -BW/2 - $F_{Ioffset, case\ 2}$ and ≥ +BW/2 + $F_{Ioffset, case\ 2}$
1, 3, 7, 8, 20, 22, 28, 31, 33, 34, 38, 40, 41, 42, 43, 65, 68, 72, 87, 88	$F_{Interferer}$	MHz	(note 2)	F_{DL_low} - 15 to F_{DL_high} + 15

NOTE 1: For certain bands, the unwanted modulated interfering signal may not fall inside the aerial UE receive band, but within the first 15 MHz below or above the aerial UE receive band.
NOTE 2: For each carrier frequency the requirement is valid for two frequencies:
a) the carrier frequency - BW/2 - FIoffset, case 1; and
b) the carrier frequency + BW/2 + FIoffset, case 1.
NOTE 3: $F_{Interferer}$ range values for unwanted modulated interfering signal are interferer centre frequencies.

Table 4.1.2.9.2-3: Out-of-band blocking parameters

Rx Parameter	Units	Channel bandwidth					
		1,4 MHz	3 MHz	5 MHz	10 MHz	15 MHz	20 MHz
Power in Transmission Bandwidth Configuration	dBm	REFSENS + channel bandwidth specific value below					
		6	6	6	6	7	9

NOTE 1: The transmitter shall be set to 4 dB below P_{CMAX_L} at the minimum uplink configuration specified in ETSI TS 136 101 [2] (Table 7.3.1-2 with P_{CMAX_L} as defined in clause 6.2.5).
NOTE 2: Reference measurement channel is clause A.3.2 of ETSI TS 136 521-1 [5].
NOTE 3: REFSENS as defined in clause 7.3.3 in ETSI TS 136 521-1 [5].
NOTE 4: For DL category M1 UE, the reference sensitivity for category M1 in ETSI TS 136 521-1 [5], Tables 7.3EA-1 and 7.3EA-2 should be used as REFSENS for the power in Transmission Bandwidth Configuration.
NOTE 5: For DL category M1 UE, the parameters for the applicable channel bandwidth apply.

Table 4.1.2.9.2-4: Out-of-band blocking

E-UTRA band	Parameter	Units	Frequency		
			Range 1	Range 2	Range 3
	$P_{Interferer}$	dBm	-44	-30	-15
1, 3, 7, 8, 20, 22, 28, 31, 33, 34, 38, 40, 41, 42 (note 2), 43 (note 2), 65, 68, 72, 87, 88	$F_{Interferer}$ (CW)	MHz	F_{DL_low} - 15 to F_{DL_low} - 60	F_{DL_low} - 60 to F_{DL_low} - 85	F_{DL_low} - 85 to 1 MHz
			F_{DL_high} + 15 to F_{DL_high} + 60	F_{DL_high} + 60 to F_{DL_high} + 85	F_{DL_high} + 85 to +12 750 MHz
NOTE 1: Range 3 shall be tested only with the highest channel bandwidth. NOTE 2: The power level of the interferer ($P_{Interferer}$) for Range 3 shall be modified to -20 dBm for $F_{Interferer}$ > 2 800 MHz and $F_{Interferer}$ < 4 400 MHz.					

Table 4.1.2.9.2-5: Narrow-band blocking

Parameter	Units	Channel Bandwidth					
		1,4 MHz	3 MHz	5 MHz	10 MHz	15 MHz	20 MHz
P_w	dBm	$P_{REFSENS}$ + channel-bandwidth specific value below					
		22	18	16	13	14	16
P_{uw} (CW)	dBm	-55	-55	-55	-55	-55	-55
F_{uw} (offset for Δf = 15 kHz)	MHz	0,9075	1,7025	2,7075	5,2125	7,7025	10,2075
NOTE 1: The transmitter shall be set a 4 dB below P_{CMAX_L} at the minimum uplink configuration specified in ETSI TS 136 101 [2] (Table 7.3.1-2 with P_{CMAX_L} as defined in clause 6.2.5). NOTE 2: Reference measurement channel is in clause A.3.2 of ETSI TS 136 521-1 [5]. NOTE 3: REFSENS as defined in clause 7.3.3 in ETSI TS 136 521-1 [5]. NOTE 4: For DL category M1 UE, the reference sensitivity for category M1 in ETSI TS 136 521-1 [5], Tables 7.3EA-1 and 7.3EA-2 should be used as $P_{REFSENS}$ for P_w. NOTE 5: For DL category M1 UE, the parameters for the applicable channel bandwidth apply.							

4.1.2.9.3 Conformance

The conformance tests for this requirement are defined in clause 5.1.3.8.

4.1.2.10 Receiver Spurious Response

4.1.2.10.1 Definition

Spurious response is a measure of the receiver's ability to receive a wanted signal on its assigned channel frequency without exceeding a given degradation due to the presence of an unwanted CW interfering signal at any other frequency at which a response is obtained i.e. for which the out-of-band blocking limit as specified in Table 4.1.2.9.2-4 is not met.

4.1.2.10.2 Limits

The throughput shall be ≥ 95 % of the maximum throughput of the reference measurement channels as specified in clauses A.2.2, A.2.3 and A.3.2 (with one sided dynamic OCNG Pattern OP.1 FDD/TDD for the DL-signal as described in clauses A.5.1.1 and A.5.2.1) in ETSI TS 136 521-1 [5] with parameters specified in Tables 4.1.2.10.2-1 and 4.1.2.10.2-2.

Table 4.1.2.10.2-1: Spurious response parameters

Rx Parameter	Units	Channel bandwidth					
		1,4 MHz	3 MHz	5 MHz	10 MHz	15 MHz	20 MHz
Power in Transmission Bandwidth Configuration	dBm	REFSENS + channel bandwidth specific value below					
		6	6	6	6	7	9
NOTE 1: The transmitter shall be set to 4 dB below P_{CMAX_L} at the minimum uplink configuration specified in ETSI TS 136 101 [2] (Table 7.3.1-2 with P_{CMAX_L} as defined in clause 6.2.5). NOTE 2: Reference measurement channel is clause A.3.2 of ETSI TS 136 521-1 [5]. NOTE 3: REFSENS as defined in clause 7.3.3 in ETSI TS 136 521-1 [5].							

Table 4.1.2.10.2-2: Spurious Response

Parameter	Units	Level
$P_{Interferer}$ (CW)	dBm	-44
$F_{Interferer}$	MHz	Spurious response frequencies

4.1.2.10.3 Conformance

The conformance tests for this requirement is defined in clause 5.1.3.9.

4.1.2.11 Receiver Intermodulation Characteristic

4.1.2.11.1 Definition

Intermodulation response rejection is a measure of the capability of the receiver to receive- a wanted signal on its assigned channel frequency in the presence of two or more interfering signals which have a specific frequency relationship to the wanted signal.

4.1.2.11.2 Limits

The throughput shall be ≥ 95 % of the maximum throughput of the reference measurement channels as specified in clauses A.2.2, A.2.3 and A.3.2 (with one sided dynamic OCNG Pattern OP.1 FDD/TDD for the DL-signal as described in clauses A.5.1.1 and A.5.2.1) in ETSI TS 136 521-1 [5] with parameters specified in Table 4.1.2.11.2-1 for the specified wanted signal mean power in the presence of two interfering signals.

Table 4.1.2.11.2-1: Test parameters for Wide band intermodulation

Rx Parameter	Units	Channel bandwidth					
		1,4 MHz	3 MHz	5 MHz	10 MHz	15 MHz	20 MHz
Power in Transmission Bandwidth Configuration	dBm	REFSENS + channel bandwidth specific value below					
		12	8	6	6	7	9
$P_{Interferer\ 1}$ (CW)	dBm	-46					
$P_{Interferer\ 2}$ (Modulated)	dBm	-46					
$BW_{Interferer\ 2}$		1,4	3	5			
$F_{Interferer\ 1}$ (Offset)	MHz	-BW/2 - 2,1 / +BW/2 + 2,1	-BW/2 - 4,5 / +BW/2 + 4,5	-BW/2 - 7,5 / +BW/2 + 7,5			
$F_{Interferer\ 2}$ (Offset)	MHz	$2 \times F_{Interferer\ 1}$					

NOTE 1: The transmitter shall be set to 4 dB below P_{CMAX_L} at the minimum uplink configuration specified in ETSI TS 136 101 [2] (Table 7.3.1-2 with P_{CMAX_L} as defined in clause 6.2.5).
NOTE 2: Reference measurement channel is clause A.3.2 of ETSI TS 136 521-1 [5].
NOTE 3: The modulated interferer shall consist of the Reference measurement channel specified in clause A.3.2 of ETSI TS 136 521-1 [5] with set-up according to clause C.3.1 of ETSI TS 136 521-1 [5].
The interfering modulated signal is a 5 MHz E-UTRA signal as described in annex C of ETSI TS 136 521-1 [5] for channel bandwidth ≥ 5 MHz.
NOTE 4: REFSENS as defined in clause 7.3.3 in ETSI TS 136 521-1 [5].
NOTE 5: For DL category M1 UE, the reference sensitivity for category M1 in ETSI TS 136 521-1 [5], Tables 7.3EA-1 and 7.3EA-2 should be used as REFSENS for the power in Transmission Bandwidth Configuration.
NOTE 6: For DL category M1 UE, the parameters for the applicable channel bandwidth apply, and BW refers to the corresponding channel bandwidth.

4.1.2.11.3 Conformance

The conformance tests for this requirement is defined in clause 5.1.3.10.

4.1.2.12 Receiver Spurious Emissions

4.1.2.12.1 Definition

The spurious emissions power is the power of emissions generated or amplified in a receiver that appear at the aerial UE antenna connector.

4.1.2.12.2 Limits

The measured spurious emissions derived in clause 5.1.3.11 shall not exceed the maximum level specified in Table 4.1.2.12.2-1.

Table 4.1.2.12.2-1: General receiver spurious emission requirements

Frequency Band	Measurement bandwidth	Maximum level
30 MHz ≤ f < 1 GHz	100 kHz	-57 dBm
1 GHz ≤ f ≤ 12,75 GHz	1 MHz	-47 dBm
NOTE: Unused PDCCH resources are padded with resource element groups with power level given by PDCCH_RA/RB as defined in ETSI TS 136 101 [2], clause C.3.1.		

4.1.2.12.3 Conformance

The conformance tests for this requirement are defined in clause 5.1.3.11.

4.1.2.13 Aerial UE Identifier

4.1.2.13.1 Definition

The aerial identifier for an aerial UE is intended to ensure that only UEs that are configured to behave as an aerial UE are allowed to carry out or receive aerial-related services and operations. Such aerial UE identifier differentiates between an aerial UE and a non-aerial UE.

4.1.2.13.2 Requirements

An aerial UE shall be provisioned with a single CAA-Level UAV ID. An aerial UE shall not be capable to connect to an E-UTRA network for aerial-related services without such a CAA-Level UAV ID.

4.1.2.13.3 Conformance

Conformance assessments are defined in clause 5.1.3.12.

4.1.2.14 Aerial UE Authorization and Authentication

4.1.2.14.1 Definition

The authentication and authorization procedure of the aerial UE are intended to ensure that the aerial UE has successfully registered with a UAS Service Supplier and has therefore been authorized for aerial-related services and operations. Hence, connectivity for aerial services is not established unless a successful aerial UE authorization and authentication procedure is performed.

4.1.2.14.2 Requirements

An aerial UE with aerial subscription shall perform the authentication and authorization procedure as defined in clause 5.2 in ETSI TS 123 256 [1]. In addition, the aerial UE shall not interfere with the authentication and authorization procedure.

4.1.2.14.3 Conformance

Conformance assessments are defined in clause 5.1.3.13.

4.2 Technical requirements specifications for NR

4.2.1 Environmental profile

Same as clause 4.1.1.

4.2.2 Conformance requirements

4.2.2.0 General

The requirements in the present document are based on the assumption that the operating band (see Table 1.1-1) is shared between systems of the IMT family (for band n3 and n8 also GSM) or systems having compatible characteristics.

The transmission bandwidth configuration in Table 5.3.2-1 of ETSI TS 138 101-1 [3] shall be supported for each of the specified channel bandwidths in Table 1.1-2.

4.2.2.1 Introduction

To meet the essential requirement under article 3.2 of Directive 2014/53/EU [i.1] and requirements given in ECC Decision (22)07 [i.18] for aerial UE, a set of essential parameters in addition to those in ETSI EN 301 908-1 [i.8] have been identified. Table 4.2.2.1-1 provides a cross reference between these essential parameters and the corresponding technical requirements for equipment within the scope of the present document.

Table 4.2.2.1-1: Cross references for NR

Essential parameter	Corresponding technical requirements	Corresponding test suite
Transmitter spectrum mask	4.2.2.4 Transmitter Spectrum emissions mask	5.2.3.3
Transmitter unwanted emissions in the out-of-band domain	4.2.2.5 Transmitter adjacent channel leakage power ratio	5.2.3.4
Transmitter unwanted emissions in the spurious domain	4.2.2.6 Transmitter spurious emissions	5.2.3.5
Transmitter power limits	4.2.2.2 Transmitter maximum output power	5.2.3.1
Transmitter Power Control (TPC)	4.2.2.3 Transmitter minimum output power	5.2.3.2
	4.2.2.13 Transmit OFF power	5.2.3.12
Transmitter power accuracy	4.2.2.2 Transmitter maximum output power	5.2.3.1
Receiver unwanted emissions in the spurious domain	4.2.2.12 Receiver spurious emissions	5.2.3.11
Receiver blocking	4.2.2.9 Receiver Blocking characteristics	5.2.3.8
Receiver desensitization		
Receiver spurious response rejection	4.2.2.10 Receiver spurious response	5.2.3.9
Receiver radio-frequency intermodulation	4.2.2.11 Receiver Intermodulation characteristics	5.2.3.10
Receiver adjacent signal selectivity	4.2.2.8 Receiver Adjacent Channel Selectivity (ACS)	5.2.3.7
Receiver sensitivity	4.2.2.7 Receiver Reference Sensitivity Level	5.2.3.6
Aerial UE identifier	4.2.2.14 Aerial UE identifier	5.2.3.13
Aerial UE Authorization and Authentication	4.2.2.15 Aerial UE Authorization and Authentication	5.2.3.14

The transmitter and receiver characteristics are specified at the antenna connector(s) of the aerial UE with a single or multiple transmit antenna(s).

4.2.2.2 Transmitter Maximum Output Power

4.2.2.2.1 Definition

The definition given in clause 4.1.2.2.1 applies.

4.2.2.2.2 Limits

The maximum output power for aerial UE with power class 3 shall be within the range in Table 4.2.2.2.2-1.

The maximum output power for aerial UE with power class 2 shall be within the range in Table 4.2.2.2.2-2.

These requirements do not take into account the maximum power reductions allowed to the aerial UE subject to certain transmission conditions specified in ETSI TS 138 101-1 [3], clauses 6.2.3 and 6.2.4.

Table 4.2.2.2.2-1: Maximum Output Power requirement for Power Class 3

NR band	Power Class 3			
	BW ≤ 40 MHz		40 MHz < BW ≤ 100 MHz	
	Lower limit (dBm)	Upper limit (dBm)	Lower limit (dBm)	Upper limit (dBm)
n1	20,3	25,7	20,0	26,0
n3	20,3 (note 2)	25,7		
n7	20,3 (note 2)	25,7		
n8	20,3 (note 2)	25,7		
n20	20,3 (note 2)	25,0		
n28	20,3	25,0		
n31	20,3	25,7		
n38	20,3	25,7		
n91	20,0 (note 2)	25,7		
n92	20,0 (note 2)	25,7		
n93	20,0 (note 2)	25,7		
n94	20,0 (note 2)	25,7		
n97	20,3	25,7		
n109	20,0 (note 2)	25,7		
NOTE 1: Power class 3 is the default power class unless otherwise stated. NOTE 2: Refers to the transmission bandwidths confined within F_{UL_low} and F_{UL_low} + 4 MHz or F_{UL_high} - 4 MHz and F_{UL_high}, the maximum output power requirement applies by reducing the lower limit by 1,5 dB.				

Table 4.2.2.2.2-2: Maximum Output Power requirement for Power Class 2

NR band	Power Class 2 (dBm)			
	BW ≤ 40 MHz		40 MHz < BW ≤ 100 MHz	
	Lower limit (dBm)	Upper limit (dBm)	Lower limit (dBm)	Upper limit (dBm)
n7	22,3 (note 2)	28,7	22,0 (note 2)	29,0
NOTE 1: Power class 3 is the default power class unless otherwise stated. NOTE 2: Refers to the transmission bandwidths confined within F_{UL_low} and F_{UL_low} + 4 MHz or F_{UL_high} - 4 MHz and F_{UL_high}, the maximum output power requirement applies by reducing the lower limit by 1,5 dB.				

4.2.2.2.3 Conformance

The conformance tests for this requirement are defined in clause 5.2.3.1.

4.2.2.3 Transmitter Minimum Output Power

4.2.2.3.1 Definition

The definition given in clause 4.1.2.3.1 applies.

4.2.2.3.2 Limits

The minimum output power shall not exceed the values specified in Table 4.2.2.3.2-1.

Table 4.2.2.3.2-1: Minimum output power

Channel bandwidth (MHz)	Minimum output power (dBm)	Measurement bandwidth (MHz)
	f ≤ 3,0 GHz	
5	-39	4,515
10	-39	9,375
15	-39	14,235
20	-39	19,095
25	-38	23,955
30	-37,2	28,815
35	-36,6	-36,3
40	-36	38,895
45	-35,2	-35,2
50	-34,7	48,615

4.2.2.3.3 Conformance

The conformance tests for this requirement are defined in clause 5.2.3.2.

4.2.2.4 Transmitter Spectrum Emission Mask

4.2.2.4.1 Definition

The Out of band emissions are unwanted emissions immediately outside the assigned channel bandwidth resulting from the modulation process and non-linearity in the transmitter but excluding spurious emissions. This out of band emission limit is specified in terms of a spectrum emission mask and an adjacent channel leakage power ratio.

4.2.2.4.2 Limits

The spectrum emission mask of the aerial UE applies to frequencies (Δf_{OOB}) starting from the ± edge of the assigned NR channel bandwidth. The power of any aerial UE emission shall fulfil requirements in Tables 4.2.2.4.2-1.

Table 4.2.2.4.2-1: General NR spectrum emission mask

Spectrum emission limit (dBm) / Channel bandwidth									
Δf_{OOB} (MHz)	**5 MHz**	**10 MHz**	**15 MHz**	**20 MHz**	**25 MHz**	**30 MHz**	**40 MHz**	**50 MHz**	**Measurement bandwidth**
± 0-1	-11,5	-11,5	-11,5	-11,5	-11,5	-11,5	-11,5		1 % channel bandwidth
± 0-1								-22,5	30 kHz
± 1-5	-8,5	-8,5	-8,5	-8,5	-8,5	-8,5	-8,5	-8,5	1 MHz
± 5-6	-11,5								
± 6-10	-23,5	-11,5	-11,5	-11,5	-11,5	-11,5	-11,5	-11,5	
± 10-15		-23,5							
± 15-20			-23,5						
± 20-25				-23,5					
± 25-30					-23,5				
± 30-35						-23,5			
± 35-40									
± 40-45							-23,5		
± 45-50									
± 50-55								-23,5	
NOTE 1: The first and last measurement position with a 30 kHz filter is at Δf_{OOB} equals to 0,015 MHz and 0,985 MHz. NOTE 2: At the boundary of spectrum emission limit, the first and last measurement position with a 1 MHz filter is the inside of +0,5 MHz and -0,5 MHz, respectively. NOTE 3: The measurements are to be performed above the upper edge of the channel and below the lower edge of the channel.									

4.2.2.4.3 Conformance

The conformance tests for this requirement are defined in clause 5.2.3.3.

4.2.2.5 Transmitter Adjacent Channel Leakage Power Ratio

4.2.2.5.1 Definition

NR Adjacent Channel Leakage power Ratio (NR ACLR) is the ratio of the filtered mean power centred on the assigned NR channel frequency to the filtered mean power centred on an adjacent NR channel frequency at nominal channel spacing.

UTRA Adjacent Channel Leakage power Ratio (UTRA ACLR) is the ratio of the filtered mean power centred on the assigned NR channel frequency to the filtered mean power centred on an adjacent(s) UTRA channel frequency.

4.2.2.5.2 Limits

4.2.2.5.2.1 NR ACLR

The assigned NR channel power and adjacent NR channel power are measured with rectangular filters with measurement bandwidths specified in Table 4.2.2.5.2.1-1.

Table 4.2.2.5.2.1-1: NR ACLR measurement bandwidth

Channel bandwidth	(MHz)	5,10,15,20,25,30,35,40,45,50	60,70,80,90,100
REF_SCS	(kHz)	15	30
NR ACLR measurement bandwidth	(MHz)	MBW=REF_SCS × (12 × N_{RB} + 1) / 1 000	
NOTE: "N_{RB}" in the formula is the maximum transmission bandwidth configuration as defined in Table 5.3.2-1 of ETSI TS 138 101-1 [3].			

If the measured adjacent channel power is greater than -50 dBm then the NR ACLR shall be higher than the value specified in Table 4.2.2.5.2.1-2.

Table 4.2.2.5.2.1-2: NR ACLR requirement

	Power class 2	Power class 3
NR ACLR	30,2 dB	29,2 dB

4.2.2.5.2.2 UTRA ACLR

UTRA ACLR requirement is applicable when signalled by the network with network signalling value indicated by the field *additionalSpectrumEmission*.

UTRA ACLR is specified for the first adjacent UTRA channel (UTRA ACLR1) which centre frequency is ± 2,5 MHz from the NR channel edge and for the 2nd adjacent UTRA channel (UTRA ACLR2) which centre frequency is ± 7,5 MHz from the NR channel edge.

The UTRA channel power is measured with an RRC filter with roll-off factor α =0,22 and bandwidth of 3,84 MHz. The assigned NR channel power is measured with a rectangular filter with measurement bandwidth specified in Table 4.2.2.5.2.1-1.

If the measured adjacent channel power is greater than -50 dBm then the UTRA ACLR1 and UTRA ACLR2 shall be higher than the value specified in Table 4.2.2.5.2.2-1.

Table 4.2.2.5.2.2-1: UTRA ACLR requirement

	Power class 3
UTRA ACLR1	32,2 dB
UTRA ACLR2	35,2 dB

4.2.2.5.3 Conformance

The conformance tests for this requirement are defined in clause 5.2.3.4.

4.2.2.6 Transmitter Spurious Emissions

4.2.2.6.1 Definition

The definition given in clause 4.1.2.6.1 applies.

4.2.2.6.2 Limits

4.2.2.6.2.1 General spurious emissions

This clause specifies the requirements for the specified NR band for Transmitter Spurious emissions requirement with frequency range as indicated in Table 4.2.2.6.2.1-2.

Unless otherwise stated, the spurious emission limits apply for the frequency ranges that are more than F_{OOB} (MHz) in Table 4.2.2.6.2.1-1 from the edge of the channel bandwidth. The spurious emission limits in Table 4.2.2.6.2.1-2 apply for all transmitter band configurations (N_{RB}) and channel bandwidths.

The measured average power of spurious emission shall not exceed the described value in Table 4.2.2.6.2.1-2.

Table 4.2.2.6.2.1-1: Boundary between NR out of band and general spurious emission domain

Channel bandwidth	OOB boundary F_{OOB} (MHz)
$BW_{Channel}$	$BW_{Channel}$ + 5

Table 4.2.2.6.2.1-2: General spurious emissions test requirements

Frequency Range	Maximum Level	Measurement bandwidth	Note
9 kHz ≤ f < 150 kHz	-36 dBm	1 kHz	
150 kHz ≤ f < 30 MHz	-36 dBm	10 kHz	
30 MHz ≤ f < 1 000 MHz	-36 dBm	100 kHz	
1 GHz ≤ f < 12,75 GHz	-30 dBm	1 MHz	
	-25 dBm	1 MHz	
12,75 GHz ≤ f < 5th harmonic of the upper frequency edge of the UL operating band in GHz	-30 dBm	1 MHz	Note
NOTE: Applies for Band for which the upper frequency edge of the UL Band is greater than 2,55 GHz and less than or equal to 5,2 GHz.			

4.2.2.6.2.2 Spurious emission for UE co-existence

This clause specifies the requirements for the specified NR band for coexistence with protected bands as indicated in Table 4.2.2.6.2.2-1.

Unless otherwise stated, the spurious emission limits apply for the frequency ranges that are more than F_{OOB} (MHz) in Table 4.2.2.6.2.1-1 from the edge of the channel bandwidth. The spurious emission limits in Table 4.2.2.6.2.2-1 apply for all transmitter band configurations (N_{RB}) and channel bandwidths.

The measured average power of spurious emission shall not exceed the described value in Table 4.2.2.6.2.2-1.

Table 4.2.2.6.2.2-1: Requirements for spurious emissions for UE co-existence

NR Band	Spurious emission for UE co-existence						
	Protected band	Frequency range (MHz)			Maximum Level (dBm)	MBW (MHz)	Note
n1	E-UTRA Band 1, 7, 8, 20, 22, 28, 31, 32, 38, 40, 41, 42, 43, 50, 51, 65, 67, 68, 69, 75, 76, NR Band n78, n79, n100, n109	FDL_low	-	FDL_high	-50	1	
	NR Band n77	FDL_low	-	FDL_high	-50	1	2
	E-UTRA Band 3, 34	FDL_low	-	FDL_high	-50	1	15
	Frequency range	1 880	-	1 895	-40	1	15, 27
	Frequency range	1 895	-	1 915	-15,5	5	15, 26, 27
	Frequency range	1 915	-	1 920	+1,6	5	15, 26, 27
n3	E-UTRA Band 1, 7, 8, 20, 28, 31, 32, 33, 34, 38, 40, 41, 43, 50, 51, 65, 67, 68, 69, 75, 76.	FDL_low	-	FDL_high	-50	1	
	E-UTRA Band 3	FDL_low	-	FDL_high	-50	1	15
	E-UTRA Band 22, 42, NR Band n77, n78	FDL_low	-	FDL_high	-50	1	2
	Frequency range	1 884,5	-	1 915,7	-41	0,3	8
n7	E-UTRA Band 1, 3, 7, 8, 20, 22, 28, 31, 32, 33, 34, 40, 42, 43, 50, 51, 65, 67, 68, 75, 76 NR Band n77, n78, n100, n101, n109	FDL_low	-	FDL_high	-50	1	
	Frequency range	2 570	-	2 575	+1,6	5	15, 21, 26
	Frequency range	2 575	-	2 595	-15,5	5	15, 21, 26
	Frequency range	2 595	-	2 620	-40	1	15, 21
n8, n93, n94	E-UTRA Band 1, 20, 28, 31, 32, 33, 34, 38, 40, 50, 51, 65, 67, 68, 69, 75, 76	FDL_low	-	FDL_high	-50	1	
	E-UTRA band 3, 7, 22, 41, 42, 43, NR Band n77, n78, n101, n109	FDL_low	-	FDL_high	-50	1	2
	E-UTRA 8	FDL_low	-	FDL_high	-50	1	15
	Frequency range	1 884,5	-	1 915,7	-41	0,3	8

NR Band	Spurious emission for UE co-existence						
	Protected band	Frequency range (MHz)			Maximum Level (dBm)	MBW (MHz)	Note
n20, n91, n92	E-UTRA Band 1, 3, 7, 8, 22, 31, 32, 33, 34, 40, 43, 50, 51, 65, 67, 68, 75, 76 NR Band n100, n101, n104, n109	FDL_low	-	FDL_high	-50	1	
	E-UTRA Band 20	FDL_low	-	FDL_high	-50	1	15
	E-UTRA Band 38, 42, 69, NR Band n77, n78	FDL_low	-	FDL_high	-50	1	2
	Frequency range	758	-	788	-50	1	
n28	E-UTRA Band 1, 22, 32, 42, 43, 50, 51, 65, 75, 76 NR Band n77, n78, n100, n101, n109	FDL_low	-	FDL_high	-50	1	2
	E-UTRA Band 1	FDL_low	-	FDL_high	-50	1	19, 25
	E-UTRA Band 3, 7, 8, 20, 31, 34, 38, 40, 41	FDL_low	-	FDL_high	-50	1	
	Frequency range	470	-	694	-42	8	15, 35
	Frequency range	470	-	710	-26,2	6	34
	Frequency range	662	-	694	-26,2	6	15
	Frequency range	758	-	773	-32	1	15
	Frequency range	773	-	803	-50	1	
	Frequency range	1 884,5	-	1 915,7	-41	0,3	8, 19
n38	E-UTRA Band 1, 3, 8, 20, 22, 28, 31, 32, 33, 34, 40, 42, 43, 50, 51, 65, 67, 68, 75, 76 NR Band n100, n101, n109	FDL_low	-	FDL_high	-50	1	
	Frequency range	2 620	-	2 645	-15,5	5	15, 22, 26
	Frequency range	2 645	-	2 690	-40	1	15, 22
n109	E-UTRA Band 22, 32, 42, 43, 65, NR Band n78, n100, n101	F_{DL_low}	-	F_{DL_high}	-50	1	2
	E-UTRA Band 1	F_{DL_low}	-	F_{DL_high}	-50	1	19, 25
	E-UTRA Band 3, 7, 8, 20, 38,40	F_{DL_low}	-	F_{DL_high}	-50	1	
	Frequency range	662	-	694	-26.2	6	15
	Frequency range	758	-	773	-32	1	15
	Frequency range	773	-	803	-50	1	
	Frequency range	470	-	694	-42	8	15, 35

NOTE 1: FDL_low and FDL_high refer to each frequency band specified in Table 1.1-1 for NR band, Table 5.5-1 in ETSI TS 136 521-1 [5] for E-UTRA band.

NOTE 2: As exceptions, measurements with a level up to the applicable requirements defined in Table 4.2.2.6.2.1-2 are permitted for each assigned NR carrier used in the measurement due to 2nd, 3rd, 4th or 5th harmonic spurious emissions. Due to the spreading of the harmonic emission, the exception is also allowed for the first 1 MHz frequency range immediately outside the harmonic emission on both sides of the harmonic emission. This results in an overall exception interval centred at the harmonic emission of (2 MHz + N × LCRB × RBsize kHz), where N is 2, 3, 4, 5 for the 2nd, 3rd, 4th or 5th harmonic respectively. The exception is allowed if the measurement bandwidth (MBW) totally or partially overlaps the overall exception interval.

NOTE 3: 15 kHz SCS is assumed when RB is mentioned in the note when channel bandwidth is less than or equal to 50 MHz, lowest SCS is assumed when channel bandwidth is larger than 50 MHz. The transmission bandwidth in terms of RB position and range is not limited to 15 kHz SCS and shall scale with SCS accordingly.

NOTES 4 to 7: N/A.

NOTE 8: Applicable when co-existence with PHS system operating in 1 884,5 MHz to 1 915,7 MHz.

NOTES 9 to 14: N/A.

NOTE 15: These requirements also apply for the frequency ranges that are less than FOOB (MHz) in Table 4.2.2.6.2.1-1 from the edge of the channel bandwidth.

NOTES 16 to 18: N/A.

NOTE 19: Applicable when the assigned NR carrier is confined within 718 MHz to 748 MHz and when the channel bandwidth used is 5 or 10 MHz.

NOTE 20: N/A.

NOTE 21: This requirement is applicable for any channel bandwidths within the range 2 500 MHz to 2 570 MHz with the following restriction: for carriers of 15 MHz bandwidth when carrier centre frequency is within the range 2 560,5 MHz to 2 562,5 MHz and for carriers of 20 MHz bandwidth when carrier centre frequency is within the range 2 552 MHz to 2 560 MHz the requirement is applicable only for an uplink transmission bandwidth less than or equal to 54 RB.

NR Band	Spurious emission for UE co-existence				
	Protected band	Frequency range (MHz)	Maximum Level (dBm)	MBW (MHz)	Note
NOTE 22: This requirement is applicable for power class 3 UE for any channel bandwidths within the range 2 570 MHz to 2 615 MHz with the following restriction: for carriers of 15 MHz bandwidth when carrier centre frequency is within the range 2 605,5 MHz to 2 607,5 MHz and for carriers of 20 MHz bandwidth when carrier centre frequency is within the range 2 597 MHz to 2 605 MHz the requirement is applicable only for an uplink transmission bandwidth less than or equal to 54 RB. For power class 2 UE for any channel bandwidths within the range 2 570 MHz to 2 615 MHz, NS_44 shall apply. For power class 2 or 3 UE for carriers with channel bandwidth overlapping the frequency range 2 615 MHz to 2 620 MHz the requirement applies with the maximum output power configured to +19 dBm in the IE P-Max. NOTE 23: N/A. NOTE 24: N/A. NOTE 25: As exceptions, measurements with a level up to the applicable requirement of -36 dBm/MHz is permitted for each assigned NR carrier used in the measurement due to 3rd harmonic spurious emissions. An exception is allowed if there is at least one individual RB within the transmission bandwidth (see Figure 5.3.1-1 of ETSI TS 138 101-1 [3]) for which the 3rd harmonic totally or partially overlaps the Measurement Bandwidth (MBW). NOTE 26: For these adjacent bands, the emission limit could imply risk of harmful interference to UE(s) operating in the protected operating band. NOTE 27: This requirement is applicable for any channel bandwidths within the range 1 920 MHz to 1 980 MHz with the following restriction: for carriers of 15 MHz bandwidth when carrier centre frequency is within the range 1 927,5 MHz to 1 929,5 MHz and for carriers of 20 MHz bandwidth when carrier centre frequency is within the range 1 930 MHz to 1 938 MHz the requirement is applicable only for an uplink transmission bandwidth less than or equal to 54 RB. NOTES 28 to 33: N/A. NOTE 34: This requirement is applicable for 5 and 10 MHz NR channel bandwidth allocated within 718 MHz to 728 MHz. For carriers of 10 MHz bandwidth, this requirement applies for an uplink transmission bandwidth less than or equal to 30 RB with RBstart > 1 and RBstart < 48. NOTE 35: This requirement is applicable in the case of a 10 MHz NR carrier confined within 703 MHz and 733 MHz, otherwise the requirement of -25 dBm with a measurement bandwidth of 8 MHz applies. NOTES 36 to 43: N/A.					

NOTE: To simplify Table 4.2.2.6.2.2-1, E-UTRA band numbers are listed for bands which are specified only for E-UTRA operation or both E-UTRA and NR operation in ETSI TS 138 101-1 [3] or ETSI TS 136 101 [2]. NR band numbers are listed for bands which are specified only for NR operation.

4.2.2.6.2.3 Additional spurious emissions

These requirements are specified in terms of an additional spectrum emission requirement. Additional spurious emission requirements are signalled by the network to indicate that the aerial UE shall meet an additional requirement for a specific deployment scenario as part of the cell handover/broadcast message.

This clause specifies the requirements for the specified NR band for an additional spectrum emission requirement with protected bands for different NS_values.

The measured UE mean power in the channel bandwidth shall fulfil requirements in Table 6.2.3.5-8 of ETSI TS 138 521-1 [4] for power class 3 aerial UE.

The power of any aerial UE emission shall not exceed the levels specified in Table 4.2.2.6.2.3-1. This requirement also applies for the frequency ranges that are less than FOOB (MHz) in Table 4.2.2.6.2.1-1 from the edge of the channel bandwidth.

Table 4.2.2.6.2.3-1: Additional requirements for "NS_18"

Frequency range (MHz)	Channel bandwidth / Spectrum emission limit (dBm)	Measurement bandwidth
	5, 10, 15, 20, 30 MHz	
692 - 698	-26,2	6 MHz

In the present document, the requirement in Table 4.2.2.6.2.3-2 is applicable for 25, 30, 40 and 50 MHz channel bandwidths.

Table 4.2.2.6.2.3-2: Additional requirements for "NS_48"

Protected band	Frequency range (MHz)			Maximum Level (dBm)	MBW (MHz)
E-UTRA band 34 NR band n34	FDL_low	-	FDL_high	-50	1
Frequency range (note)	1 900	-	1 915	-15,5	5
Frequency range (note)	1 915	-	1 920	+1,6	5
NOTE: For these adjacent bands, the emission limit could imply risk of harmful interference to UE(s) operating in the protected operating band.					

When "NS_49" is indicated in the cell (applicable to bands n1):

The measured aerial UE mean power in the channel bandwidth shall fulfil requirements in Table 6.2.3.5-33 of ETSI TS 138 521-1 [4] for power class 3 aerial UE.

The power of any aerial UE emission shall not exceed the levels specified in Table 4.2.2.6.2.3-3.In the present release the requirement in Table 4.2.2.6.2.3-3 is applicable for 25, 30, 40 and 50 MHz channel bandwidths.

Table 4.2.2.6.2.3-3: Additional requirements for "NS_49"

Protected band	Frequency range (MHz)			Maximum Level (dBm)	MBW (MHz)
E-UTRA band 34 NR band n34	FDL_low	-	FDL_high	-50	1
Frequency range	1 880	-	1 895	-40	1
Frequency range (note)	1 895		1 915	-15,5	5
Frequency range (note)	1 915	-	1 920	1,6	5
NOTE: For these adjacent bands, the emission limit could imply risk of harmful interference to UE(s) operating in the protected operating band.					

When "NS_44" is indicated in the cell (applicable to bands n38):

The measured aerial UE mean power in the channel bandwidth shall fulfil requirements in Table 6.2K.3.5-1of ETSI TS 138 521-1 [4] for power class 3 aerial UE.

The power of any aerial UE emission shall not exceed the levels specified in Table 4.2.2.6.2.3-4.

Table 4.2.2.6.2.3-4: Additional spurious emissions limits (network signalled value "NS_UAV_44")

Frequency range (MHz)	Channel bandwidth (MHz) / Spectrum emission limit (dBm)	Measurement bandwidth	Note
	5, 10, 15, 20, 25, 30, 40 MHz		
2 620 ≤ f < 2 645	-15,5	5 MHz	Note
2 645 ≤ f < 2 690	-40	1 MHz	
2 690 ≤ f ≤ 2 900	-50	1 MHz	
NOTE: For these adjacent bands, the emission limit could imply risk of harmful interference to UE(s) operating in the protected operating band.			

When "NS_70" is indicated in the cell (applicable to bands n3):

The measured aerial UE mean power in the channel bandwidth shall fulfil requirements in Table 6.2K.3.5-2 of ETSI TS 138 521-1 [4] for power class 3 aerial UE.

The power of any aerial UE emission shall not exceed the levels specified in Table 4.2.2.6.2.3-5.

Table 4.2.2.6.2.3-5: Additional spurious emissions limits (network signalled value "NS_UAV_70")

Frequency range (MHz)	Channel bandwidth (MHz) / Spectrum emission limit (dBm)	Measurement bandwidth
	5, 10, 15, 20, 25, 30, 35, 40, 45, 50 MHz	
1 675 ≤ f ≤ 1 710	-40	1 MHz

4.2.2.6.3 Conformance

The conformance tests for this requirement are defined in clause 5.2.3.5.

4.2.2.7 Receiver Reference Sensitivity Level

4.2.2.7.1 Definition

The definition given in clause 4.1.2.7.1 applies.

In later subclauses of clause 4.2.2.7 where the value of REFSENS is used as a reference to set the corresponding requirement:

- in all bands, the aerial UE shall be verified against those requirements by applying the REFSENS value in Table 4.2.2.7.2-1 with 2 Rx antenna ports tested;
- for bands where the aerial UE is required to be equipped with 4 Rx antenna ports, the aerial UE shall additionally be verified against those requirements by applying the resulting REFSENS value derived from the requirement in Table 4.2.2.7.2-2 with 4 Rx antenna ports tested.

4.2.2.7.2 Limits

The throughput shall be ≥ 95 % of the maximum throughput of the reference measurement channels as specified in ETSI TS 138 101-1 [3], clauses A.2.2.2, A.2.3.2, A3.2 and A.3.3 (with one sided dynamic OCNG Pattern OP.1 FDD/TDD for the DL-signal as described in ETSI TS 138 101-1 [3], clauses A.5.1.1/A.5.2.1) with parameters specified in Table 4.2.2.7.2-1.

Table 4.2.2.7.2-1: Two antenna port reference sensitivity

Operating Band	SCS kHz	5 MHz (dBm)	10 MHz (dBm)	15 MHz (dBm)	20 MHz (dBm)	25 MHz (dBm)	30 MHz (dBm)	35 MHz (dBm)	40 MHz (dBm)	50 MHz (dBm)	Duplex Mode
n1	15	-99,3	-96,1	-94,3	-93,1	-92,0	-91,2		-89,9	-88,9	FDD
	30		-96,4	-94,4	-93,3	-92,1	-91,3		-90,0	-89,0	
	60		-96,8	-94,7	-93,5	-92,3	-91,4		-90,2	-89,0	
n3	15	-96,3	-93,1	-91,3	-90,1	-89	-88,2	-85.5	-81,6	-79,0	FDD
	30		-93,4	-91,4	-90,3	-89,1	-88,3	-85.6	-81,7	-79,1	
	60		-93,8	-91,7	-90,5	-89,3	-88,4	-85.7	-81,9	-79,2	
n7 (note)	15	-97,3	-94,1	-92,3	-91,1	-90,0	-89,2		-87,9	-80,8	FDD
	30		-94,4	-92,4	-91,3	-90,1	-89,3		-88,0	-80,8	
	60		-94,8	-92,7	-91,5	-90,3	-89,4		-88,2	-80,8	
n8	15	-96,3	-93,1	-90,7	-85,1	-82,9	-80,6	-77,7			FDD
	30		-93,4	-91	-86,5	-84,0	-80,7	-77,8			
	60										
n20	15	-96,3	-93,1	-90,3	-89,1						FDD
	30		-93,4	-90,4	-89,3						
	60										
n28	15	-97,8	-94,8	-92,8	-90,1		-77,8		-65,6		FDD
	30		-94,9	-92,9	-90,3		-77,9		-65,6		
	60										
n38 (note)	15	-99,3	-96,1	-94,3	-93,0	-92,0	-91,2		-89,9		TDD
	30		-96,4	-94,4	-93,1	-92,1	-91,3		-89,9		
	60		-96,8	-94,7	-93,4	-92,3	-91,4		-90,1		
	30										
	60										

Operating Band	SCS kHz	5 MHz (dBm)	10 MHz (dBm)	15 MHz (dBm)	20 MHz (dBm)	25 MHz (dBm)	30 MHz (dBm)	35 MHz (dBm)	40 MHz (dBm)	50 MHz (dBm)	Duplex Mode
n91	15	-99,3									FDD
	30										
	60										
n92	15	-99,3	-96,1	-94,3							FDD
	30		-96,4	-94,4							
	60										
n93	15	-99,3									FDD
	30										
	60										
n94	15	-99,3	-96,1	-94,3	-93,0						FDD
	30		-96,4	-94,4	-93,1						
	60										
n109	15	-99,3	-96,1	-94,3	-93,0	-92,0	-91,2		-89,8	-89,0	FDD
	30	-99,8	-96,4	-94,4	-93,1	-92,1	-91,3		-89,8	-89,0	
	60										
NOTE: Four Rx antenna ports shall be the baseline for this operating band.											

For power class 2 aerial UEs, certain degradation of the reference sensitivity in Table 4.2.2.7.2-1 is allowed. The maximum amount of degradation is specified in Table 4.2.2.7.2-1a for an aerial UE that does not indicates *txDiversity-r16*.

Table 4.2.2.7.2-1a: Reference Sensitivity Degradation from PC3 to PC2 for FDD bands for UE not supporting Tx Diversity

Operating Band	5 MHz (dB)	10 MHz (dB)	15 MHz (dB)	20 MHz (dB)	25 MHz (dB)	30 MHz (dB)	35 MHz (dB)	40 MHz (dB)	45 MHz (dB)	50 MHz (dB)
n7	0,5	0,5	0,5	0,5	0,5	0,5	-	0,5	-	2,0
NOTE: The transmitter shall be set to P_{UMAX} as defined in clause 6.2.4 of ETSI TS 138 101-1 [3].										

For aerial UE(s) equipped with 4 Rx antenna ports, reference sensitivity for 2Rx antenna ports in Table 4.2.2.7.2-1 shall be modified by the amount given in $\Delta R_{IB,4R}$ in Table 4.2.2.7.2-2 for the applicable operating bands.

Table 4.2.2.7.2-2: Four antenna port reference sensitivity allowance $\Delta R_{IB,4R}$

Operating band	$\Delta R_{IB,4R}$ (dB)
n1, n3, n7, n38	-2,7

The REFSENS requirement specified in Table 4.2.2.7.2-1 and Table 4.2.2.7.2-2 shall be met with uplink transmission bandwidth less than or equal to that specified in Table 4.2.2.7.2-3.

Table 4.2.2.7.2-3: Uplink configuration for reference sensitivity

	Operating band / SCS / Channel bandwidth / Duplex mode											
Operating Band	SCS kHz	5 MHz	10 MHz	15 MHz	20 MHz	25 MHz	30 MHz	35 MHz	40 MHz	45 MHz	50 MHz	Duplex Mode
n1	15	25	50 Note 1	75 Note 1	100 Note 1	128 Note 1	128 Note 1		128 Note 1	128 Note 1	128 Note 1	FDD
	30		24	36 Note 1	50 Note 1	64 Note 1	64 Note 1		64 Note 1	64 Note 1	64 Note 1	
	60		10 Note 1	18	24	30 Note 1	30 Note 1		30 Note 1	30 Note 1	30 Note 1	
n3	15	25	50 Note 1	50 Note 1	50 Note 1	50 Note 1	50 Note 1	50 Note 1	50 Note 1	50 Note 1	50 Note 1	FDD
	30		24	24 Note 1	24 Note 1	24 Note 1	24 Note 1	24 Note 1	24 Note 1	24 Note 1	24 Note 1	
	60		10 Note 1	10 Note 1	10 Note 1	10 Note 1	10 Note 1	10 Note 1	10 Note 1	10 Note 1	10 Note 1	
n7	15	25	50 Note 1	75 Note 1	75 Note 1	72 Note 1	64 Note 1		45 Note 1		45 Note 1	FDD

	Operating band / SCS / Channel bandwidth / Duplex mode											
Operating Band	**SCS kHz**	**5 MHz**	**10 MHz**	**15 MHz**	**20 MHz**	**25 MHz**	**30 MHz**	**35 MHz**	**40 MHz**	**45 MHz**	**50 MHz**	**Duplex Mode**
	30		24	36 Note 1	36 Note 1	36 Note 1	32 Note 1		20 Note 1		20 Note 1	
	60		10 Note 1	18	18 Note 1	18 Note 1	16 Note 1		10 Note 1		10 Note 1	
n8	15	25	25 Note 1	25 Note 1	25 Note 1	Note 5	Note 5	Note 5				FDD
	30		10 Note 1	10 Note 1	10 Note 1	Note 5	Note 5	Note 5				
	60											
n20	15	25	20 Note 1	20 Note 1	20 Note 1							FDD
	30		10 Note 1	10 Note 2	10 Note 2							
	60											
n28	15	25	25 Note 1	25 Note 1	25 Note 1		25 Note 1		25 Note 1			FDD
	30		10 Note 1	10 Note 1	10 Note 1		10 Note 1		10 Note 1			
	60											
n38	15	25	50	75	100	128	160		216			TDD
	30		24	36	50	64	75		100			
	60		10	18	24	30	36		50			
n91	15	25 Note 4	20 Notes 1 and 4									FDD
	30											
	60											
n92	15	25	20 Note 1	20 Note 1								FDD
	30		10 Note 1	10 Note 1								
	60											
n93	15	25 Note 4	25 Notes 1 and 4									FDD
	30											
	60											
n94	15	25	25 Note 1	20 Note 1								FDD
	30		12 Note 1	10 Note 1								
	60											
n109	15	25	Note 5	Note 5	Note 5	Note 5	Note 5		Note 5			FDD
	30		24	Note 5	Note 5	Note 5	Note 5		Note 5			
	60											

NOTE 1: UL resource blocks shall be located as close as possible to the downlink operating band but confined within the transmission bandwidth configuration for the channel bandwidth (Table 5.3.2-1 of ETSI TS 138 101-1 [3]).
NOTE 2: For Band 20; for 15 kHz SCS, in the case of 15 MHz channel bandwidth, the UL resource blocks shall be located at RB_{start} 11 and in the case of 20 MHz channel bandwidth, the UL resource blocks shall be located at RB_{start} 16; for 30 kHz SCS, in the case of 15 MHz channel bandwidth, the UL resource blocks shall be located at RB_{start} 6 and in the case of 20 MHz channel bandwidth, the UL resource blocks shall be located at RB_{start} 8; for 60 kHz SCS, in the case of 15 MHz channel bandwidth, the UL resource blocks shall be located at RB_{start} 3 and in the case of 20 MHz channel bandwidth, the UL resource blocks shall be located at RBstart 4.
NOTE 3: For band n91 and n93, largest supported UL bandwidth configuration shall be used.
NOTE 4: The UL resource blocks shall be located at RBstart 10.
NOTE 5: For this DL channel bandwidth, the UL configuration of 5 MHz for 15 kHz SCS and 10 MHz for 30 kHz shall be used.
NOTE 6: in the case of 5 MHz channel bandwidth, the UL resource blocks shall be located at RB_{start} 10.

4.2.2.7.3 Conformance

The conformance tests for this requirement are defined in clause 5.2.3.6.

4.2.2.8 Receiver Adjacent Channel Selectivity (ACS)

4.2.2.8.1 Definition

ACS is a measure of a receiver's ability to receive an NR signal at its assigned channel frequency in the presence of an adjacent channel signal at a given frequency offset from the centre frequency of the assigned channel. ACS is the ratio of the receive filter attenuation on the assigned channel frequency to the receive filter attenuation on the adjacent channel(s).

4.2.2.8.2 Limits

The aerial UE shall fulfil the minimum requirements specified in Table 4.2.2.8.2-1 for NR bands. These requirements apply for all values of an adjacent channel interferer up to -25 dBm and for any SCS specified for the channel bandwidth of the wanted signal. However, it is not possible to directly measure the ACS; instead, the lower and upper range of test parameters are chosen as in Table 4.2.2.8.2-2 and Table 4.2.2.8.2-3 for verification of the requirements specified in Table 4.2.2.8.2-1. For these test parameters, the throughput shall be $\geq$ 95 % of the maximum throughput of the reference measurement channels as specified in ETSI TS 138 101-1 [3], clauses A.2.2, A.2.3, A.3.2 and A.3.3 (with one sided dynamic OCNG Pattern OP.1 FDD/TDD for the DL-signal as described in ETSI TS 138 101-1 [3], clauses A.5.1.1/A.5.2.1).

Table 4.2.2.8.2-1: ACS for NR bands

RX parameter	Units	Channel bandwidth (MHz)		
		5, 10	15	20, 25, 30, 35, 40, 45, 50, 60, 70, 80, 90, 100
ACS	dB	33	30	27 - $10\log_{10}(BW_{Channel}/20)$
NOTE: ACS value is rounded to the next higher 0,5 dB value.				

Table 4.2.2.8.2-2: Test parameters for NR bands, case 1

RX parameter	Units	Channel bandwidth (MHz)		
		5, 10	15	20, 25, 30, 35, 40, 45, 50, 60, 70, 80, 90, 100
Power in transmission bandwidth configuration	dBm	REFSENS + 14 dB		
$P_{interferer}$ (note 4)	dBm	REFSENS + 45,5 dB	REFSENS + 42,5 dB	REFSENS + 39,5 - $10\log_{10}(BW_{Channel}/20)$
$BW_{interferer}$	MHz	5		
$F_{interferer}$ (offset)	MHz	$BW_{Channel}/2 + 2,5$ / $-(BW_{Channel}/2 + 2,5)$		
NOTE 1: The transmitter shall be set to 4 dB below $P_{CMAX_L,f,c}$ at the minimum UL configuration specified in Table 4.2.2.7.2-3 with $P_{CMAX_L,f,c}$ defined in clause 6.2.4 of ETSI TS 138 101-1 [3].				
NOTE 2: The absolute value of the interferer offset $F_{interferer}$ (offset) shall be further adjusted to $(\lceil \lvert F_{\text{interferer}} \rvert / SCS \rceil + 0{,}5)SCS$ MHz with SCS the sub-carrier spacing of the wanted signal in MHz. The interferer is an NR signal with 15 kHz SCS.				
NOTE 3: The interferer consists of the NR interferer RMC specified in ETSI TS 138 101-1 [3], clauses A.3.2.2 and A.3.3.2 with one sided dynamic OCNG Pattern OP.1 FDD/TDD for the DL-signal as described in ETSI TS 138 101-1 [3], clauses A.5.1.1/A.5.2.1.				
NOTE 4: $10\log_{10}(x)$ is rounded to the next higher 0,5 dB value.				

Table 4.2.2.8.2-3: Test parameters for NR bands, case 2

RX parameter	Units	Channel bandwidth (MHz)		
		5, 10	15	20, 25, 30, 35, 40, 45, 50, 60, 70, 80, 90, 100
Power in transmission bandwidth configuration (note 4)	dBm	-56,5	-53,5	$-50,5 + 10\log_{10}(BW_{Channel}/20)$
$P_{interferer}$	dBm	-25		
$BW_{interferer}$	MHz	5		
$F_{interferer}$ (offset)	MHz	$BW_{Channel}/2 + 2,5$ / $-(BW_{Channel}/2 + 2,5)$		

NOTE 1: The transmitter shall be set to 24 dB below $P_{CMAX_L,f,c}$ at the minimum UL configuration specified in Table 4.2.2.7.2-3 with $P_{CMAX_L,f,c}$ defined in clause 6.2.4 of ETSI TS 138 101-1 [3].
NOTE 2: The absolute value of the interferer offset $F_{interferer}$ (offset) shall be further adjusted to $(\lceil |F_{\text{interferer}}|/SCS \rceil + 0,5)SCS$ MHz with SCS the sub-carrier spacing of the wanted signal in MHz. The interferer is an NR signal with 15 kHz SCS.
NOTE 3: The interferer consists of the RMC specified in ETSI TS 138 101-1 [3], clauses A.3.2.2 and A.3.3.2 with one sided dynamic OCNG Pattern OP.1 FDD/TDD for the DL-signal as described in ETSI TS 138 101-1 [3], clauses A.5.1.1/A.5.2.1.
NOTE 4: $10\log_{10}(x)$ is rounded to the next higher 0,5 dB value.

4.2.2.8.3 Conformance

The conformance tests for this requirement are defined in clause 5.2.3.7.

4.2.2.9 Receiver Blocking Characteristics

4.2.2.9.1 Definition

The definition given in clause 4.1.2.9.1 applies.

4.2.2.9.2 Limits

4.2.2.9.2.1 In-band blocking

In-Band Blocking (IBB) is defined for an unwanted interfering signal within the range of 15 MHz below and 15 MHz above the aerial UE receive band.

The throughput of the wanted signal shall be ≥ 95 % of the maximum throughput of the reference measurement channels as specified in ETSI TS 138 101-1 [3], clauses A.2.2, A.2.3, A.3.2 and A.3.3 (with one sided dynamic OCNG Pattern OP.1 FDD/TDD for the DL-signal as described in ETSI TS 138 101-1 [3], clauses A.5.1.1/A.5.2.1) with parameters specified in Table 4.2.2.9.2.1-1 and Table 4.2.2.9.2.1-2. The relative throughput requirement shall be met for any SCS specified for the channel bandwidth of the wanted signal. For operating bands with an unpaired DL part (as noted in Table 1.1-1), the requirements only apply for carriers assigned in the paired part.

Table 4.2.2.9.2.1-1: In-band blocking parameters for NR bands

RX parameter	Units	Channel bandwidth (MHz)		
		5, 10	15	20, 25, 30, 35, 40, 45, 50, 60, 70, 80, 90, 100
Power in transmission bandwidth configuration (note 3)	dBm	REFSENS + 6 dB	REFSENS + 7 dB	REFSENS + (9 + $10\log_{10}(BW_{Channel}/20)$) dB
$BW_{interferer}$	MHz	5		
$F_{loffset,\ case\ 1}$	MHz	7,5		
$F_{loffset,\ case\ 2}$	MHz	12,5		
NOTE 1: The transmitter shall be set to 4 dB below $P_{CMAX_L,f,c}$ at the minimum UL configuration specified in Table 4.2.2.7.2-3 with $P_{CMAX_L,f,c}$ defined in clause 6.2.4 of ETSI TS 138 101-1 [3]. NOTE 2: The interferer consists of the RMC specified in ETSI TS 138 101-1 [3], clauses A.3.2.2 and A.3.3.2 with one sided dynamic OCNG Pattern OP.1 FDD/TDD for the DL-signal as described in ETSI TS 138 101-1 [3], clauses A.5.1.1/A.5.2.1 and 15 kHz SCS. NOTE 3: $10\log_{10}(x)$ is rounded to the next higher 0,5 dB value.				

Table 4.2.2.9.2.1-2: In-band blocking for NR bands

NR band	Parameter	Unit	Case 1	Case 2	Case 3	Case 4	Case 5
	$P_{interferer}$	dBm	-56	-44	-15	-38	-224
	$F_{interferer}$ (offset)	MHz	$-BW_{Channel}/2 - F_{loffset,\ case\ 1}$ and $BW_{Channel}/2 + F_{loffset,\ case\ 1}$	$\leq -BW_{Channel}/2 - F_{loffset,\ case\ 2}$ and $\geq BW_{Channel}/2 + F_{loffset,\ case\ 2}$		$-BW_{Channel}/2-11$	
n1, n3, n7, n8, n20, n28, n31, n38, n91, n92, n93, n94	$F_{interferer}$	MHz	Note 2	F_{DL_low} - 15 to F_{DL_high} + 15			
NOTE 1: The absolute value of the interferer offset Finterferer (offset) shall be further adjusted to $(\lceil \lvert F_{interferer} \rvert / SCS \rceil + 0.5) SCS$ MHz with SCS the sub-carrier spacing of the wanted signal in MHz. The interferer is an NR signal with 15 kHz SCS. NOTE 2: For each carrier frequency, the requirement applies for two interferer carrier frequencies: a: $-BW_{Channel}/2 - F_{loffset,\ case\ 1}$; b: $BW_{Channel}/2 + F_{loffset,\ case\ 1}$.							

4.2.2.9.2.2 Out-of-band blocking

Out-of-band blocking is defined for an unwanted CW interfering signal falling outside a frequency range 15 MHz below or above the aerial UE receive band. The throughput of the wanted signal shall be ≥ 95 % of the maximum throughput of the reference measurement channels as specified in ETSI TS 138 101-1 [3], clauses A.2.2, A.2.3, A.3.2 and A.3.3 (with one sided dynamic OCNG Pattern OP.1 FDD/TDD for the DL-signal as described in ETSI TS 138 101-1 [3], clauses A.5.1.1 and A.5.2.1) with parameters specified in Table 4.2.2.9.2.2-1 and Table 4.2.2.9.2.2-2. The relative throughput requirement shall be met for any SCS specified for the channel bandwidth of the wanted signal.

Table 4.2.2.9.2.2-1: Out-of-band blocking parameters for NR bands with F_{DL_high} < 2 700 MHz and F_{UL_high} < 2 700 MHz

Channel bandwidth	Power in transmission bandwidth configuration [dBm]
5 MHz	REFSENS + 6,0 dB
10 MHz	REFSENS + 6,0 dB
15 MHz	REFSENS + 7,0 dB
20 MHz	REFSENS + 9,0 dB
25 MHz	REFSENS + 10,0 dB
30 MHz	REFSENS + 11,0 dB
35 MHz	REFSENS + 11,5 dB
40 MHz	REFSENS + 12,0 dB
45 MHz	REFSENS + 12,5 dB
50 MHz	REFSENS + 13,0 dB
60 MHz	REFSENS + 14,0 dB
70 MHz	REFSENS + 14,5 dB
80 MHz	REFSENS + 15,0 dB
90 MHz	REFSENS + 15,5 dB
100 MHz	REFSENS + 16,0 dB
NOTE: The transmitter shall be set to 4 dB below $P_{CMAX_L,f,c}$ at the minimum UL configuration specified in Table 4.2.2.7.2-3 with $P_{CMAX_L,f,c}$ defined in clause 6.2.4 of ETSI TS 138 101-1 [3].	

Table 4.2.2.9.2.2-2: Out of-band blocking for NR bands

NR band	Parameter	Unit	Range 1	Range 2	Range 3
n1, n3, n7, n8, n20, n28, n31, n38, n91, n92, n93, n94	$P_{interferer}$	dBm	-44	-30	-15
	$F_{interferer}$ (CW)	MHz	-60 < f – F_{DL_low} < -15 or 15 < f – F_{DL_high} < 60	-85 < f – F_{DL_low} ≤ -60 or 60 ≤ f – F_{DL_high} < 85	1 ≤ f ≤ F_{DL_low} - 85 or F_{DL_high} + 85 ≤ f ≤ 12 750
NOTE: The power level of the interferer ($P_{Interferer}$) for Range 3 shall be modified to -20 dBm for $F_{Interferer}$ > 6 000 MHz.					

For interferer frequencies across ranges 1, 2 and 3 in Table 4.2.2.9.2.2-2, a maximum of:

$$\lfloor max\{24{,}6 \cdot \lceil n \cdot N_{RB}/6 \rceil\}/min\{\lfloor n \cdot N_{RB}/10 \rfloor, 5\} \rfloor$$

exceptions are allowed for spurious response frequencies in each assigned frequency channel when measured using a step size of $\min(BW_{channel}/2, 5)$ MHz with N_{RB} the number of resource blocks in the downlink transmission bandwidth configuration, $BW_{Channel}$ is the bandwidth of the frequency channel in MHz and n = 1, 2, 3 for SCS = 15, 30, 60 kHz, respectively. For these exceptions, the requirements in clause 4.2.2.10 apply.

4.2.2.9.2.3 Narrow band blocking

This requirement is the measure of a receiver's ability to receive a NR signal at its assigned channel frequency in the presence of an unwanted narrow band CW interferer at a frequency, which is less than the nominal channel spacing.

The relative throughput shall be ≥ 95 % of the maximum throughput of the reference measurement channels as specified in ETSI TS 138 101-1 [3], clauses A.2.2, A.2.3, A.3.2 and A.3.3 (with one sided dynamic OCNG Pattern OP.1 FDD/TDD for the DL-signal as described in ETSI TS 138 101-1 [3], clauses A.5.1.1/A.5.2.1) with parameters specified in Table 4.2.2.9.2.3-1. For operating bands with an unpaired DL part (as noted in Table 1.1-1), the requirements only apply for carriers assigned in the paired part.

Table 4.2.2.9.2.3-1: Narrow Band Blocking

NR band	Parameter	Unit	Channel Bandwidth (MHz)					
			5	10	15	20	25, 30, 35, 40, 45, 50	60, 70, 80, 90, 100
n1, n3, n7, n8, n20, n28, , n31, n38	P_w	dBm	$P_{REFSENS}$ + channel-bandwidth specific value below					
		dB	16	13	14	16	16	16
	P_{uw} (CW)	dBm	-55					
	F_{uw} (offset SCS= 15 kHz) Note 4	MHz	$\left(\left\lfloor \frac{BW_{Channel}}{2} + 0.2}{SCS} + 0.5 \right\rfloor + 0.5\right) SCS$				$\left(\left\lfloor \frac{BW_{Channel}}{2} + BW_{GB,Channe}}{SCS} + 0.5 \right\rfloor + 0.5\right) SCS$	NA
	F_{uw} (offset SCS= 30 kHz) Note 4	MHz	NA					$\left(\left\lfloor \frac{BW_{Channel}}{2} + BW_{GB,Channe}}{SCS} + 0.5 \right\rfloor + 0.5\right) SCS$

NOTE 1: The transmitter shall be set a 4 dB below $P_{CMAX_L,f,c}$ at the minimum UL configuration specified in Table 4.2.2.7.2-3 with $P_{CMAX_L,f,c}$ defined in clause 6.2.4 of ETSI TS 138 101-1 [3].
NOTE 2: Reference measurement channel is specified in ETSI TS 138 101-1 [3], clauses A.3.2 and A.3.3 with one sided dynamic OCNG Pattern OP.1 FDD/TDD as described in ETSI TS 138 101-1 [3], clauses A.5.1.1/A.5.2.1.
NOTE 3: The $P_{REFSENS}$ power level is specified in Table 4.2.2.7.2-1 and Table 4.2.2.7.2-2 for two and four antenna ports, respectively.
NOTE 4: F_{uw} shall be rounded to half of SCS.

4.2.2.9.3 Conformance

The conformance tests for this requirement are defined in clause 5.2.3.8.

4.2.2.10 Receiver Spurious Response

4.2.2.10.1 Definition

The definition given in clause 4.1.2.10.1 applies.

4.2.2.10.2 Limits

The throughput shall be ≥ 95 % of the maximum throughput of the reference measurement channels as specified in ETSI TS 138 101-1 [3], clauses A.2.2, A.2.3 and A.3.2 (with one sided dynamic OCNG Pattern OP.1 FDD/TDD for the DL-signal as described in ETSI TS 138 101-1 [3], clauses A.5.1.1/A.5.2.1) with parameters for the wanted signal as specified in Table 4.2.2.10.2-1 and for the interferer as specified in Table 4.2.2.10.2-2. The relative throughput requirement shall be met for any SCS specified for the channel bandwidth of the wanted signal. For operating bands with an unpaired DL part (as noted in Table 1.1-1), the requirements only apply for carriers assigned in the paired part.

Table 4.2.2.10.2-1: Spurious response parameters for NR bands

RX parameter	Units	Channel bandwidth (MHz)		
		5, 10	15	20, 25, 30, 35, 40, 45, 50, 60, 70, 80, 90, 100
Power in transmission bandwidth configuration (note 2)	dBm	REFSENS + 6 dB	REFSENS +7 dB	REFSENS + (9 + $10\log_{10}(BW_{Channel}/20)$) dB
NOTE 1: The transmitter shall be set to 4 dB below $P_{CMAX_L,f,c}$ at the minimum UL configuration specified in Table 4.2.2.7.2-3 with $P_{CMAX_L,f,c}$ defined in clause 6.2.4 in ETSI TS 138 101-1 [3]. NOTE 2: 10log10(x) is rounded to the next higher 0,5 dB value.				

Table 4.2.2.10.2-2: Spurious response

Parameter	Unit	Level
$P_{Interferer}$ (CW)	dBm	-44
$F_{Interferer}$	MHz	Spurious response frequencies

4.2.2.10.3 Conformance

The conformance tests for this requirement are defined in clause 5.2.3.9.

4.2.2.11 Receiver Intermodulation Characteristic

4.2.2.11.1 Definition

The definition given in clause 4.1.2.11.1 applies.

4.2.2.11.2 Limits

The wideband intermodulation requirement is defined using a CW carrier and modulated NR signal as interferer 1 and interferer 2 respectively.

The throughput shall be ≥ 95 % of the maximum throughput of the reference measurement channels as specified in ETSI TS 138 101-1 [3], clauses A.2.2, A.2.3, A.3.2 and A.3.3 (with one sided dynamic OCNG Pattern OP.1 FDD/TDD for the DL-signal as described in ETSI TS 138 101-1 [3], clauses A.5.1.1/A.5.2.1) with parameters specified in Table 4.2.2.11.2-1. The relative throughput requirement shall be met for any SCS specified for the channel bandwidth of the wanted signal. For operating bands with an unpaired DL part (as noted in Table 1.1-1), the requirements only apply for carriers assigned in the paired part.

Table 4.2.2.11.2-1: Wide band intermodulation parameters for NR bands

Rx parameter	Units	Channel bandwidth (MHz)		
		5, 10	15	20, 25, 30, 35, 40, 45, 50, 60, 70, 80, 90, 100
P_w in Transmission Bandwidth Configuration, per CC Note 5	dBm	REFSENS + 6 dB	REFSENS + 7 dB	REFSENS + (9 + $10\log_{10}(BW_{Channel}$ /20)) dB
$P_{Interferer\ 1}$ (CW)	dBm	-46		
$P_{Interferer\ 2}$ (Modulated)	dBm	-46		
$BW_{Interferer\ 2}$	MHz	5		
$F_{Interferer\ 1}$ (Offset)	MHz	$-BW_{channel}/2$ - 7,5 / $+BW_{channel}/2$ + 7,5		
$F_{Interferer\ 2}$ (Offset)	MHz	$2*F_{Interferer\ 1}$		
NOTE 1: The transmitter shall be set to 4 dB below $P_{CMAX_L,f,c}$ at the minimum UL configuration specified in Table 4.2.2.7.2-3 with $P_{CMAX_L,f,c}$ defined in clause 6.2.4 of ETSI TS 138 101-1 [3]. NOTE 2: Reference measurement channel is specified in ETSI TS 138 101-1 [3], clauses A.2.2, A.2.3, A.3.2 and A.3.3 (with one sided dynamic OCNG Pattern OP.1 FDD/TDD for the DL-signal as described in ETSI TS 138 101-1 [3] clauses A.5.1.1/A.5.2.1). NOTE 3: The modulated interferer consists of the Reference measurement channel specified in ETSI TS 138 101-1 [3], clauses A.3.2.2 and A.3.3.2 with one sided dynamic OCNG Pattern OP.1 FDD/TDD for the DL-signal as described in ETSI TS 138 101-1 [3], clauses A.5.1.1/A.5.2.1 and 15 kHz SCS. NOTE 4: The $F_{interferer\ 1}$ (offset) is the frequency separation of the centre frequency of the carrier closest to the interferer and the centre frequency of the CW interferer and $F_{interferer\ 2}$ (offset) is the frequency separation of the centre frequency of the carrier closest to the interferer and the centre frequency of the modulated interferer. NOTE 5: $10\log_{10}(x)$ is rounded to the next higher 0,5 dB value.				

4.2.2.11.3 Conformance

The conformance tests for this requirement are defined in clause 5.2.3.10.

4.2.2.12 Receiver Spurious Emissions

4.2.2.12.1 Definition

The definition given in clause 4.1.2.12.1 applies.

4.2.2.12.2 Limits

The power of any spurious emission shall not exceed the maximum level specified in Table 4.2.2.12.2-1.

Table 4.2.2.12.2-1: General receiver spurious emission requirements

Frequency range	Measurement bandwidth	Maximum level	Note
30 MHz ≤ f < 1 GHz	100 kHz	-57 dBm	
1 GHz ≤ f ≤ 12,75 GHz	1 MHz	-47 dBm	
12,75 GHz ≤ f ≤ 5th harmonic of the upper frequency edge of the DL operating band in GHz	1 MHz	-47 dBm	2
12,75 GHz to 26 GHz	1 MHz	-47 dBm	3
NOTE 1: Unused PDCCH resources are padded with resource element groups with power level given by PDCCH as defined in ETSI TS 138 101-1 [3], clause C.3.1. NOTE 2: Applies for Band that the upper frequency edge of the DL Band more than 2,69 GHz. NOTE 3: Applies for Band that the upper frequency edge of the DL Band more than 5,2 GHz.			

4.2.2.12.3 Conformance

The conformance tests for this requirement are defined in clause 5.2.3.11.

4.2.2.13 Transmit OFF Power

4.2.2.13.1 Definition

Transmit OFF power is defined as the mean power in the channel bandwidth when the transmitter is OFF. The transmitter is considered OFF when the aerial UE is not allowed to transmit on any of its ports. An excess transmit OFF power potentially increases the Rise Over Thermal (RoT) and therefore reduces the cell coverage area for other UEs.

The transmit OFF power is defined as the mean power in a duration of at least one sub-frame (1 ms) excluding any transient periods.

4.2.2.13.2 Requirements

The transmit OFF power shall not exceed the values specified in Table 4.2.2.13.2-1.

Table 4.2.2.13.2-1: Transmit OFF power

Channel bandwidth (MHz)	Transmit OFF power (dBm) f ≤ 3,0 GHz	Measurement bandwidth (MHz)
5	-48,5	4,515
10	-48,5	9,375
15	-48,5	14,235
20	-48,5	19,095
25	-48,5	23,955
30	-48,5	28,815
35	-48,5	-48,2
40	-48,5	38,895
45	-48,3	-48,2
50	-48,3	48,615

4.2.2.13.3 Conformance

The conformance tests for this requirement are defined in clause 5.2.3.12 of the present document.

4.2.2.14 Aerial UE Identifier

4.2.2.14.1 Definition

The definition given in clause 4.1.2.13.1 applies.

4.2.2.14.2 Requirements

An aerial UE shall be provisioned with a single CAA-Level UAV ID. An aerial UE shall not be capable to connect to an NR network for aerial-related services without such a CAA-Level UAV ID.

4.2.2.14.3 Conformance

Conformance assessments are defined in clause 5.2.3.13.

4.2.2.15 Aerial UE Authorization and Authentication

4.2.2.15.1 Definition

The definition given in clause 4.1.2.14.1 applies.

4.2.2.15.2 Requirements

The requirement given in clause 4.1.2.14.2 applies.

4.2.2.15.3 Conformance

Conformance assessments are defined in clause 5.2.3.14.

5 Testing for compliance with technical requirements

5.1 Testing for compliance with technical requirements for E-UTRA

5.1.1 Environmental conditions for testing

Tests defined in the present document shall be carried out at representative points within the boundary limits of the operational environmental profile defined by its intended use.

Where technical performance varies subject to environmental conditions, tests shall be carried out under a sufficient variety of environmental conditions (within the boundary limits of the operational environmental profile defined by its intended use) to give confidence of compliance for the affected technical requirements.

Test environments are specified in annex B.

For guidance on the use of other conditions to be used in order to show compliance reference can be made to ETSI TS 136 521-1 [5], clause F.1.1.

For each operating frequency band of the aerial UE, the tests in the present document are performed with appropriate frequencies defined in ETSI TS 136 508 [6].

5.1.2 Interpretation of the measurement results

The interpretation of the results recorded in a test report for the measurements described in the present document shall be as follows:

- the measured value related to the corresponding limit will be used to decide whether an equipment meets the requirements of the present document;
- the value of the measurement uncertainty for the measurement of each parameter shall be included in the test report.

Values for the maximum measurement uncertainty based on this expansion factor can be found in clause D.1.

5.1.3 Essential radio test suites

5.1.3.0 General

This clause describes the test suites for E-UTRA FDD and TDD.

5.1.3.1 Transmitter Maximum Output Power

5.1.3.1.1 Method of test

5.1.3.1.1.1 Initial conditions

Test environment: normal, TL/VL, TL/VH, TH/VL, TH/VH; as specified in annex B.

Frequencies to be tested: low range, mid-range, high range; as specified in ETSI TS 136 508 [6], clause 4.3.1.

Channel bandwidths to be tested: lowest, 5 MHz and highest channel bandwidth as defined in ETSI TS 136 508 [6], clause 4.3.1.

Uplink/Downlink configurations: as specified in ETSI TS 136 521-1 [5], clause 6.2.2.4:

1) Connect the SS to the aerial UE antenna connectors as shown in ETSI TS 136 508 [6], annex A.

2) The parameter settings for the cell are set up according to ETSI TS 136 508 [6], clause 4.4.3.

3) Downlink signals are initially set up according to ETSI TS 136 521-1 [5], clauses C.0, C.1 and C.3.0 and uplink signals according to clauses H.1 and H.3.0.

4) The UL Reference Measurement channels are set according to ETSI TS 136 521-1 [5], clause 6.2.2.4.1.

5) Propagation conditions are set according to ETSI TS 136 521-1 [5], clause B.0.

6) Ensure the aerial UE is in State 3A-RF according to ETSI TS 136 508 [6], clause 5.2A.2.

NOTE: When reference is made to test set up, call set up and test mode, guidance on the applicability of these can be found in ETSI TS 136 521-1 [5], ETSI TS 136 508 [6] and ETSI TS 136 509 [i.19] respectively.

5.3.1.1.1.2 Procedure

1) SS sends uplink scheduling information for each UL HARQ process via PDCCH DCI format 0 for C_RNTI to schedule the UL RMC according to Table 6.2.2.1.4.1-1 of ETSI TS 136 521-1 [5]. Since the aerial UE has no payload and no loopback data to send the aerial UE sends uplink MAC padding bits on the UL RMC.

2) Send continuously uplink power control "up" commands in every uplink scheduling information to the aerial UE; allow at least 200 ms for the aerial UE to reach P_{UMAX} level.

3) Measure the mean power of the aerial UE in the channel bandwidth of the radio access mode. The period of measurement shall be at least the continuous duration of one sub-frame (1 ms). For TDD slots with transient periods are not under test.

4) Repeat for applicable test frequencies, channel bandwidths, operating bands and environmental conditions.

Details of the test method can be found in ETSI TS 136 521-1 [5], clause 6.2.2.

5.1.3.1.2 Test requirements

The results obtained shall be compared to the limits in clause 4.1.2.2.2 in order to show compliance.

5.1.3.2 Transmitter Minimum Output Power

5.1.3.2.1 Method of test

5.1.3.2.1.1 Initial conditions

Test Environment: normal, TL/VL, TL/VH, TH/VL, TH/VH; as specified in annex B.

Frequencies to be tested: low range, mid-range and high range; see ETSI TS 136 508 [6], clause 4.3.1.

Channel bandwidths to be tested: lowest, 5 MHz and highest channel bandwidth, as specified in ETSI TS 136 508 [6], clause 4.3.1.

Uplink/Downlink configurations: as specified in ETSI TS 136 521-1 [5], clause 6.3.2.4:

1) Connect the SS to the aerial UE antenna connectors as shown in ETSI TS 136 508 [6], annex A.

2) The parameter settings for the cell are set up according to ETSI TS 136 508 [6], clause 4.4.3.

3) Downlink signals are initially set up according to ETSI TS 136 521-1 [5], clauses C.0, C.1 and C.3.0 and uplink signals according to clauses H.1 and H.3.0.

4) The UL Reference Measurement channels are set according to ETSI TS 136 521-1 [5], clause 6.3.2.4.1.

5) Propagation conditions are set according to ETSI TS 136 521-1 [5], clause B.0.

6) Ensure the aerial UE is in State 3A-RF according to ETSI TS 136 508 [6], clause 5.2A.2.

NOTE: When reference is made to test set up, call set up and test mode, guidance on the applicability of these can be found in ETSI TS 136 521-1 [5], ETSI TS 136 508 [6] and ETSI TS 136 509 [i.19] respectively.

5.1.3.2.1.2 Procedure

1) SS sends uplink scheduling information for each UL HARQ process via PDCCH DCI format 0 for C_RNTI to schedule the UL RMC according to ETSI TS 136 521-1 [5], Table 6.3.2.4.1-1. Since the aerial UE has no payload and no loopback data to send the aerial UE sends uplink MAC padding bits on the UL RMC.

2) Send continuous uplink power control "down" commands in the uplink scheduling information to the aerial UE to ensure that the aerial UE transmits at its minimum output power.

3) Measure the mean power of the aerial UE in the associated measurement bandwidth specified in Table 4.1.2.3.2-1 for the specific channel bandwidth under test. The period of measurement shall be the continuous duration of one sub-frame (1 ms). For TDD slots with transient periods are not under test.

4) Repeat for applicable test frequencies, channel bandwidths, operating bands and environmental conditions.

Details of the test method can be found in ETSI TS 136 521-1 [5], clause 6.3.2.

5.1.3.2.2 Test requirements

The results obtained shall be compared to the limits in clause 4.1.2.3.2 in order to show compliance.

5.1.3.3 Transmitter Spectrum Emission Mask

5.1.3.3.1 Method of test

5.1.3.3.1.1 Initial conditions

Test environment: normal, as specified in annex B.

Frequencies to be tested: low range, mid-range and high range; as specified in ETSI TS 136 508 [6], clause 4.3.1.

Channel bandwidths to be tested: lowest, 5 MHz, 10 MHz and highest channel bandwidth as defined in ETSI TS 136 508 [6], clause 4.3.1.

Uplink/Downlink configurations: as specified in ETSI TS 136 521-1 [5], clause 6.6.2.1.4:

1) Connect the SS to the UE antenna connectors as shown in ETSI TS 136 508 [6], annex A.

2) The parameter settings for the cell are set up according to ETSI TS 136 508 [6], clause 4.4.3.

3) Downlink signals are initially set up according to ETSI TS 136 521-1 [5], clauses C.0, C.1 and C.3.0 and uplink signals according to clauses H.1 and H.3.0.

4) The UL Reference Measurement channels are set according to ETSI TS 136 521-1 [5], clause 6.6.2.1.4.1.

5) Propagation conditions are set according to ETSI TS 136 521-1 [5], clause B.0.

6) Ensure the aerial UE is in State 3A-RF according to ETSI TS 136 508 [6], clause 5.2A.2.

NOTE: When reference is made to test set up, call set up and test mode, guidance on the applicability of these can be found in ETSI TS 136 521-1 [5], ETSI TS 136 508 [6] and ETSI TS 136 509 [i.19] respectively.

5.1.3.3.1.2 Procedure

1) SS sends uplink scheduling information via PDCCH DCI format 0 for C_RNTI to schedule the UL RMC according to ETSI TS 136 521-1 [5], Table 6.6.2.1.4.1-1. Since the aerial UE has no payload data to send, the aerial UE transmits uplink MAC padding bits on the UL RMC.

2) Send continuously uplink power control "up" commands in the uplink scheduling information to the aerial UE until the aerial UE transmits at P_{UMAX} level.

3) Measure the power of the transmitted signal with a measurement filter of bandwidths according to Table 4.1.2.4.2-1. The centre frequency of the filter shall be stepped in continuous steps according to the same table. The measured power shall be recorded for each step. The measurement period shall capture the active TSs.

4) Repeat for applicable test frequencies, channel bandwidths and operating bands.

Details of the test method can be found in ETSI TS 136 521-1 [5], clause 6.6.2.1.

5.1.3.3.2 Test requirements

The results obtained shall be compared to the limits in clause 4.1.2.4.2 in order to show compliance.

5.1.3.4 Transmitter Adjacent Channel Leakage Power Ratio

5.1.3.4.1 Method of test

5.1.3.4.1.1 Initial conditions

Test Environment: normal, TL/VL, TL/VH, TH/VL and TH/VH, as specified in annex B.

Frequencies to be tested: low range, mid-range and high range; see ETSI TS 136 508 [6], clause 4.3.1.

Channel bandwidth to be tested: lowest, 5 MHz, 10 MHz and highest channel bandwidth as defined in ETSI TS 136 508 [6], clause 4.3.1.

Uplink/Downlink configurations: as specified in ETSI TS 136 521-1 [5], clause 6.6.2.3.4:

1) Connect the SS to the aerial UE to the aerial UE antenna connectors as shown in ETSI TS 136 508 [6], annex A.

2) The parameter settings for the cell are set up according to ETSI TS 136 508 [6], clause 4.4.3.

3) Downlink signals are initially set up according to ETSI TS 136 521-1 [5], clauses C.0, C.1, C.3.0 and uplink signals according to clauses H.1 and H.3.0.

4) The UL Reference Measurement channels are set according to ETSI TS 136 521-1 [5].

5) Propagation conditions are set according to ETSI TS 136 521-1 [5], clause B.0.

6) Ensure the aerial UE is in State 3A-RF according to ETSI TS 136 508 [6], clause 5.2A.2.

NOTE: When reference is made to test set up, call set up and test mode, guidance on the applicability of these can be found in ETSI TS 136 521-1 [5], ETSI TS 136 508 [6] and ETSI TS 136 509 [i.19] respectively.

5.1.3.4.1.2 Procedure

1) SS sends uplink scheduling information for each UL HARQ process via PDCCH DCI format 0 for C_RNTI to schedule the UL RMC according to ETSI TS 136 521-1 [5], Table 6.6.2.3.4.1-1. Since the aerial UE has no payload data to send, the aerial UE transmits uplink MAC padding bits on the UL RMC.

2) Send continuous uplink power control "up" commands in the uplink scheduling information to the aerial UE to ensure that the aerial UE transmits at P_{UMAX} level.

3) Measure the mean power of the aerial UE in the channel bandwidth of the radio access mode according to the test configuration, which shall meet the requirements described in Tables 4.1.2.5.2-1 or 4.1.2.5.2-2, as applicable. The period of the measurement shall be at least the continuous duration of one sub-frame (1 ms). For TDD slots with transient periods are not under test.

4) Measure the filtered mean power for E-UTRA.

5) Measure the filtered mean power of the first E-UTRA adjacent channel.

6) Measure the RRC filtered mean power of the first and the second UTRA adjacent channel.

Calculate the ratio of the power between the values measured in step 4) over step 5) for E-UTRA ACLR.

Calculate the ratio of the power between the values measured in step 4) over step 6) for UTRA ACLR1, UTRA ACLR2.

7) Repeat for applicable test frequencies, channel bandwidths, operating bands and environmental conditions.

Details of the test method can be found in ETSI TS 136 521-1 [5], clause 6.6.2.3.

5.1.3.4.2 Test requirements

The results obtained shall be compared to the limits in clause 4.1.2.5.2 in order to show compliance.

5.1.3.5 Transmitter Spurious Emissions

5.1.3.5.1 Method of test

5.1.3.5.1.1 Initial conditions

Test Environment: normal, as specified in annex B.

Frequencies to be tested: low range, mid-range and high range; see ETSI TS 136 508 [6], clause 4.3.1.

Channel bandwidths to be tested: lowest, 5 MHz and highest channel bandwidth, as specified in ETSI TS 136 508 [6].

Uplink/Downlink configurations: as specified in ETSI TS 136 521-1 [5], clause 6.6.3.1.4:

1) Connect the SS to the UE antenna connectors as shown in ETSI TS 136 508 [6], annex A.

2) The parameter settings for the cell are set up according to ETSI TS 136 508 [6], clause 4.4.3.

3) Downlink signals are initially set up according to ETSI TS 136 521-1 [5], clauses C.0, C.1 and C.3.0 and uplink signals according to clauses H.1 and H.3.0.

4) The UL Reference Measurement channels are set according to ETSI TS 136 521-1 [5], clause 6.3.2.4.1.

5) Propagation conditions are set according to ETSI TS 136 521-1 [5], clause B.0.

6) Ensure the aerial UE is in State 3A-RF according to ETSI TS 136 508 [6], clause 5.2A.2.

NOTE: When reference is made to test set up, call set up and test mode, guidance on the applicability of these can be found in ETSI TS 136 521-1 [5], ETSI TS 136 508 [6] and ETSI TS 136 509 [i.19] respectively.

5.1.3.5.1.2 Procedure

1) SS sends uplink scheduling information for each UL HARQ process via PDCCH DCI format 0 for C_RNTI to schedule the UL RMC according to ETSI TS 136 521-1 [5], Table 6.3.2.4.1-1. Since the aerial UE has no payload and no loopback data to send the aerial UE sends uplink MAC padding bits on the UL RMC.

2) Send continuous uplink power control "down" commands in the uplink scheduling information to the aerial UE to ensure that the aerial UE transmits at its minimum output power.

3) For each applicable requirement in Tables from 4.1.2.6.2-2 to 4.1.2.6.2-; measure the power of the transmitted signal with a measurement filter of bandwidths according to Tables from 4.1.2.6.2-2 to 4.1.2.6.2-6. The centre frequency of the filter shall be stepped in contiguous steps according to the tables. The measured power shall be verified for each step. The measurement period shall capture the active time slots.

4) Repeat for applicable test frequencies, channel bandwidths, operating bands and environmental conditions.

NOTE 1: For measurement conditions at the edge of each frequency range, the lowest frequency of the measurement position in each frequency range should be set at the lowest boundary of the frequency range plus MBW/2. The highest frequency of the measurement position in each frequency range should be set at the highest boundary of the frequency range minus MBW/2. MBW denotes the measurement bandwidth defined for the protected band.

Details of the test method can be found in ETSI TS 136 521-1 [5], clauses 6.6.3.1, 6.6.3.2 and 6.6.3.3.

NOTE 2: No test points are defined in the present document for limits given in Tables 4.1.2.6.2-5 and 4.1.2.6.2-6.

5.1.3.5.2 Test requirements

The results obtained shall be compared to the limits in clause 4.1.2.6.2 in order to show compliance.

5.1.3.6 Receiver Reference Sensitivity Level

5.1.3.6.1 Method of test

5.1.3.6.1.1 Initial conditions

Test Environment: normal, TL/VL, TL/VH, TH/VL, TH/VH; as specified in annex B.

Frequencies to be tested: low range, mid-range, high range see ETSI TS 136 508 [6], clause 4.3.1.

Channel bandwidth to be tested: lowest, 5 MHz and highest channel bandwidth as defined in ETSI TS 136 508 [6], clause 4.3.1.

Uplink/Downlink configurations: as specified in ETSI TS 136 521-1 [5], clause 7.3.4:

1) Connect the SS and interfering source to the aerial UE antenna connectors as shown in ETSI TS 136 508 [6], annex A.

2) The parameter settings for the cell are set up according to ETSI TS 136 508 [6], clause 4.4.3.

3) Downlink signals are initially set up according to ETSI TS 136 521-1 [5], clauses C.0, C.1, C.3.1 and uplink signals according to clauses H.1 and H.3.1.

4) The UL and DL Reference Measurement channels are set according to ETSI TS 136 521-1 [5], Table 7.3.4.1-1.

5) Propagation conditions are set according to ETSI TS 136 521-1 [5], clause B.0.

6) Ensure the aerial UE is in State 3A-RF according to ETSI TS 136 508 [6], clause 5.2A.2.

NOTE: When reference is made to test set up, call set up and test mode, guidance on the applicability of these can be found in ETSI TS 136 521-1 [5], ETSI TS 136 508 [6] and ETSI TS 136 509 [i.19] respectively.

5.1.3.6.1.2 Procedure

1) SS transmits PDSCH via PDCCH DCI format 1A for C_RNTI to transmit the DL RMC according to ETSI TS 136 521-1 [5], Table 7.3.4.1-1. The SS sends downlink MAC padding bits on the DL RMC.

2) SS sends uplink scheduling information for each UL HARQ process via PDCCH DCI format 0 for C_RNTI to schedule the UL RMC according to ETSI TS 136 521-1 [5], Table 7.3.4.1-1. Since the aerial UE has no payload data to send, the aerial UE transmits uplink MAC padding bits on the UL RMC.

3) Set the Downlink signal level to the appropriate REFSENS value defined in Table 4.1.2.7.2-1. Send continuously uplink power control "up" commands in the uplink scheduling information to the aerial UE to ensure the aerial UE transmits P_{UMAX} level for at least the duration of the Throughput measurement. (obtain correct UE output power as specified in ETSI TS 136 521-1 [5]).

4) Measure the average throughput according to clause G.2 of ETSI TS 136 521-1 [5].

5) Repeat for applicable test frequencies, channel bandwidths and operating bands.

Details of the test method can be found in ETSI TS 136 521-1 [5], clause 7.3.

5.1.3.6.2 Test requirements

The results obtained shall be compared to the limits in clause 4.1.2.7.2 in order to show compliance.

5.1.3.7 Receiver Adjacent Channel Selectivity (ACS)

5.1.3.7.1 Method of test

5.1.3.7.1.1 Initial conditions

Test Environment: normal, as specified in annex B.

Frequencies to be tested: mid-range see ETSI TS 136 508 [6], clause 4.3.1.

Channel bandwidth to be tested: lowest, 5 MHz and highest channel bandwidth as defined in ETSI TS 136 508 [6], clause 4.3.1.

Uplink/Downlink configurations: as specified in ETSI TS 136 521-1 [5], clause 7.5.4:

1) Connect the SS and interfering source to the aerial UE antenna connectors as shown in ETSI TS 136 508 [6], annex A.

2) The parameter settings for the cell are set up according to ETSI TS 136 508 [6], clause 4.4.3.

3) Downlink signals are initially set up according to ETSI TS 136 521-1 [5], clauses C.0, C.1 and C.3.1 and uplink signals according to clauses H.1 and H.3.1.

4) The UL and DL Reference Measurement channels are set according to ETSI TS 136 521-1 [5], Table 7.5.4.1-1.

5) Propagation conditions are set according to ETSI TS 136 521-1 [5], clause B.0.

6) Ensure the aerial UE is in State 3A-RF according to ETSI TS 136 508 [6], clause 5.2A.2.

NOTE: When reference is made to test set up, call set up and test mode, guidance on the applicability of these can be found in ETSI TS 136 521-1 [5], ETSI TS 136 508 [6] and ETSI TS 136 509 [i.19] respectively.

5.1.3.7.1.2 Procedure

1) SS transmits PDSCH via PDCCH DCI format 1A for C_RNTI to transmit the DL RMC according to ETSI TS 136 521-1 [5], Table 7.5.4.1-1. The SS sends downlink MAC padding bits on the DL RMC.

2) SS sends uplink scheduling information for each UL HARQ process via PDCCH DCI format 0 for C_RNTI to schedule the UL RMC according to ETSI TS 136 521-1 [5], Table 7.5.4.1-1. Since the aerial UE has no payload data to send, the aerial UE transmits uplink MAC padding bits on the UL RMC.

3) Set the Downlink signal level to the value as defined in Table 4.1.2.8.2-2 (Case 1). Send Uplink power control commands to the aerial UE (less or equal to 1 dB step size should be used), to ensure that the aerial UE output power is within +0, -3,4 dB of the target level in Table 4.1.2.8.2-2 (Case 1) for carrier frequency $f \leq 3,0$ GHz or within +0, -4,0 dB of the target level for carrier frequency 3,0 GHz < $f \leq 4,2$ GHz, for at least the duration of the Throughput measurement (obtain correct UE output power as specified in ETSI TS 136 521-1 [5], Table 7.5.3-2).

4) Set the Interferer signal level to the value as defined in Table 4.1.2.8.2-2 (Case 1) and frequency below the wanted signal, using a modulated interferer as defined in ETSI TS 136 521-1 [5], annex D.

5) Measure the average throughput according to clause G.2 of ETSI TS 136 521-1 [5].

6) Set the Downlink signal level to the value as defined in Table 4.1.2.8.2-3 (Case 2). Send Uplink power control commands to the aerial UE (less or equal to 1 dB step size should be used), to ensure that the aerial UE output power is within +0, -3,4 dB of the target level in Table 4.2.2.8.2-3 (Case 2) for carrier frequency $f \leq 3,0$ GHz or within +0, -4,0 dB of the target level for carrier frequency 3,0 GHz < $f \leq 4,2$ GHz, for at least the duration of the throughput measurement (obtain correct UE output power as specified in ETSI TS 136 521-1 [5], Table 7.5.3-3.

7) Set the Interferer signal level to the value as defined in Table 4.1.2.8.2-3 (Case 2) and frequency below the wanted signal, using a modulated interferer as defined in ETSI TS 136 521-1 [5], annex D.

8) Measure the average throughput according to ETSI TS 136 521-1 [5], annex G.

9) Repeat for applicable channel bandwidths in both Case 1 and Case 2.

10) Repeat for applicable test frequencies, channel bandwidths and operating bands.

Details of the test method can be found in ETSI TS 136 521-1 [5], clause 7.5

5.1.3.7.2 Test requirements

5.1.3.8 Receiver Blocking Characteristics

5.1.3.8.1 Method of test

5.1.3.8.1.1 Initial conditions

Test Environment: normal, as specified in annex B.

For In-band blocking, the frequencies to be tested are mid-range as defined in ETSI TS 136 508 [6], clause 4.3.1.

For out-of-band blocking, the frequency to be tested is low or high range as defined in ETSI TS 136 508 [6], clause 4.3.1.

For Narrow-band blocking, the frequencies to be tested are mid-range as defined in ETSI TS 136 508 [6], clause 4.3.1.

Channel bandwidth to be tested: lowest, 5 MHz and highest channel bandwidth as defined in ETSI TS 136 508 [6], clause 4.3.1. Range 3 of out-of-band blocking is tested only with the highest bandwidth.

Uplink/Downlink configurations: as specified in ETSI TS 136 521-1 [5]:

1) Connect the SS to the aerial UE antenna connectors as shown in ETSI TS 136 508 [6], annex A.

2) The parameter settings for the cell are set up according to ETSI TS 136 508 [6], clause 4.4.3.

3) Downlink signals are initially set up according to ETSI TS 136 521-1 [5], clauses C.0, C.1 and C.3.1 and uplink signals according to clauses H.1 and H.3.1.

4) The UL and DL Reference Measurement channels are set according to ETSI TS 136 521-1[5], Table 7.6.1.4.1-1, or Table 7.6.2.4.1-1, or Table 7.6.2.4.1-1 as applicable.

5) Propagation conditions are set according to ETSI TS 136 521-1 [5], clause B.0.

6) Ensure the aerial UE is in State 3A-RF according to ETSI TS 136 508 [6], clause 5.2A.2.

NOTE: When reference is made to test set up, call set up and test mode, guidance on the applicability of these can be found in ETSI TS 136 521-1 [5], ETSI TS 136 508 [6] and ETSI TS 136 509 [i.19] respectively.

5.3.6.1.1.2 In-Band Procedure

1) SS transmits PDSCH via PDCCH DCI format 1A for C_RNTI to transmit the DL RMC according to ETSI TS 136 521-1 [5], Table 7.6.1.4.1-1. The SS sends downlink MAC padding bits on the DL RMC.

2) SS sends uplink scheduling information for each UL HARQ process via PDCCH DCI format 0 for C_RNTI to schedule the UL RMC according to ETSI TS 136 521-1 [5], Table 7.6.1.4.1-1. Since the aerial UE has no payload data to send, the aerial UE transmits uplink MAC padding bits on the UL RMC.

3) Set the parameters of the signal generator for an interfering signal below the wanted signal in Case 1 according to Tables 4.1.2.9.2-1 and 4.1.2.9.2-2.

4) Set the downlink signal level according to the Table 4.1.2.9.2-1. Send uplink power control commands to the aerial UE (less or equal to 1 dB step size should be used), to ensure that the aerial UE output power is within +0, -3,4 dB of the target level in Table 4.1.2.9.2-1 for carrier frequency $f \leq 3,0$ GHz or within +0, -4,0 dB of the target level for carrier frequency 3,0 GHz $< f \leq 4,2$ GHz, for at least the duration of the throughput measurement as specified in ETSI TS 136 521-1 [5].

5) Measure the average throughput according to clause G.2 of ETSI TS 136 521-1 [5].

6) Repeat steps from 3 to 5, using an interfering signal above the wanted signal in Case 1 at step 3).

7) Repeat steps from 3 to 6, using interfering signals in Case 2 at steps 3) and 6). The ranges of case 2 are covered in steps equal to the interferer bandwidth. The test frequencies are chosen in analogy to ETSI TS 136 521-1 [5], Table 7.6.1.4.2-1.

8) Repeat for applicable test frequencies, channel bandwidths and operating bands.

Details of the test method can be found in ETSI TS 136 521-1 [5], clause 7.6.1.

5.3.6.1.1.3 Out-Of-Band Procedure

1) SS transmits PDSCH via PDCCH DCI format 1A for C_RNTI to transmit the DL RMC according to ETSI TS 136 521-1 [5], Table 7.6.2.4.1-1. The SS sends downlink MAC padding bits on the DL RMC.

2) SS sends uplink scheduling information for each UL HARQ process via PDCCH DCI format 0 for C_RNTI to schedule the UL RMC according to ETSI TS 136 521-1 [5], Table 7.6.2.4.1-1. Since the aerial UE has no payload data to send, the aerial UE transmits uplink MAC padding bits on the UL RMC.

3) Set the parameters of the CW signal generator for an interfering signal according to Table 4.1.2.9.2-4. The frequency step size is 1 MHz.

4) Set the downlink signal level according to Table 4.1.2.9.2-3. Send uplink power control commands to the aerial UE (less or equal to 1 dB step size should be used), to ensure that the aerial UE output power is within +0, -3,4 dB of the target level in Table 4.1.2.9.2-3 for carrier frequency $f \leq 3,0$ GHz or within +0, -4,0 dB of the target level for carrier frequency 3,0 GHz $< f \leq 4,2$ GHz, for at least the duration of the throughput measurement as specified in ETSI TS 136 521-1 [5].

5) Measure the average throughput according to clause G.2 of ETSI TS 136 521-1 [5].

6) For Table 4.1.2.9.2-4 record the frequencies for which the throughput does not meet the requirements.

7) Repeat for applicable test frequencies, channel bandwidths and operating bands.

Details of the test method can be found in ETSI TS 136 521-1 [5], clause 7.6.2.

5.3.6.1.1.4 Narrow-Band Procedure

1) SS transmits PDSCH via PDCCH DCI format 1A for C_RNTI to transmit the DL RMC according to ETSI TS 136 521-1 [5], Table 7.6.3.4.1-1. The SS sends downlink MAC padding bits on the DL RMC.

2) SS sends uplink scheduling information for each UL HARQ process via PDCCH DCI format 0 for C_RNTI to schedule the UL RMC according to ETSI TS 136 521-1 [5], Table 7.6.3.4.1-1. Since the aerial UE has no payload data to send, the aerial UE transmits uplink MAC padding bits on the UL RMC.

3) Set the parameters of the CW signal generator for an interfering signal below the wanted signal according to Table 4.1.2.9.2-5.

4) Set the downlink signal level according to Table 4.1.2.9.2-5. Send uplink power control commands to the aerial UE (less or equal to 1 dB step size should be used), to ensure that the aerial UE output power is within +0, -3,4 dB of the target level in Table 4.1.2.9.2-5 for carrier frequency $f \leq 3,0$ GHz or within +0, -4,0 dB of the target level for carrier frequency 3,0 GHz $< f \leq 4,2$ GHz, for at least the duration of the throughput measurement as specified in ETSI TS 136 521-1 [5].

5) Measure the average throughput according to clause G.2 of ETSI TS 136 521-1 [5].

6) Repeat steps from 3 to 5, using an interfering signal above the wanted signal at step 3).

7) Repeat for applicable test frequencies, channel bandwidths and operating bands.

Details of the test method can be found in ETSI TS 136 521-1 [5], clause 7.6.3.

5.1.3.8.2 Test requirements

The results obtained shall be compared to the limits in clause 4.1.2.9.2 in order to show compliance.

5.1.3.9 Receiver Spurious Response

5.1.3.9.1 Method of test

5.1.3.9.1.1 Initial conditions

The initial conditions shall be the same as for those in out-of-band blocking in clause 5.1.3.8.1 in order to test spurious responses obtained in clause 5.1.3.8.2 under the same conditions.

5.1.3.9.1.1 Procedure

1) SS transmits PDSCH via PDCCH DCI format 1A for C_RNTI to transmit the DL RMC according to ETSI TS 136 521-1 [5], Table 7.6.2.4.1-1. The SS sends downlink MAC padding bits on the DL RMC.

2) SS sends uplink scheduling information for each UL HARQ process via PDCCH DCI format 0 for C_RNTI to schedule the UL RMC according to ETSI TS 136 521-1 [5], Table 7.6.2.4.1-1. Since the aerial UE has no payload data to send, the aerial UE transmits uplink MAC padding bits on the UL RMC.

3) Set the parameters of the CW signal generator for an interfering signal according to Table 4.1.2.10.2-2. The spurious frequencies are taken from step 5) records in clause 5.1.3.8.1.2.

4) Set the downlink signal level according to Table 4.1.2.10.2-1. Send uplink power control commands to the aerial UE (less or equal to 1 dB step size should be used), to ensure that the aerial UE output power is within +0, -3,4 dB of the target level in Table 4.1.2.10.2-1 for carrier frequency $f \leq 3,0$ GHz or within +0, -4,0 dB of the target level for carrier frequency 3,0 GHz $< f \leq 4,2$ GHz, for at least the duration of the throughput measurement as specified in ETSI TS 136 521-1 [5].

5) For the spurious frequency, measure the average throughput.

Details of the test method can be found in ETSI TS 136 521-1 [5], clause 7.7.

5.1.3.9.2 Test requirements

The results obtained shall be compared to the limits in clause 4.1.2.10.2 in order to show compliance.

5.1.3.10 Receiver Intermodulation Characteristic

5.1.3.10.1 Method of test

5.1.3.10.1.1 Initial conditions

Test Environment: normal, as specified in annex B.

Frequencies to be tested: mid-range; see ETSI TS 136 508 [6], clause 4.3.1.

Channel bandwidths to be tested: lowest, 5 MHz and highest channel bandwidth as defined in ETSI TS 136 508 [6], clause 4.3.1.

Uplink/Downlink configurations: as specified in ETSI TS 136 521-1 [5], clause 7.8.1.4:

1) Connect the SS and interfering sources to the aerial UE antenna connectors as shown in ETSI TS 136 508 [6], annex A.

2) The parameter settings for the cell are set up according to ETSI TS 136 508 [6], clause 4.4.3.

3) Downlink signals are initially set up according to ETSI TS 136 521-1 [5], clauses C.0, C.1, C.3.1 and uplink signals according to clauses H.1 and H.3.1.

4) The UL and DL Reference Measurement channels are set according to ETSI TS 136 521-1 [5], Table 7.8.4.1-1.

5) Propagation conditions are set according to ETSI TS 136 521-1 [5], clause B.0.

6) Ensure the aerial UE is in State 3A-RF according to ETSI TS 136 508 [6], clause 5.2A.2.

NOTE: When reference is made to test set up, call set up and test mode, guidance on the applicability of these can be found in ETSI TS 136 521-1 [5], ETSI TS 136 508 [6] and ETSI TS 136 509 [i.19] respectively.

5.1.3.10.1.2 Procedure

1) SS transmits PDSCH via PDCCH DCI format 1A for C_RNTI to transmit the DL RMC according to ETSI TS 136 521-1 [5], Table 7.8.1.4.1-1. The SS sends downlink MAC padding bits on the DL RMC.

2) SS sends uplink scheduling information for each UL HARQ process via PDCCH DCI format 0 for C_RNTI to schedule the UL RMC according to ETSI TS 136 521-1 [5], Table 7.8.1.4.1-1. Since the aerial UE has no payload data to send, the aerial UE transmits uplink MAC padding bits on the UL RMC.

3) Set the Downlink signal level to the value as defined in Table 4.1.2.11.2-1. Send uplink power control commands to the aerial UE (less or equal to 1 dB step size should be used), to ensure that the aerial UE output power is within +0, -3,4 dB of the target level in Table 4.1.2.11.2-1 for carrier frequency $f \leq 3{,}0$ GHz or within +0, -4,0 dB of the target level for carrier frequency 3,0 GHz $< f \leq 4{,}2$ GHz, for at least the duration of the throughput measurement as specified in ETSI TS 136 521-1 [5].

4) Set the Interfering signal levels to the values as defined in Table 4.1.2.11.2-1, using a modulated interferer bandwidth as defined in annex D of ETSI TS 136 521-1 [5].

5) Measure the average throughput for according to clause G.2 of ETSI TS 136 521-1 [5].

6) Repeat for applicable test frequencies, channel bandwidths and operating bands.

Details of the test method can be found in ETSI TS 136 521-1 [5], clause 7.8.

5.1.3.10.2 Test requirements

The results obtained shall be compared to the limits in clause 4.1.2.11.2 in order to show compliance.

5.1.3.11 Receiver Spurious Emissions

5.1.3.11.1 Method of test

5.1.3.11.1.1 Initial conditions

Test Environment: normal, as specified in annex B.

Frequencies to be tested: low range, mid-range and high range; as specified in ETSI TS 136 508 [6], clause 4.3.1.

Channel bandwidth to be tested: highest channel bandwidth as defined in ETSI TS 136 508 [6], clause 4.3.1.

Uplink/Downlink configurations: as specified in ETSI TS 136 521-1 [5], clause 7.9.4:

1) Connect a spectrum analyser (or other suitable test equipment) to the aerial UE antenna connectors as shown in ETSI TS 136 508 [6], annex A.
2) The parameter settings for the cell are set up according to ETSI TS 136 508 [6], clause 4.4.3.
3) Downlink signals are initially set up according to ETSI TS 136 521-1 [5], clauses C.0, C.1 and C.3.1.
4) The DL Reference Measurement channels are set according to ETSI TS 136 521-1 [5], clause 7.9.4.1.
5) Propagation conditions are set according to ETSI TS 136 521-1 [5], clause B.0.
6) Ensure the aerial UE is in State 3A-RF according to ETSI TS 136 508 [6], clause 5.2A.2.

NOTE: When reference is made to test set up, call set up and test mode, guidance on the applicability of these can be found in ETSI TS 136 521-1 [5], ETSI TS 136 508 [6] and ETSI TS 136 509 [i.19] respectively.

5.1.3.11.1.2 Procedure

1) Sweep the spectrum analyser over a frequency range from 30 MHz to 12,75 GHz and measure the average power of the spurious emissions.
2) Repeat step 1) for all E-UTRA Rx antennas of the aerial UE.
3) Repeat for applicable test frequencies, channel bandwidths and operating bands.

Details of the test method can be found in ETSI TS 136 521-1 [5], clause 7.9.

5.1.3.11.2 Test requirements

The results obtained shall be compared to the limits in clause 4.1.2.12 in order to show compliance.

5.1.3.12 Aerial UE Identifier

5.1.3.12.1 Method of test

The test description and procedure given in clause 10.10.1.3 in ETSI TS 136 523-1 [8] applies.

5.1.3.12.2 Test requirements

If the aerial UE successfully passes the test given in ETSI TS 136 523-1 [8], clause 10.10.1, then the aerial UE is successfully provisioned with CAA-level UAV ID, as the abovementioned test requires as a pre-test condition that the aerial UE is capable of UAS operation and provisioned with CAA-level UAV ID.

5.1.3.13 Aerial UE Authorization and Authentication

5.1.3.13.1 Method of test

The test description and procedure given in clause 10.10.1.3 in ETSI TS 136 523-1 [8] applies.

5.1.3.13.2 Test requirements

If the aerial UE successfully passes the test given in ETSI TS 136 523-1 [8], clause 10.10.1, then the aerial UE has successfully performed the UUAA procedure as defined in clause 5.2 in ETSI TS 123 256 [1].

5.2 Testing for compliance with technical requirements for NR

5.2.1 Environmental conditions for testing

Tests defined in the present document shall be carried out at representative points within the boundary limits of the operational environmental profile defined by its intended use.

Where technical performance varies subject to environmental conditions, tests shall be carried out under a sufficient variety of environmental conditions (within the boundary limits of the operational environmental profile defined by its intended use) to give confidence of compliance for the affected technical requirements.

5.2.2 Interpretation of the measurement results

The interpretation of the results recorded in a test report for the measurements described in the present document shall be as follows:

- the measured value related to the corresponding limit will be used to decide whether an equipment meets the requirements of the present document;
- the value of the measurement uncertainty for the measurement of each parameter shall be included in the test report.

For the test methods, according to the present document, the measurement uncertainty figures shall be calculated and shall correspond to an expansion factor (coverage factor) k = 1,96 (which provide a confidence level of respectively 95 % in the case where the distributions characterizing the actual measurement uncertainties are normal (Gaussian)). Principles for the calculation of measurement uncertainty are contained in ETSI TR 100 028 [i.20], in particular in annex C of the ETSI TR 100 028-2 [i.20]. For guidance on other measurement conditions reference can be made to annexes A to H of ETSI TS 138 521-1 [4].

Values for the maximum measurement uncertainty based on this expansion factor can be found in clause D.2.

5.2.3 Essential radio test suites

5.2.3.0 General

5.2.3.1 Transmitter Maximum Output Power

5.2.3.1.1 Method of test

5.2.3.1.1.1 Initial conditions

Initial conditions are a set of test configurations the aerial UE needs to be tested in and the steps for the SS to take with the aerial UE to reach the correct measurement state.

The initial test configurations consist of environmental conditions, test frequencies, test channel bandwidths and sub-carrier spacing based on NR operating bands specified in Table 1.1-2. All of these configurations shall be tested with applicable test parameters for each combination of test channel bandwidth and sub-carrier spacing, and are shown in Table 5.2.3.1.1.1-1. The details of the uplink Reference Measurement Channels (RMCs) are specified in clause A.2 of ETSI TS 138 521-1 [4]. Configurations of PDSCH and PDCCH before measurement are specified in clause C.2 of ETSI TS 138 521-1 [4].

Table 5.2.3.1.1.1-1: Test Configuration Table

Initial Conditions			
Test Environment as specified in ETSI TS 138 508-1 [7], clause 4.1		Normal, TL/VL, TL/VH, TH/VL, TH/VH	
Test Frequencies as specified in ETSI TS 138 508-1 [7], clause 4.3.1		Low range, Mid range, High range	
Test Channel Bandwidths as specified in ETSI TS 138 508-1 [7], clause 4.3.1		Lowest, Mid, Highest (note 3)	
Test SCS as specified in Table 1.1-2		Lowest, Highest	
Test Parameters			
Test ID	**Downlink Configuration**	**Uplink Configuration**	
	N/A for maximum output power test case	**Modulation (note 2)**	**RB allocation (note 1)**
1		DFT-s-OFDM PI/2 BPSK	Inner Full
2		DFT-s-OFDM PI/2 BPSK	Inner 1RB Left
3		DFT-s-OFDM PI/2 BPSK	Inner 1RB Right
4		DFT-s-OFDM QPSK	Inner Full
5		DFT-s-OFDM QPSK	Inner 1RB Left
6		DFT-s-OFDM QPSK	Inner 1RB Right
NOTE 1: The specific configuration of each RB allocation is defined in Table 6.1-1 of ETSI TS 138 521-1 [4]. NOTE 2: DFT-s-OFDM PI/2 BPSK test applies only for aerial UEs which supports half Pi BPSK. NOTE 3: For band n28, the Highest test channel bandwidth is replaced by 20 MHz due to MPR is always larger than 0 dB for 30 MHz bandwidth.			

1) Connect the SS to the aerial UE antenna connectors as shown in ETSI TS 138 508-1 [7], annex A, Figure A.3.1.1.1 for TE diagram and clause A.3.2 for UE diagram.

2) The parameter settings for the cell are set up according to ETSI TS 138 508-1 [7], clause 4.4.3.

3) Downlink signals are initially set up according to clauses C.0, C.1, C.2 of ETSI TS 138 521-1 [4], and uplink signals according to clauses G.0, G.1, G.2, G.3.0 of ETSI TS 138 521-1 [4].

4) The UL Reference Measurement Channel is set according to Table 5.2.3.1.1.1-1.

5) Propagation conditions are set according to clause B.0 of ETSI TS 138 521-1 [4].

6) Ensure the aerial UE is in state RRC_CONNECTED with generic procedure parameters Connectivity *NR*, Connected without release *On*, Test Mode *On* and Test Loop Function *On* according to ETSI TS 138 508-1 [7], clause 4.5. Message contents are defined in clause 6.2.1.4.3 of ETSI TS 138 521-1 [4].

5.2.3.1.1.2 Procedure

1) SS sends uplink scheduling information for each UL HARQ process via PDCCH DCI format 0_1 for C_RNTI to schedule the UL RMC according to Table 5.2.3.1.1.1-1. Since the aerial UE has no payload and no loopback data to send the aerial UE sends uplink MAC padding bits on the UL RMC.

2) Send continuously uplink power control "up" commands in every uplink scheduling information to the aerial UE; allow at least 200 ms starting from the first TPC command in this step for the aerial UE to reach P_{UMAX} level.

3) Measure the mean power of the aerial UE in the channel bandwidth of the radio access mode. The period of measurement shall be at least the continuous duration of one active sub-frame (1 ms) and in the uplink symbols. For TDD symbols with transient periods are not under test.

4) For UEs supporting Power Class 2, repeat steps 1~3 on the applicable bands with message exception of P-Max defined in Table 6.2.1.4.3-2 in ETSI TS 138 521-1 [4].

5.2.3.1.2 Test requirements

The results obtained shall be compared to the limits in clause 4.2.2.2.2 in order to show compliance.

5.2.3.2 Transmitter Minimum Output Power

5.2.3.2.1 Method of test

5.2.3.2.1.1 Initial conditions

Initial conditions are a set of test configurations the aerial UE needs to be tested in and the steps for the SS to take with the aerial UE to reach the correct measurement state.

The initial test configurations consist of environmental conditions, test frequencies, test channel bandwidths and sub-carrier spacing based on NR operating bands specified in Table 1.1-2. All of these configurations shall be tested with applicable test parameters for each combination of test channel bandwidth and sub-carrier spacing and are shown in Table 5.2.3.2.1.1-1. The details of the uplink reference measurement channels are specified in clauses A.2 of ETSI TS 138 521-1 [4]. Configurations of PDSCH and PDCCH before measurement are specified in clause C.2 of ETSI TS 138 521-1 [4].

Table 5.2.3.2.1.1-1: Test Configuration Table

Initial Conditions			
Test Environment as specified in ETSI TS 138 508-1 [7] clause 4.1		Normal, TL/VL, TL/VH, TH/VL, TH/VH	
Test Frequencies as specified in ETSI TS 138 508-1 [7] clause 4.3.1		Low range, Mid range, High range (note 2)	
Test Channel Bandwidths as specified in ETSI TS 138 508-1 [7] clause 4.3.1		Lowest, Mid, Highest	
Test SCS as specified in Table 1.1-2		Highest	
Test Parameters for Channel Bandwidths			
Test ID	**Downlink Configuration**	**Uplink Configuration**	
	N/A for minimum output power test case	**Modulation**	**RB allocation (note 1)**
1		DFT-s-OFDM QPSK	Outer Full
NOTE 1: The specific configuration of each RB allocation is defined in Table 6.1-1 of ETSI TS 138 521-1 [4]. NOTE 2: For NR band n28, 30 MHz test channel bandwidth is tested with Low range and High range test frequencies.			

1) Connect the SS to the aerial UE antenna connectors as shown in ETSI TS 138 508-1 [7], annex A, Figure A.3.1.1.1 for TE diagram and clause A.3.2 for UE diagram.

2) The parameter settings for the cell are set up according to ETSI TS 138 508-1 [7], clause 4.4.3.

3) Downlink signals are initially set up according to clauses C.0, C.1, C.2 of ETSI TS 138 521-1 [4], and uplink signals according to clauses G.0, G.1, G.2, G.3.0 of ETSI TS 138 521-1 [4].

4) The UL Reference Measurement Channel is set according to Table 5.2.3.2.1.1-1.

5) Propagation conditions are set according to clause B.0 of ETSI TS 138 521-1 [4].

6) Ensure the aerial UE is in State RRC_CONNECTED with generic procedure parameters Connectivity *NR*, Connected without release *On*, Test Mode *On* and Test Loop Function *On* according to ETSI TS 138 508-1 [7], clause 4.5. Message contents are defined in clause 6.3.1.4.3 of ETSI TS 138 521-1 [4].

5.2.3.2.1.2 Procedure

1) SS sends uplink scheduling information for each UL HARQ process via PDCCH DCI format 0_1 for C_RNTI to schedule the UL RMC according to Table 5.2.3.2.1.1-1. Since the aerial UE has no payload and no loopback data to send the aerial UE sends uplink MAC padding bits on the UL RMC.

2) Send continuously uplink power control "down" commands in every uplink scheduling information to the aerial UE; allow at least 200 ms starting from the first TPC command in this step to ensure that the aerial UE transmits at its minimum output power.

3) Measure the mean power of the aerial UE in the associated measurement channel bandwidth specified in Table 4.2.2.3.2-1 for the specific channel bandwidth under test. The period of measurement shall be at least the continuous duration of one active sub-frame (1 ms) and in the uplink symbols. For TDD slots with transient periods are not under test.

5.2.3.2.2 Test requirements

The results obtained shall be compared to the limits in clause 4.2.2.3.2 in order to show compliance.

5.2.3.3 Transmitter Spectrum Emission Mask

5.2.3.3.1 Method of test

5.2.3.3.1.1 Initial conditions

Initial conditions are a set of test configurations the aerial UE needs to be tested in and the steps for the SS to take with the aerial UE to reach the correct measurement state.

The initial test configurations consist of environmental conditions, test frequencies, test channel bandwidths and sub-carrier spacing based on NR operating bands specified in Table 1.1-2. All of these configurations shall be tested with applicable test parameters for each combination of channel bandwidth and sub-carrier spacing, which are shown in Tables 5.2.3.3.1.1-1, 5.2.3.3.1.1-2, and 5.2.3.3.1.1-3. The details of the uplink Reference Measurement Channels (RMCs) are specified in clause A.2 of ETSI TS 138 521-1 [4]. Configurations of PDSCH and PDCCH before measurement are specified in clause C.2 of ETSI TS 138 521-1 [4].

Table 5.2.3.3.1.1-1: Test Configuration Table for power class 3 (contiguous allocation)

Test Environment as specified in ETSI TS 138 508-1 [7], clause 4.1					Normal	
Test Frequencies as specified in ETSI TS 138 508-1 [7], clause 4.3.1					Low range, High range	
Test Channel Bandwidths as specified in ETSI TS 138 508-1 [7], clause 4.3.1					Lowest, Highest	
Test SCS as specified in Table 1.1-2					Lowest, Highest	
Test Parameters for Channel Bandwidths						
Test ID	**Freq**	**ChBw**	**SCS**	**Downlink Configuration**	**Uplink Configuration**	
		Default	Default	N/A for Spectrum Emission Mask test case	**Modulation (note 2)**	**RB allocation (note 1)**
1 note 3	Low				DFT-s-OFDM PI/2 BPSK	Edge_1RB_Left
2 note 3	High				DFT-s-OFDM PI/2 BPSK	Edge_1RB_Right
3 note 3	Default				DFT-s-OFDM PI/2 BPSK	Outer_Full
4 note 4	Low				DFT-s-OFDM PI/2 BPSK	Edge_1RB_Left
5 note 4	High				DFT-s-OFDM PI/2 BPSK	Edge_1RB_Right
6 note 4	Default				DFT-s-OFDM PI/2 BPSK	Outer_Full
	Low				DFT-s-OFDM QPSK	Edger_1RB_Left
	High				DFT-s-OFDM QPSK	Edge_1RB_Right
	Default				DFT-s-OFDM QPSK	Outer_Full
	Low				DFT-s-OFDM 16 QAM	Edge_1RB_Left
	High				DFT-s-OFDM 16 QAM	Edge_1RB_Right
	Default				DFT-s-OFDM 16 QAM	Outer_Full
	Low				DFT-s-OFDM 64 QAM	Edge_1RB_Left
	High				DFT-s-OFDM 64 QAM	Edge_1RB_Right
	Default				DFT-s-OFDM 64 QAM	Outer_Full
	Low				DFT-s-OFDM 256 QAM	Edge_1RB_Left
	High				DFT-s-OFDM 256 QAM	Edge_1RB_Right
	Default				DFT-s-OFDM 256 QAM	Outer_Full
	Low				CP-OFDM QPSK	Edge_1RB_Left

Test Parameters for Channel Bandwidths						
Test ID	**Freq**	**ChBw**	**SCS**	**Downlink Configuration**	**Uplink Configuration**	
	High				CP-OFDM QPSK	Edge_1RB_Right
	Default				CP-OFDM QPSK	Outer_Full
	Low				CP-OFDM 16 QAM	Edge_1RB_Left
	High				CP-OFDM 16 QAM	Edge_1RB_Right
	Default				CP-OFDM 16 QAM	Outer_Full
	Low				CP-OFDM 64 QAM	Edge_1RB_Left
	High				CP-OFDM 64 QAM	Edge_1RB_Right
	Default				CP-OFDM 64 QAM	Outer_Full
	Low				CP-OFDM 256 QAM	Edge_1RB_Left
	High				CP-OFDM 256 QAM	Edge_1RB_Right
	Default				CP-OFDM 256 QAM	Outer_Full
31 note 5	Low				DFT-s-OFDM Pi/2 BPSK w Pi/2 BPSK DMRS	Edge_1RB_Left
32 note 5	High				DFT-s-OFDM Pi/2 BPSK w Pi/2 BPSK DMRS	Edge_1RB_Right
33 note 5	Default				DFT-s-OFDM Pi/2 BPSK w Pi/2 BPSK DMRS	Outer Full

NOTE 1: The specific configuration of each RF allocation is defined in Table 6.1-1 of ETSI TS 138 521-1 [4].
NOTE 2: DFT-s-OFDM PI/2 BPSK test applies only for aerial UEs which supports half Pi BPSK.
NOTE 3: UE operating in TDD mode with Pi/2 BPSK modulation and UE indicates support for UE capability *powerBoosting-pi2BPSK* and the IE *powerBoostPi2BPSK* is set to 1 for bands n40, n41, n77 andn78.
NOTE 4: UE operating in FDD mode, or in TDD mode in bands other than n40, n41, n77 and n78, or in TDD mode the IE *powerBoostPi2BPSK* is set to 0 for bands n40, n41, n77 and n78.
NOTE 5: For Power Class 3 testing, UE operating in FDD mode, or in TDD mode in bands other than n40, n41, n77 and n78, or in TDD mode the IE *powerBoostPi2BPSK* is set to 0 for bands n40, n77 and n78.
NOTE 6: Applicable to UEs indicating support for UE capability *lowPAPR-DMRS-PUSCHwithPrecoding-r16*.
NOTE 7: Only test frequencies within the frequency range operated in Europe as defined in Table 1.1-1 apply.

Table 5.2.3.3.1.1-2: Test Configuration Table for power class 2 (contiguous allocation)

Initial Conditions				
Test Environment as specified in ETSI TS 138 508-1 [7], clause 4.1			Normal	
Test Frequencies as specified in ETSI TS 138 508-1 [7], clause 4.3.1			Low range, High range (note 4)	
Test Channel Bandwidths as specified in ETSI TS 138 508-1 [7], clause 4.3.1			Lowest, Highest	
Test SCS as specified in Table 1.1-2			Lowest, Highest	
Test Parameters for Channel Bandwidths				
Test ID	**Freq**	**Downlink Configuration**	**Uplink Configuration**	
		N/A	**Modulation (note 2)**	**RB allocation (note 1)**
1	Low		DFT-s-OFDM Pi/2 BPSK	Edge_1RB_Left
2	High		DFT-s-OFDM Pi/2 BPSK	Edge_1RB_Right
3	Default		DFT-s-OFDM Pi/2 BPSK	Outer Full
4	Low		DFT-s-OFDM QPSK	Edge_1RB_Left
5	High		DFT-s-OFDM QPSK	Edge_1RB_Right
6	Default		DFT-s-OFDM QPSK	Outer Full
7	Low		DFT-s-OFDM 16 QAM	Edge_1RB_Left
8	High		DFT-s-OFDM 16 QAM	Edge_1RB_Right
9	Default		DFT-s-OFDM 16 QAM	Outer Full
10	Low		DFT-s-OFDM 64 QAM	Edge_1RB_Left
11	High		DFT-s-OFDM 64 QAM	Edge_1RB_Right
12	Default		DFT-s-OFDM 64 QAM	Outer Full
13	Low		DFT-s-OFDM 256 QAM	Edge_1RB_Left
14	High		DFT-s-OFDM 256 QAM	Edge_1RB_Right
15	Default		DFT-s-OFDM 256 QAM	Outer Full
16	Low		CP-OFDM QPSK	Edge_1RB_Left
17	High		CP-OFDM QPSK	Edge_1RB_Right
18	Default		CP-OFDM QPSK	Outer Full
19	Low		CP-OFDM 16 QAM	Edge_1RB_Left
20	High		CP-OFDM 16 QAM	Edge_1RB_Right
21	Default		CP-OFDM 16 QAM	Outer Full
22	Low		CP-OFDM 64 QAM	Edge_1RB_Left
23	High		CP-OFDM 64 QAM	Edge_1RB_Right
24	Default		CP-OFDM 64 QAM	Outer Full
25	Low		CP-OFDM 256 QAM	Edge_1RB_Left
26	High		CP-OFDM 256 QAM	Edge_1RB_Right
27	Default		CP-OFDM 256 QAM	Outer Full

NOTE 1: The specific configuration of each RB allocation is defined in Table 6.1-1 of ETSI TS 138 521-1 [4].
NOTE 2: DFT-s-OFDM Pi/2 BPSK test applies only for aerial UEs which supports half Pi BPSK.
NOTE 3: It is essential that all test points in this table also exist in Table 6.2.2.4.1-2 of ETSI TS 138 521-1 [4].
NOTE 4: Only test frequencies within the frequency range operated in Europe as defined in Table 1.1-1 apply.

Table 5.2.3.3.1.1-3: Test Configuration Table for power class 2&3 (almost contiguous allocation)

Initial Conditions				
Test Environment as specified in ETSI TS 138 508-1 [7], clause 4.1			Normal	
Test Frequencies as specified in ETSI TS 138 508-1 [7], clause 4.3.1			Low range, High range (note 4)	
Test Channel Bandwidths as specified in ETSI TS 138 508-1 [7], clause 4.3.1			Highest	
Test SCS as specified in Table 1.1-2			Lowest, Highest	
Test Parameters for Channel Bandwidths				
Test ID	**Freq**	**Downlink Configuration**	**Uplink Configuration**	
		N/A	**Modulation**	**RB allocation (note 1)**
1	Default		CP-OFDM QPSK	Inner Full
2	Default		CP-OFDM QPSK	Outer Full
3	Default		CP-OFDM 16 QAM	Inner Full
4	Default		CP-OFDM 16 QAM	Outer Full
5	Default		CP-OFDM 64 QAM	Outer Full
6	Default		CP-OFDM 256 QAM	Outer Full
NOTE 1: The specific configuration of each RB allocation is defined in Table 6.5.2.2.4.1-4 of ETSI TS 138 521-1 [4].				
NOTE 2: It is essential that all test points in this Table also exist in Table 6.2.2.4.1-3 of ETSI TS 138 521-1 [4].				
NOTE 3: Test applies only for UEs which support almost contiguous UL CP-OFDM transmissions.				
NOTE 4: Only test frequencies within the frequency range operated in Europe as defined in Table 1.1-1 apply.				

1) Connect the SS to the aerial UE antenna connectors as shown in ETSI TS 138 508-1 [7], annex A, Figure A.3.1.1.1 for TE diagram and clause A.3.2 for UE diagram.

2) The parameter settings for the cell are set up according to ETSI TS 138 508-1 [7], clause 4.4.3.

3) Downlink signals are initially set up according to clauses C.0, C.1, C.2 of ETSI TS 138 521-1 [4], and uplink signals according to clauses G.0, G.1, G.2, G.3.0 of ETSI TS 138 521-1 [4].

4) The UL Reference Measurement channels are set according to Table 5.2.3.3.1.1-1.

5) Propagation conditions are set according to clause B.0 of ETSI TS 138 521-1 [4].

6) Ensure the aerial UE is in state RRC_CONNECTED with generic procedure parameters Connectivity *NR*, Connected without release *On*, Test Mode *On* and Test Loop Function *On* according to ETSI TS 138 508-1 [7], clause 4.5. Message contents are defined in clause 6.5.2.2.4.3 of ETSI TS 138 521-1 [4].

5.2.3.3.1.2 Procedure

1) SS sends uplink scheduling information for each UL HARQ process via PDCCH DCI format 0_1 for C_RNTI to schedule the UL RMC according to Tables 5.2.3.3.1.1-1, 5.2.3.3.1.1-2 and 5.2.3.3.1.1-3. Since the UL has no payload and no loopback data to send the aerial UE sends uplink MAC padding bits on the UL RMC.

2) Send continuously power control "up" commands to the aerial UE until the aerial UE transmits at the P_{UMAX} level. Allow at least 200 ms for the aerial UE to reach the P_{UMAX} level.

3) Measure the mean power of the aerial UE in the channel bandwidth of the radio access mode according to the test configuration, which shall meet the requirements described in Tables 6.2.2.5-1 to 6.2.2.5-9 of ETSI TS 138 521-1 [4]. The period of the measurement shall be at least the continuous duration of 1 ms over consecutive active uplink slots. For TDD, only slots consisting of only UL symbols are under test.

4) Measure the power of the transmitted signal with a measurement filter of bandwidths according to Table 4.2.2.4.2-1. The centre frequency of the filter shall be stepped in continuous steps according to the same table. The measured power shall be recorded for each step. The measurement period shall capture the active TSs.

NOTE 1: When switching to DFT-s-OFDM waveform, as specified in the test configuration Table 5.2.3.3.1.1-1 and Table 5.2.3.3.1.1-2, send an NR RRCReconfiguration message according to ETSI TS 138 508-1 [7] clause 4.6.3, Table 4.6.3-118 PUSCH-Config with TRANSFORM_PRECODER_ENABLED condition.

NOTE 2: For measurement conditions at the edge of each frequency range, the lowest frequency of the measurement position in each frequency range should be set at the lowest boundary of the frequency range plus MBW/2. The highest frequency of the measurement position in each frequency range should be set at the highest boundary of the frequency range minus MBW/2. MBW denotes the measurement bandwidth defined for the protected band.

NOTE 3: To improve measurement accuracy, sensitivity and efficiency, the resolution bandwidth may be smaller than the measurement bandwidth. When the resolution bandwidth is smaller than the measurement bandwidth, the result should be integrated over the measurement bandwidth in order to obtain the equivalent noise bandwidth of the measurement bandwidth.

5.2.3.3.2 Test requirements

The results obtained shall be compared to the limits in clause 4.2.2.4.2 in order to show compliance.

5.2.3.4 Transmitter Adjacent Channel Leakage Power Ratio

5.2.3.4.1 Method of test

5.2.3.4.1.1 NR ACLR

5.2.3.4.1.1.1 Initial conditions

Initial conditions are a set of test configurations the aerial UE needs to be tested in and the steps for the SS to take with the aerial UE to reach the correct measurement state.

The initial test configurations consist of environmental conditions, test frequencies, test channel bandwidths and sub-carrier spacing based on NR operating bands specified in Table 1.1-2. All of these configurations shall be tested with applicable test parameters for each combination of channel bandwidth and sub-carrier spacing, are shown in the test configuration tables in clause 6.2.2.4.1 of ETSI TS 138 521-1 [4]. The details of the uplink Reference Measurement Channels (RMCs) are specified in clause A.2 of ETSI TS 138 521-1 [4]. Configurations of PDSCH and PDCCH before measurement are specified in clause C.2 of ETSI TS 138 521-1 [4]:

1) Connect the SS to the aerial UE antenna connectors as shown in ETSI TS 138 508-1 [7] annex A, Figure A.3.1.1.1 for TE diagram and clause A.3.2 for UE diagram.

2) The parameter settings for the cell are set up according to ETSI TS 138 508-1 [7], clause 4.4.3.

3) Downlink signals are initially set up according to clauses C.0, C.1, C.2 of ETSI TS 138 521-1 [4], and uplink signals according to clauses G.0, G.1, G.2, G.3.0 of ETSI TS 138 521-1 [4].

4) The UL Reference Measurement channels are set according to the test configuration tables in clause 6.2.2.4.1 of ETSI TS 138 521-1 [4].

5) Propagation conditions are set according to clause B.0 of ETSI TS 138 521-1 [4].

6) Ensure the aerial UE is in state RRC_CONNECTED with generic procedure parameters Connectivity *NR*, Connected without release *On*, Test Mode *On* and Test Loop Function *On* according to ETSI TS 138 508-1 [7], clause 4.5. Message contents are defined in clause 6.5.2.4.1.4.3 of ETSI TS 138 521-1 [4].

5.1.3.4.1.1.1.2 Procedure

1) SS sends uplink scheduling information for each UL HARQ process via PDCCH DCI format 0_1 for C_RNTI to schedule the UL RMC according to the test configuration tables in clause 6.2.2.4.1 of ETSI TS 138 521-1 [4]. Since the UL has no payload and no loopback data to send the aerial UE sends uplink MAC padding bits on the UL RMC.

2) Send continuously power control "up" commands to the aerial UE until the aerial UE transmits at the P_{UMAX} level. Allow at least 200 ms for the aerial UE to reach the P_{UMAX} level.

3) Measure the mean power of the aerial UE in the channel bandwidth of the radio access mode according to the test configuration, as measured in step 3 of 6.2.2.4.2 of ETSI TS 138 521-1 [4], which shall meet the requirements described in clause 6.2.2.5 of ETSI TS 138 521-1 [4] as appropriate.

4) Measure the rectangular filtered mean power for the assigned NR channel.

5) Measure the rectangular filtered mean power of the first NR adjacent channel on both the lower and upper side of the assigned NR channel, respectively.

6) Calculate the ratios of the power between the values measured in step 4 over step 5 for lower and upper NR ACLR, respectively.

7) For UEs supporting Power Class 2, repeat steps 1~6 for Test ID 22 and 36 in Table 6.2.2.4.1-1 of ETSI TS 138 521-1 [4] on the applicable bands with message exception of P-Max defined in Table 6.5.2.4.1.4.3-1 of ETSI TS 138 521-1 [4].

NOTE: When switching to DFT-s-OFDM waveform, as specified in the test configuration tables in clause 6.2.2.4.1 of ETSI TS 138 521-1 [4], send an NR RRCReconfiguration message according to ETSI TS 138 508-1 [7], clause 4.6.3, Table 4.6.3-118 PUSCH-Config with TRANSFORM_PRECODER_ENABLED condition.

5.2.3.4.1.2 UTRA ACLR

5.2.3.4.1.2.1 Initial conditions

Same as in clause 6.2.3.4.1 of ETSI TS 138 521-1 [4] with the following exceptions: Message contents in step 6 are defined in clause 6.5.2.4.2.4.3 of ETSI TS 138 521-1 [4].

5.2.3.4.1.2.2 Procedure

1) SS sends uplink scheduling information for each UL HARQ process via PDCCH DCI format 0_1 for C_RNTI to schedule the UL RMC according to the applicable test configuration table in clause 6.2.3.4.1 of ETSI TS 138 521-1 [4]. Since the UL has no payload and no loopback data to send the aerial UE sends uplink MAC padding bits on the UL RMC.

2) Send continuously power control "up" commands to the aerial UE until the aerial UE transmits at the P_{UMAX} level. Allow at least 200 ms for the aerial UE to reach the P_{UMAX} level.

3) Measure the mean power of the aerial UE in the channel bandwidth of the radio access mode according to the test configuration, which shall meet the requirements described in 6.2.3.5 of ETSI TS 138 521-1 [4] as appropriate. The period of the measurement shall be at least the continuous duration of one active sub-frame (1 ms) and in the uplink symbols. For TDD slots with transient periods are not under test.

4) Measure the rectangular filtered mean power for the assigned NR channel.

5) Measure the RRC filtered mean power of the first and the second UTRA adjacent channel on both the lower and upper side of the NR channel, respectively.

6) Calculate the ratios of the power between the values measured in step 4 over step 5 for lower and upper UTRA ACLR, respectively.

NOTE: When switching to DFT-s-OFDM waveform, as specified in the test configuration Table 6.2.3.4.1-1 of ETSI TS 138 521-1 [4], send an NR RRCReconfiguration message according to ETSI TS 138 508-1 [7] clause 4.6.3, Table 4.6.3-118 PUSCH-Config with TRANSFORM_PRECODER_ENABLED condition.

5.2.3.4.2 Test requirements

The results obtained shall be compared to the limits in clause 4.2.2.5.2 in order to show compliance.

5.2.3.5 Transmitter Spurious Emissions

5.2.3.5.1 Method of test

5.2.3.5.1.1 General spurious emissions

5.2.3.5.1.1.1 Initial conditions

Initial conditions are a set of test configurations the aerial UE needs to be tested in and the steps for the SS to take with the aerial UE to reach the correct measurement state.

The initial test configurations consist of environmental conditions, test frequencies, test channel bandwidths and sub-carrier spacing based on NR operating bands specified in Table 1.1-2. All of these configurations shall be tested with applicable test parameters for each combination of channel bandwidth and sub-carrier spacing, which are shown in Table 5.2.3.5.1.1.1-1. The details of the uplink Reference Measurement Channels (RMCs) are specified in clause A.2 of ETSI TS 138 521-1 [4]. Configurations of PDSCH and PDCCH before measurement are specified in clause C.2 of ETSI TS 138 521-1 [4].

Table 5.2.3.5.1.1.1-1: Test Configuration Table

Initial Conditions			
Test Environment as specified in ETSI TS 138 508-1 [7], clause 4.1.		Normal	
Test Frequencies as specified in ETSI TS 138 508-1 [7], clause 4.3.1.		Low range, Mid range, High range (notes 2 and 3)	
Test Channel Bandwidths as specified in ETSI TS 138 508-1 [7], clause 4.3.1.		Lowest, Mid, Highest	
Test SCS as specified in Table 1.1-2		Lowest	
Test Parameters			
Test ID	**Downlink Configuration**	**Uplink Configuration**	
		Modulation	**RB allocation (note 1)**
1	N/A for Spurious Emissions testing	CP-OFDM QPSK	OuterFull
2		CP-OFDM QPSK	Edge_1RB_Left
3		CP-OFDM QPSK	Edge_1RB_Right
NOTE 1: The specific configuration of each RB allocation is defined in Table 6.1-1 of ETSI TS 138 521-1 [4] Common UL configuration. NOTE 2: For NR band n28, 30 MHz test channel bandwidth is tested with Low range and High range test frequencies. NOTE 3: Only test frequencies within the frequency range operated in Europe as defined in Table 1.1-1 apply.			

1) Connect the SS to the aerial UE antenna connectors as shown in ETSI TS 138 508-1 [7], annex A, Figure A.3.1.1.1 for TE diagram and clause A.3.2 for UE diagram.

2) The parameter settings for the cell are set up according to ETSI TS 138 508-1 [7], clause 4.4.3.

3) Downlink signals are initially set up according to clauses C.0, C.1 and C.2 of ETSI TS 138 521-1 [4], and uplink signals according to clauses G.0, G.1, G.2, G.3.0 of ETSI TS 138 521-1 [4].

4) The UL Reference Measurement channels are set according to Table 5.2.3.5.1.1.1-1.

5) Propagation conditions are set according to clause B.0 of ETSI TS 138 521-1 [4].

6) Ensure the aerial UE is in state RRC_CONNECTED with generic procedure parameters Connectivity *NR*, Connected without release *On*, Test Mode *On* and Test Loop Function *On* according to ETSI TS 138 508-1 [7] clause 4.5. Message contents are defined in clause 6.5.3.1.4.3 of ETSI TS 138 521-1 [4] with no exceptions.

5.2.3.5.1.1.2 Procedure

1) SS sends uplink scheduling information for each UL HARQ process via PDCCH DCI format 0_1 for C_RNTI to schedule the UL RMC according to Table 5.2.3.5.1.1.1-1. Since the aerial UE has no payload data to send, the aerial UE transmits uplink MAC padding bits on the UL RMC.

2) Send continuously uplink power control "up" commands in the uplink scheduling information to the aerial UE until the aerial UE transmits at the P_{UMAX} level.

3) Measure the power of the transmitted signal with a measurement filter of bandwidths according to Table 4.2.2.6.2.1-2. The centre frequency of the filter shall be stepped in contiguous steps according to Table 4.2.2.6.2.1-2. The measured power shall be verified for each step. The measurement period shall capture the active time slots.

For measurement conditions at the edge of each frequency range, the lowest frequency of the measurement position in each frequency range shall be set at the lowest boundary of the frequency range plus MBW/2. The highest frequency of the measurement position in each frequency range shall be set at the highest boundary of the frequency range minus MBW/2. MBW denotes the measurement bandwidth defined for the protected band.

NOTE: To improve measurement accuracy, sensitivity and efficiency, the resolution bandwidth may be smaller than the measurement bandwidth. When the resolution bandwidth is smaller than the measurement bandwidth, the result should be integrated over the measurement bandwidth in order to obtain the equivalent noise bandwidth of the measurement bandwidth.

5.2.3.5.1.2 Spurious emission for UE co-existence

5.2.3.5.1.2.1 Initial conditions

Initial conditions are a set of test configurations the aerial UE needs to be tested in and the steps for the SS to take with the aerial UE to reach the correct measurement state.

The initial test configurations consist of environmental conditions, test frequencies, test channel bandwidths and sub-carrier spacing based on NR operating bands specified in Table 1.1-2. All of these configurations shall be tested with applicable test parameters for each combination of channel bandwidth and sub-carrier spacing, are shown in Table 5.2.3.5.1.2.1-1. The details of the uplink Reference Measurement Channels (RMCs) are specified in clause A.2 of ETSI TS 138 521-1 [4]. Configurations of PDSCH and PDCCH before measurement are specified in clause C.2 of ETSI TS 138 521-1 [4].

Table 5.2.3.5.1.2.1-1: Test Configuration Table

Initial Conditions			
Test Environment as specified in ETSI TS 138 508-1 [7], clause 4.1.		Normal	
Test Frequencies as specified in ETSI TS 138 508-1 [7], clause 4.3.1.		Low range, Mid range, High range (see note 2)	
Test Channel Bandwidths as specified in ETSI TS 138 508-1 [7], clause 4.3.1.		Lowest, Mid, Highest	
Test SCS as specified in Table 1.1-2		Lowest	
Test Parameters			
Test ID	**Downlink Configuration**	**Uplink Configuration**	
	N/A	**Modulation**	**RB allocation (note 1)**
1		CP-OFDM QPSK	Outer_Full
2		CP-OFDM QPSK	Edge_1RB_Left
3		CP-OFDM QPSK	Edge_1RB_Right
NOTE 1: The specific configuration of each RB allocation is defined in ETSI TS 138 521-1 [4], Table 6.1-1 Common UL configuration. NOTE 2: For NR band n28, 30 MHz test channel bandwidth is tested with Low range and High range test frequencies.			

1) Connect the SS to the aerial UE antenna connectors as shown in ETSI TS 138 508-1 [7], annex A, Figure A.3.1.1.1 for TE diagram and clause A.3.2 for UE diagram.

2) The parameter settings for the cell are set up according to ETSI TS 138 508-1 [7], clause 4.4.3.

3) Downlink signals are initially set up according to clauses C.0, C.1, C.2 of ETSI TS 138 521-1 [4], and uplink signals according to clauses G.0, G.1, G.2, G.3.0 of ETSI TS 138 521-1 [4].

4) The UL Reference Measurement channels are set according to Table 5.2.3.5.1.2.1-1.

5) Propagation conditions are set according to clause B.0 of ETSI TS 138 521-1 [4].

6) Ensure the aerial UE is in state RRC_CONNECTED with generic procedure parameters Connectivity *NR*, Connected without release *On*, Test Mode *On* and Test Loop Function *On* according to ETSI TS 138 508-1 [7] clause 4.5. Message contents are defined in clause 6.5.3.2.4.3 of ETSI TS 138 521-1 [4].

5.2.3.5.1.2.2 Procedure

1) SS sends uplink scheduling information for each UL HARQ process via PDCCH DCI format 0_1 for C_RNTI to schedule the UL RMC according to Table 5.2.3.5.1.2.1-1. Since the aerial UE has no payload data to send, the aerial UE transmits uplink MAC padding bits on the UL RMC.

2) Send continuously uplink power control "up" commands in the uplink scheduling information to the aerial UE until the aerial UE transmits at the P_{UMAX} level.

3) Measure the power of the transmitted signal with a measurement filter of bandwidths according to Table The centre frequency of the filter shall be stepped in contiguous steps according to Table 4.2.2.6.2.2-1. The measured power shall be verified for each step. The measurement period shall capture the active time slots.

5.2.3.5.1.3 Additional spurious emissions

5.2.3.5.1.3.1 Initial conditions

Initial conditions are a set of test configurations the aerial UE needs to be tested in and the steps for the SS to take with the aerial UE to reach the correct measurement state.

The initial test configurations consist of environmental conditions, test frequencies, test channel bandwidths and sub-carrier spacing based on NR operating bands specified in Table 1.1-2. All these configurations shall be tested with applicable test parameters for each channel bandwidth and sub-carrier spacing, are shown in this clause for different NS values. The details of the uplink RMCs are specified in clause A.2.2 of ETSI TS 138 521-1 [4]. Configurations of PDSCH and PDCCH before measurement are specified in clause C.2 of ETSI TS 138 521-1 [4].

For network signalled value NS_18 the same test configuration as listed in Table 6.2.3.4.1-11 of ETSI TS 138 521-1 [4] shall be used with the following exceptions:

- Test Channel Bandwidths shall be: 5, 10, 20 and 30 MHz.

For network signalled value NS_48 the same test configuration as listed in Table 6.2.3.4.1-19 of ETSI TS 138 521-1 [4] shall be used with the following exceptions:

- Test Channel Bandwidths shall be: 25, 30, 40, 45 and 50 MHz.
- Test SCS shall be: Lowest.

For network signalled value NS_49 the same test configuration as listed in Table 6.2.3.4.1-29 of ETSI TS 138 521-1 [4] shall be used with the following exceptions:

- Test Channel Bandwidths shall be: 25, 30, 40, 45 and 50 MHz.
- Test SCS shall be: Lowest.

For network signalled value NS_UAV_44 the same test configuration as listed in Table 6.5K.3.3.4.1-1 of ETSI TS 138 521-1 [4] shall be used with the following exceptions:

- Test Channel Bandwidths shall be: 25, 30 and 40 MHz.
- Test SCS shall be: Lowest and highest.

For network signalled value NS_UAV_70 the same test configuration as listed in Table 6.5K.3.3.4.1-3 of ETSI TS 138 521-1 [4] shall be used with the following exceptions:

- Test Channel Bandwidths shall be: 5 and 50 MHz.
- Test SCS shall be: Lowest and highest.

1) Connect the SS to the aerial UE to the aerial UE antenna connectors as shown in ETSI TS 138 508-1 [7], annex A, Figure A.3.1.1.1 for TE diagram and clause A.3.2 for UE diagram.

2) The parameter settings for the cell are set up according to ETSI TS 138 508-1 [7], clause 4.4.3.

3) Downlink signals are initially set up according to clauses C.0, C.1 and C.2 of ETSI TS 138 521-1 [4], and uplink signals according to clauses G.0, G.1, G.2 and G.3.0 of ETSI TS 138 521-1 [4].

4) The UL Reference Measurement channels are set according to clause 5.2.3.5.1.2.1-1.

5) Propagation conditions are set according to clause B.0 of ETSI TS 138 521-1 [4].

6) Ensure the aerial UE is in state RRC_CONNECTED with generic procedure parameters Connectivity *NR*, Connected without release *On*, Test Mode *On* and Test Loop Function *On* according to ETSI TS 138 508-1 [7], clause 4.5. Message contents are defined in clause 6.5.3.3.4.3 of ETSI TS 138 521-1 [4].

5.2.3.5.1.3.2 Procedure

1) SS sends uplink scheduling information for each UL HARQ process via PDCCH DCI format 0-1 for C_RNTI to schedule the UL RMC according to the test configurations provided in clause 5.2.3.5.1.3.1. Since the aerial UE has no payload data to send, the aerial UE transmits uplink MAC padding bits on the UL RMC.

2) Send continuously uplink power control "up" commands in the uplink scheduling information to the aerial UE until the aerial UE transmits at the P_{UMAX} level.

3) Measure the mean power of the aerial UE in the channel bandwidth of the radio access mode according to the test configurations provided in clause 5.2.3.5.1.3.1 as appropriate, which shall meet the requirements in clause 6.5.3.3.5 of ETSI TS 138 521-1 [4] with allowed A-MPR values specified in Tables 6.2.3.5-1 through 6.2.3.5-27 and Table 6.5K.3.3.3.1-1, Table 6.5K.3.3.3.2-1 and Table 6.5K.3.3.3.3-1 of ETSI TS 138 521-1 [4] as appropriate per test condition specified in Tables 6.2.3.4.1-1 through 6.2.3.4.1-30 and Tables 6.5K.3.3.4.1-1 through 6.5K.3.3.4.1-3 of ETSI TS 138 521-1 [4] as appropriate. The measured power shall be verified for each step. The measurement period shall capture the active time slots.

4) Measure the power of the transmitted signal with a measurement filter of bandwidths according to Table 6.5.3.3.3.1-1 to Table 6.5.3.3.3.1-27 of ETSI TS 138 521-1 [4] as appropriate. The centre frequency of the filter shall be stepped in contiguous steps according to the same table.

For measurement conditions at the edge of each frequency range, the lowest frequency of the measurement position in each frequency range shall be set at the lowest boundary of the frequency range plus MBW/2. The highest frequency of the measurement position in each frequency range shall be set at the highest boundary of the frequency range minus MBW/2. MBW denotes the measurement bandwidth defined for the protected band.

Table 5.2.3.5.1.3.2-1: Uplink power control window

Carrier centre frequency, F_C	$BW_{Channel}$	Uplink power control window (dB)
$F_C \leq 3{,}0$ GHz	$BW_{Channel} \leq 40$ MHz	-0,7 dB to 3,1 dB
	40 MHz < $BW_{Channel} \leq 100$ MHz	-1,4 dB to 4,1 dB

5.2.3.5.2 Test requirements

The results obtained shall be compared to the limits in clause 4.2.2.6 in order to show compliance.

5.2.3.6 Receiver Reference Sensitivity Level

5.2.3.6.1 Method of test

5.2.3.6.1.1 Initial conditions

Initial conditions are a set of test configurations the aerial UE needs to be tested in and the steps for the SS to take with the aerial UE to reach the correct measurement state.

The initial test configurations consist of environmental conditions, test frequencies, and channel bandwidths based on NR operating bands specified in Table 1.1-2. All of these configurations shall be tested with applicable test parameters for each channel bandwidth, and are shown in Table 5.2.3.6.1.1-1, Table 5.2.3.6.1.1-2 and Table 5.2.3.6.1.1-3. The details of the uplink RMCs are specified in ETSI TS 138 521-1 [4], clause A.2.2. Configurations of PDSCH and PDCCH before measurement are specified in ETSI TS 138 521-1 [4], clause C.2.

Table 5.2.3.6.1.1-1: Test Configuration Table

Initial Conditions				
Test Environment as specified in ETSI TS 138 508-1 [7], clause 4.1			Normal, TL/VL, TL/VH, TH/VL, TH/VH	
Test Frequencies as specified in ETSI TS 138 508-1 [7], clause 4.3.1			Low range, Mid range, High range (notes 4 and 6)	
Test Channel Bandwidths as specified in ETSI TS 138 508-1 [7], clause 4.3.1			Lowest, Mid, Highest (note 4) Lowest UL / Lowest DL, Lowest UL / Highest DL (note 3)	
Test SCS as specified in Table 1.1-2			Lowest	
Test Parameters				
Test ID	**Downlink Configuration**		**Uplink Configuration**	
	Modulation	**RB allocation**	**Modulation**	**RB allocation**
1	CP-OFDM QPSK	Full RB (note 1)	DFT-s-OFDM QPSK	REFSENS (note 2)
NOTE 1: Full RB allocation shall be used per each SCS and channel BW as specified in Table 5.2.3.6.1.1-2. NOTE 2: REFSENS refers to Table 5.2.3.6.1.1-3 which defines uplink RB configuration and start RB location for each SCS, channel BW and NR band. NOTE 3: Additional test points selected according to asymmetric channel bandwidths specified in clause 5.3.6 of ETSI TS 138 521-1 [4]. DL channel bandwidth shall be selected first. NOTE 4: For NR band n28, 30 MHz test channel bandwidth is tested with Low range and High range test frequencies. NOTE 5: In a band where UE supports 4Rx, the test needs to be repeated with only 2Rx antennas connected and the other antennas terminated. NOTE 6: Only test frequencies within the frequency range operated in Europe as defined in Table 1.1-1 apply.				

Table 5.2.3.6.1.1-2: Downlink Configuration of each RB allocation

Channel Bandwidth	**SCS (kHz)**	**LCRBmax**	**Outer RB allocation / Normal RB allocation**
5 MHz	15	25	25@0
	30	11	11@0
	60	N/A	N/A
10 MHz	15	52	52@0
	30	24	24@0
	60	11	11@0
15 MHz	15	79	79@0
	30	38	38@0
	60	18	18@0
20 MHz	15	106	106@0
	30	51	51@0
	60	24	24@0
25 MHz	15	133	133@0
	30	65	65@0
	60	31	31@0
30 MHz	15	160	160@0
	30	78	78@0
	60	38	38@0
40 MHz	15	216	216@0
	30	106	106@0
	60	51	51@0
50 MHz	15	270	270@0
	30	133	133@0
	60	65	65@0
NOTE: Test Channel Bandwidths are checked separately for each NR band, the applicable channel bandwidths are specified in Table 1.1-2.			

Table 5.2.3.6.1.1-3: Uplink configuration for reference sensitivity, LCRB @ Restart format

Operating Band	SCS kHz	5 MHz	10 MHz	15 MHz	20 MHz	25 MHz	30 MHz	40 MHz	Duplex Mode
n1	15	25@0	50@2 (note 1)	75@4 (note 1)	100@6 (note 1)		128@32 (note 1)	128@88 (note 1)	FDD
	30		24@0	36@2 (note 1)	50@1 (note 1)		64@14 (note 1)	64@42 (note 1)	
	60		10@1 (note 1)	18@0	24@0		30@8 (note 1)	30@21 (note 1)	
n3	15	25@0	50@2 (note 1)	50@29 (note 1)	50@56 (note 1)	50@83 (note 1)	50@110 (note 1)		FDD
	30		24@0	24@14 (note 1)	24@27 (note 1)	24@41 (note 1)	24@54 (note 1)		
	60		10@1 (note 1)	10@8 (note 1)	10@14 (note 1)	10@21 (note 1)	10@28 (note 1)		
n7	15	25@0	50@2 (note 1)	75@4 (note 1)	75@31 (note 1)				FDD
	30		24@0	36@2 (note 1)	36@15 (note 1)				
	60		10@1 (note 1)	18@0	18@6 (note 1)				
n8	15	25@0	25@27 (note 1)	25@54 (note 1)	25@81 (note 1)				FDD
	30		10@14 (note 1)	10@28 (note 1)	10@41 (note 1)				
	60								
n20	15	25@0	20@0 (note 1)	20@11 (note 2)	20@16 (note 2)				FDD
	30		10@0 (note 1)	10@6 (note 2)	10@8 (note 2)				
	60								
n28	15	25@0	25@27 (note 1)	25@54 (note 1)	25@81 (note 1)				FDD
	30		10@14	10@28 (note 1)	10@41 (note 1)				
	60								
n38	15	25@0	50@0	75@0	100@0				TDD
	30		24@0	36@0	50@0				
	60		10@0	18@0	24@0				

NOTE 1: UL resource blocks shall be located as close as possible to the downlink operating band but confined within the transmission bandwidth configuration for the channel bandwidth (Table 5.3.2-1 of ETSI TS 138 101-1 [3]).

NOTE 2: For Band 20; for 15 kHz SCS, in the case of 15 MHz channel bandwidth, the UL resource blocks shall be located at RB_{start} 11 and in the case of 20 MHz channel bandwidth, the UL resource blocks shall be located at RB_{start} 16; for 30 kHz SCS, in the case of 15 MHz channel bandwidth, the UL resource blocks shall be located at RB_{start} 6 and in the case of 20 MHz channel bandwidth, the UL resource blocks shall be located at RB_{start} 8; for 60 kHz SCS, in the case of 15 MHz channel bandwidth, the UL resource blocks shall be located at RB_{start} 3 and in the case of 20 MHz channel bandwidth, the UL resource blocks shall be located at RB_{start} 4.

1) Connect the SS to the aerial UE antenna connectors as shown in [7], annex A, Figure A.3.1.1.1 for TE diagram and clause A.3.2 for UE diagram.

2) The parameter settings for the cell are set up according to [7], clause 4.4.3.

3) Downlink signals are initially set up according to ETSI TS 138 521-1 [4], clauses C.0, C.1, C.2, and C.3.1, and uplink signals according to ETSI TS 138 521-1 [4], clauses G.0, G.1, G.2, and G.3.1.

4) The UL and Reference Measurement Channel is set according to Table 5.2.3.6.1.1-1, Table 5.2.3.6.1.1-2, and Table 5.2.3.6.1.1-3.

5) Propagation conditions are set according to ETSI TS 138 521-1 [4], clause B.0.

6) Ensure the aerial UE is in State RRC_CONNECTED with generic procedure parameters Connectivity NR, connected without release On, Test Mode On and Test Loop Function On according to [7], clause 4.5. Message contents are defined in ETSI TS 138 521-1 [4], clause 7.3.2.4.3.

5.2.3.6.1.2 Test procedure

1) SS transmits PDSCH via PDCCH DCI format 1_1 for C_RNTI to transmit the DL RMC according to Table 5.2.3.6.1.1-1. The SS sends downlink MAC padding bits on the DL RMC.

2) SS sends uplink scheduling information for each UL HARQ process via PDCCH DCI format 0_1 for C_RNTI to schedule the UL RMC according to Table 5.2.3.6.1.1-1. Since the aerial UE has no payload data to send, the aerial UE transmits uplink MAC padding bits on the UL RMC.

3) Set the Downlink signal level to the appropriate REFSENS value defined in Table 4.2.2.7.2-1 or Table 4.2.2.7.2-2. Send continuously uplink power control "up" commands in the uplink scheduling information to the aerial UE to ensure the aerial UE transmits PUMAX level for at least the duration of the Throughput measurement.

4) Measure the average throughput according to ETSI TS 138 521-1 [4], clause H.2.

5.2.3.6.2 Test requirements

The results obtained in step 4 shall be compared to the limits in Table 4.2.2.7.2-1 for two antenna port and Table 4.2.2.7.2-2 for four antenna port in order to show compliance.

5.2.3.7 Receiver Adjacent Channel Selectivity (ACS)

5.2.3.7.1 Method of test

5.2.3.7.1.1 Initial Conditions

Initial conditions are a set of test configurations the aerial UE needs to be tested in and the steps for the SS to take with the aerial UE to reach the correct measurement state.

The initial test configurations consist of environmental conditions, test frequencies, test channel bandwidths and sub-carrier spacing based on NR operating bands specified in Table 1.1-2. All of these configurations shall be tested with applicable test parameters for each combination of channel bandwidth and sub-carrier spacing, are shown in Table 5.2.3.7.1.1-1. The details of the uplink and downlink RMC are specified in ETSI TS 138 521-1 [4], clauses A.2 and A.3. Configurations of PDSCH and PDCCH before measurement are specified in ETSI TS 138 521-1 [4], clause C.2.

Table 5.2.3.7.1.1-1: Test Configuration Table

Default Conditions				
Test Environment as specified in ETSI TS 138 508-1 [7], clause 4.1			Normal	
Test Frequencies as specified in ETSI TS 138 508-1 [7], clause 4.3.1			Mid range (notes 3 and 4)	
Test Channel Bandwidths as specified in ETSI TS 138 508-1 [7], clause 4.3.1			Lowest, Mid, Highest	
Test SCS as specified in Table 1.1-2			Lowest	
Test Parameters				
	Downlink Configuration		**Uplink Configuration**	
Test ID	**Mod'n**	**RB allocation**	**Mod'n**	**RB allocation**
1	CP-OFDM QPSK	note 1	DFT-s-OFDM QPSK	note 1
NOTE 1: The specific configuration of uplink and downlink are defined in Table 5.2.3.6.1.1-1.				
NOTE 2: In a band where aerial UE supports 4Rx, the test shall be performed only with 4Rx antennas ports connected and 4Rx REFSENS requirement (Table 4.2.2.7.2-2) is used in the test requirements.				
NOTE 3: For NR band n28, 30 MHz test channel bandwidth is tested with Low range test frequencies.				
NOTE 4: Only test frequencies within the frequency range operated in Europe as defined in Table 1.1-1 apply.				

1) Connect the SS to the aerial UE antenna connectors as shown in ETSI TS 138 508-1 [7], annex A, in Figure A.3.1.4.1 for TE diagram and clause A.3.2 for UE diagram.

2) The parameter settings for the cell are set up according to ETSI TS 138 508-1 [7], clause 4.4.3.

3) Downlink signals are initially set up according to ETSI TS 138 521-1 [4], clauses C.0, C.1, C.2, C.3.1, and uplink signals according to ETSI TS 138 521-1 [4], clauses G.0, G.1, G.2, G.3.1.

4) The DL and UL Reference Measurement channels are set according to Table 5.2.3.7.1.1-1.

5) Propagation conditions are set according to ETSI TS 138 521-1 [4], annex B.0.

6) Ensure the aerial UE is in state RRC_CONNECTED with generic procedure parameters Connectivity *NR*, Connected without release *On*, Test Mode *On* and Test Loop Function *On* according to ETSI TS 138 508-1 [7], clause 4.5. Message content is defined in ETSI TS 138 521-1 [4], clause 7.5.4.3.

5.1.3.7.1.1.2 Test Procedure

1) SS transmits PDSCH via PDCCH DCI format [1_1] for C_RNTI to transmit the DL RMC according to Table 5.2.3.7.1.1-1. The SS sends downlink MAC padding bits on the DL RMC.

2) SS sends uplink scheduling information for each UL HARQ process via PDCCH DCI format [0_1] for C_RNTI to schedule the UL RMC according to Table 5.2.3.7.1.1-1. Since the UL has no payload and no loopback data to send the aerial UE sends uplink MAC padding bits on the UL RMC.

3) Set the Downlink signal level to the value as defined in Table 4.2.2.8.2-2 as appropriate (Case 1). Send uplink power control commands to the aerial UE using 1 dB power step size to ensure that the aerial UE output power measured by the test system is within the Uplink power control window in Table 5.2.3.5.1.3.2-1 of the target power level in Table 4.2.2.8.2-2 for at least the duration of the Throughput measurement.

4) Set the Interferer signal level to the value as defined in Table 4.2.2.8.2-2 as appropriate (Case 1) and frequency below the wanted signal, using a modulated interferer bandwidth as defined in ETSI TS 138 521-1 [4], annex D.

5) Measure the average throughput according to ETSI TS 138 521-1 [4], annex H.

6) Repeat steps from 3 to 5, using an interfering signal above the wanted signal in Case 1 at step 4).

7) Set the Downlink signal level to the value as defined in Table 4.2.2.8.2-3 as appropriate (Case 2). Send uplink power control commands to the aerial UE using 1 dB power step size to ensure that the aerial UE output power measured by the test system is within the Uplink power control window in Table 5.2.3.5.1.3.2-1 of the target power level in Table 4.2.2.8.2-3 for at least the duration of the Throughput measurement.

8) Set the Interferer signal level to the value as defined in Table 4.2.2.8.2-3 as appropriate (Case 2) and frequency below the wanted signal, using a modulated interferer bandwidth as defined in ETSI TS 138 521-1 [4], annex D.

9) Measure the average throughput according to ETSI TS 138 521-1 [4], annex H.

10) Repeat steps from 7 to 9, using an interfering signal above the wanted signal in Case 2 at step 8.

11) Repeat for applicable channel bandwidths and operating band combinations in both Case 1 and Case 2.

NOTE: The purpose of the Uplink power control window is to ensure that the actual aerial UE output power is no greater than the target power level and as close as possible to the target power level. The relationship between the Uplink power control window, the target power level and the corresponding possible actual UE Uplink power window is illustrated in ETSI TS 138 521-1 [4], clause F.4.3.

5.2.3.7.2 Test requirements

The results obtained shall be compared to the limits in Tables 4.2.2.8.2-1, 4.2.2.8.2-2, and 4.2.2.8.2-3 in order to show compliance.

5.2.3.8 Receiver Blocking Characteristics

5.2.3.8.1 Method of test

5.2.3.8.1.1 In-band Blocking

5.2.3.8.1.1.1 Initial Conditions

Initial conditions are a set of test configurations the aerial UE needs to be tested in and the steps for the SS to take with the aerial UE to reach the correct measurement state.

The initial test configurations consist of environmental conditions, test frequencies, test channel bandwidths and sub-carrier spacing based on NR operating bands specified in Table 1.1-2. All of these configurations shall be tested with applicable test parameters for each combination of channel bandwidth and sub-carrier spacing, are shown in Table 5.2.3.8.1.1.1-1. The details of the uplink and downlink RMC are specified in ETSI TS 138 521-1 [4], clauses A.2 and A.3. Configurations of PDSCH and PDCCH before measurement are specified in ETSI TS 138 521-1 [4], clause C.2.

Table 5.2.3.8.1.1.1-1: Test Configuration Table

Default Conditions				
Test Environment as specified in ETSI TS 138 508-1 [7], clause 4.1			Normal	
Test Frequencies as specified in ETSI TS 138 508-1 [7], clause 4.3.1			Mid range (notes 3 and 4)	
Test Channel Bandwidths as specified in ETSI TS 138 508-1 [7], clause 4.3.1			Lowest, Mid, Highest	
Test SCS as specified in Table 1.1-2			Lowest	
Test Parameters				
	Downlink Configuration		**Uplink Configuration**	
Test ID	**Mod'n**	**RB allocation**	**Mod'n**	**RB allocation**
1	**CP-OFDM QPSK**	note 1	DFT-s-OFDM QPSK	note 1
NOTE 1: The specific configuration of uplink and downlink are defined in Table 5.2.3.6.1.1-1.				
NOTE 2: In a band where UE supports 4Rx, the test shall be performed only with 4Rx antennas ports connected and 4Rx REFSENS requirement (Tables 4.2.2.7.2-1 and 4.2.2.7.2-2) is used in the test requirements.				
NOTE 3: For NR band n28, 30 MHz test channel bandwidth is tested with Low range test frequencies.				
NOTE 4: Only test frequencies within the frequency range operated in Europe as defined in Table 1.1-1 apply.				

1) Connect the SS to the aerial UE antenna connectors as shown in ETSI TS 138 508-1 [7], annex A, in Figure A.3.1.4.1 for TE diagram and clause A.3.2 for UE diagram.

2) The parameter settings for the cell are set up according to ETSI TS 138 508-1 [7], clause 4.4.3.

3) Downlink signals are initially set up according to ETSI TS 138 521-1 [4], clauses C.0, C.1, C.2, C.3.1, and uplink signals according to ETSI TS 138 521-1 [4], clauses G.0, G.1, G.2, G.3.1.

4) The DL and UL Reference Measurement channels are set according to Table 5.2.3.8.1.1.1-1.

5) Propagation conditions are set according to ETSI TS 138 521-1 [4], clause B.0.

6) Ensure the aerial UE is in state RRC_CONNECTED with generic procedure parameters Connectivity *NR*, Connected without release *On*, Test Mode *On* and Test Loop Function *On* according to ETSI TS 138 508-1 [7], clause 4.5. Message contents are defined in ETSI TS 138 521-1 [4], clause 7.6.2.4.3.

5.2.3.8.1.1.2 Test Procedure

1) SS transmits PDSCH via PDCCH DCI format 1_1 for C_RNTI to transmit the DL RMC according to Table 5.2.3.8.1.1.1-1. The SS sends downlink MAC padding bits on the DL RMC.

2) SS sends uplink scheduling information for each UL HARQ process via PDCCH DCI format 0_1 for C_RNTI to schedule the UL RMC according to Table 5.2.3.8.1.1.1-1. Since the UL has no payload and no loopback data to send the aerial UE sends uplink MAC padding bits on the UL RMC.

3) Set the parameters of the signal generator for an interfering signal below the wanted signal in Case 1 according to Table 4.2.2.9.2.1-1 and 4.2.2.9.2.1-2.

4) Set the downlink signal level according to Table 4.2.2.9.2.1-1. Send uplink power control commands to the aerial UE using 1 dB power step size to ensure that the aerial UE output power measured by the test system is within the Uplink power control window in Table 5.2.3.5.1.3.2-1 of the target power level in Table 4.2.2.9.2.1-1 for at least the duration of the Throughput measurement.

5) Measure the average throughput according to ETSI TS 138 521-1 [4], annex H.

6) Repeat steps from 3 to 5, using an interfering signal above the wanted signal in Case 1 at step 3.

7) Repeat steps from 3 to 6, using interfering signals in Case 2 at step 3 and 6. The ranges of case 2 are covered in steps equal to the interferer bandwidth. Interferer frequencies should be chosen starting with an offset nearest to the centre frequency and sweep outwards towards the band edges. In order to ensure that full range is tested for interferer frequency, run last test steps at frequency equal to $F_{Interferer}$ range limit defined at the corresponding band edge.

8) If applicable based on NR band, repeat steps from 3 to 5, using interfering signals in Case 3 at step 3.

9) If applicable based on NR band, repeat steps from 3 to 5, using interfering signals in Case 4 at step 3

NOTE: The purpose of the Uplink power control window is to ensure that the actual UE output power is no greater than the target power level and as close as possible to the target power level. The relationship between the Uplink power control window, the target power level and the corresponding possible actual UE Uplink power window is illustrated in ETSI TS 138 521-1 [4], clause F.4.3.

5.2.3.8.1.2 Out-of-band blocking

5.2.3.8.1.2.1 Initial Conditions

Initial conditions are a set of test configurations the aerial UE needs to be tested in and the steps for the SS to take with the aerial UE to reach the correct measurement state.

The initial test configurations consist of environmental conditions, test frequencies, and channel bandwidths based on NR operating bands specified in Table 1.1-2. All of these configurations shall be tested with applicable test parameters for each combination of channel bandwidth and sub-carrier spacing, and are shown in Table 5.2.3.8.1.2.1-1. The details of the uplink and downlink RMCs are specified in ETSI TS 138 521-1 [4], clauses A.2 and A.3 respectively. The details of the OCNG patterns used are specified in ETSI TS 138 521-1 [4], clause A.5. Configurations of PDSCH and PDCCH before measurement are specified in ETSI TS 138 521-1 [4], clause C.3.

Table 5.2.3.8.1.2.1-1: Test Configuration Table

Default Conditions				
Test Environment as specified in ETSI TS 138 508-1 [7], clause 4.1			Normal	
Test Frequencies as specified in ETSI TS 138 508-1 [7], clause 4.3.1			One frequency chosen arbitrarily from low or high range (note 3)	
Test Channel Bandwidths as specified in ETSI TS 138 508-1 [7], clause 4.3.1			Lowest, Mid, Highest	
Test SCS as specified in ETSI TS 138 508-1 [7], clause 4.3.1			Lowest	
Test Parameters				
	Downlink Configuration		**Uplink Configuration**	
Test ID	**Mod'n**	**RB allocation**	**Mod'n**	**RB allocation**
1	CP-OFDM QPSK	note 1	DFT-s-OFDM QPSK	note 1
NOTE 1: The specific configuration of uplink and downlink are defined in Table 5.2.3.6.1.1-1. NOTE 2: In a band where UE supports 4Rx, the test shall be performed only with 4Rx antennas ports connected and 4Rx REFSENS requirement (Table 4.2.2.7.2-2) is used in the test requirements. NOTE 3: Only test frequencies within the frequency range operated in Europe as defined in Table 1.1-1 apply.				

1) Connect the SS to the aerial UE antenna connectors as shown in ETSI TS 138 508-1 [7], annex A, in Figure A.3.1.4. 2 for TE diagram and clause A.3.2 for UE diagram.

2) The parameter settings for the cell are set up according to ETSI TS 138 508-1 [7], clause 4.4.3.

3) Downlink signals are initially set up according to ETSI TS 138 521-1 [4], clauses C.0, C.1, C.2, C.3.1, and uplink signals according to ETSI TS 138 521-1 [4], clauses G.0, G.1, G.2 and G.3.1.

4) The UL and DL Reference Measurement channels are set according to Table 5.2.3.8.1.2.1-1.

5) Propagation conditions are set according to ETSI TS 138 521-1 [4], clause B.0.

6) Ensure the aerial UE is in State RRC_CONNECTED with generic procedure parameters Connectivity *NR*, Connected without release *On*, Test Mode *On* and Test Loop Function *On* according to ETSI TS 138 508-1 [7], clause 4.5. Message contents are defined in ETSI TS 138 521-1 [4], clause 7.6.3.4.3.

5.2.3.8.1.2.2 Test Procedure

1) SS transmits PDSCH via PDCCH DCI format 1_1 for C_RNTI to transmit the DL RMC according to Table 5.2.3.8.1.2.1-1. The SS sends downlink MAC padding bits on the DL RMC.

2) SS sends uplink scheduling information for each UL HARQ process via PDCCH DCI format 0_1 for C_RNTI to schedule the UL RMC according to Table 5.2.3.8.1.2.1-1. Since the aerial UE has no payload data to send, the aerial UE transmits uplink MAC padding bits on the UL RMC.

3) Set the parameters of the CW signal generator for an interfering signal below the wanted signal according to Table 4.2.2.9.2.2-2. The frequency step size is $\min(\lfloor BW_{channel} / 2 \rfloor, 5)$ MHz.

4) Set the downlink signal level according to Table 4.2.2.9.2.2-1. Send uplink power control commands to the aerial UE using 1 dB power step size to ensure that the aerial UE output power measured by the test system is within the Uplink power control window in Table 5.2.3.5.1.3.2-1 of the target power level in Table 4.2.2.9.2.2-1 for at least the duration of the Throughput measurement.

5) Measure the average throughput according to ETSI TS 138 521-1 [4], annex H.

6) Record the frequencies for which the throughput does not meet the requirements.

7) Repeat steps from 3 to 6, using an interfering signal above the wanted signal at step 3.

NOTE: The purpose of the Uplink power control window is to ensure that the actual UE output power is no greater than the target power level and as close as possible to the target power level. The relationship between the Uplink power control window, the target power level and the corresponding possible actual UE Uplink power window is illustrated in ETSI TS 138 521-1 [4], clause F.4.3.

5.2.3.8.1.3 Narrow-band blocking

5.2.3.8.1.3.1 Initial Conditions

Initial conditions are a set of test configurations the aerial UE needs to be tested in and the steps for the SS to take with the aerial UE to reach the correct measurement state.

The initial test configurations consist of environmental conditions, test frequencies, and channel bandwidths based on NR operating bands specified in Table 1.1-2. All of these configurations shall be tested with applicable test parameters for each combination of channel bandwidth and sub-carrier spacing and are shown in Table 5.2.3.8.1.3.1-1. The details of the uplink and downlink RMCs are specified in ETSI TS 138 521-1 [4], clauses A.2 and A.3 respectively. The details of the OCNG patterns used are specified in ETSI TS 138 521-1 [4], clause A.5. Configurations of PDSCH and PDCCH before measurement are specified in ETSI TS 138 521-1 [4], clause C.2.

Table 5.2.3.8.1.3.1-1: Test Configuration Table

Default Conditions				
Test Environment as specified in ETSI TS 138 508-1 [7], clause 4.1			Normal	
Test Frequencies as specified in ETSI TS 138 508-1 [7], clause 4.3.1			Mid range (notes 3 and 4)	
Test Channel Bandwidths as specified in ETSI TS 138 508-1 [7], clause 4.3.1			Lowest, Mid and Highest	
Test SCS as specified in ETSI TS 138 508-1 [7], clause 4.3.1			According to CH BW SCS in Table 4.2.2.9.2.3-1	
Test Parameters				
	Downlink Configuration		**Uplink Configuration**	
Test ID	**Mod'n**	**RB allocation**	**Mod'n**	**RB allocation**
1	CP-OFDM QPSK	note 1	DFT-s-OFDM QPSK	note 1
NOTE 1: The specific configuration of uplink and downlink are defined in Table 5.2.3.6.1.1-1.				
NOTE 2: In a band where UE supports 4Rx, the test shall be performed only with 4Rx antennas ports connected and 4Rx REFSENS requirement (Table 4.2.2.7.2-2) is used in the test requirements.				
NOTE 3: For NR band n28, 30 MHz test channel bandwidth is tested with Low range test frequencies.				
NOTE 4: Only test frequencies within the frequency range operated in Europe as defined in Table 1.1-1 apply.				

1) Connect the SS to the aerial UE antenna connectors as shown in ETSI TS 138 508-1 [7], annex A, in Figure A.3.1.4.2 for TE diagram and clause A.3.2 for UE diagram.

2) The parameter settings for the cell are set up according to ETSI TS 138 508-1 [7], clause 4.4.3.

3) Downlink signals are initially set up according to ETSI TS 138 521-1 [4], clauses C.0, C.1, C.2 and C.3.1, and uplink signals according to ETSI TS 138 521-1 [4], clauses G.0, G.1, G.2 and G.3.1.

4) The UL and DL Reference Measurement channels are set according to Table 5.2.3.8.1.3.1-1.

5) Propagation conditions are set according to ETSI TS 138 521-1 [4], clause B.0.

6) Ensure the aerial UE is in State RRC_CONNECTED with generic procedure parameters Connectivity *NR* according to ETSI TS 138 508-1 [7], clause 4.5. Message contents are defined in ETSI TS 138 521-1 [4], clause 7.6.4.4.3.

5.2.3.8.1.3.1.2 Test Procedure

1) SS transmits PDSCH via PDCCH DCI format 1_1 for C_RNTI to transmit the DL RMC according to Table 5.2.3.8.1.3.1-1. The SS sends downlink MAC padding bits on the DL RMC.

2) SS sends uplink scheduling information for each UL HARQ process via PDCCH DCI format 0_1 for C_RNTI to schedule the UL RMC according to Table 5.2.3.8.1.3.1-1. Since the aerial UE has no payload and no loopback data to send, the aerial UE transmits uplink MAC padding bits on the UL RMC.

3) Set the parameters of the CW signal generator for an interfering signal below the wanted signal according to Table 4.2.2.9.2.3-1.

4) Set the downlink signal level according to Table 4.2.2.9.2.3-1. Send uplink power control commands to the aerial UE using 1 dB power step size to ensure that the aerial UE output power measured by the test system is within the Uplink power control window in Table 5.2.3.5.1.3.2-1 of the target power level in Table 4.2.2.9.2.3-1 for at least the duration of the Throughput measurement.

5) Measure the average throughput according to ETSI TS 138 521-1 [4], clause H.2.

6) Repeat steps from 3 to 5, using an interfering signal above the wanted signal at step 3.

NOTE: The purpose of the Uplink power control window is to ensure that the actual UE output power is no greater than the target power level and as close as possible to the target power level. The relationship between the Uplink power control window, the target power level and the corresponding possible actual UE Uplink power window is illustrated in ETSI TS 138 521-1 [4], clause F.4.3.

5.2.3.8.2 Test requirements

The results obtained shall be compared to the limits in clause 4.2.2.9.2 in order to show compliance.

5.2.3.9 Receiver Spurious Response

5.2.3.9.1 Method of test

5.2.3.9.1.1 Initial Conditions

The initial conditions shall be the same as in clause 5.2.3.8.1.2.1 in order to test spurious responses obtained in clause 5.2.3.9.2 under the same conditions.

5.2.3.9.1.2 Test Procedure

1) SS transmits PDSCH via PDCCH DCI format [1_1] for C_RNTI to transmit the DL RMC according to Table 5.2.3.8.1.2.1-1. The SS sends downlink MAC padding bits on the DL RMC.

2) SS sends uplink scheduling information for each UL HARQ process via PDCCH DCI format [0_1] for C_RNTI to schedule the UL RMC according to Table 5.2.3.8.1.2.1-1. Since the aerial UE has no payload and no loopback data to send, the aerial UE transmits uplink MAC padding bits on the UL RMC.

3) Set the parameters of the CW signal generator for an interfering signal according to Table 4.2.2.10.2-1. The spurious frequencies are taken from records in step 6 of test procedures in clause 5.2.3.8.1.2.1.

4) Set the downlink signal level according to Table 4.2.2.10.2-1. Send uplink power control commands to the aerial UE using 1 dB power step size to ensure that the aerial UE output power measured by the test system is within the Uplink power control window in Table 5.2.3.5.1.3.2-1 of the target power level in Table 4.2.2.10.2-1 for at least the duration of the Throughput measurement.

5) For the spurious frequency, measure the average throughput according to ETSI TS 138 521-1 [4], clause H.2.

NOTE: The purpose of the Uplink power control window is to ensure that the actual aerial UE output power is no greater than the target power level, and as close as possible to the target power level. The relationship between the Uplink power control window, the target power level and the corresponding possible actual UE Uplink power window is illustrated in ETSI TS 138 521-1 [4], clause F.4.3.

5.2.3.9.2 Test requirements

The results obtained in step 5 shall be compared to the limits in Table 4.2.2.10.2-1 and Table 4.2.2.10.2-2 in order to show compliance.

5.2.3.10 Receiver Intermodulation Characteristic

5.2.3.10.1 Method of test

5.2.3.10.1.1 Initial Conditions

Initial conditions are a set of test configurations the aerial UE needs to be tested in and the steps for the SS to take with the aerial UE to reach the correct measurement state.

The initial test configurations consist of environmental conditions, test frequencies, test channel bandwidths and sub-carrier spacing based on NR operating bands specified in Table 1.1-2. All of these configurations shall be tested with applicable test parameters for each combination of channel bandwidth and sub-carrier spacing, are shown in Table 5.2.3.10.1.1-1. The details of the uplink and downlink RMC are specified in ETSI TS 138 521-1 [4], clauses A.2 and A.3. Configurations of PDSCH and PDCCH before measurement are specified in ETSI TS 138 521-1 [4], clause C.2.

Table 5.2.3.10.1.1-1: Test Configuration Table

Default Conditions				
Test Environment as specified in ETSI TS 138 508-1 [7], clause 4.1			Normal	
Test Frequencies as specified in ETSI TS 138 508-1 [7], clause 4.3.1			Mid range (notes 3 and 4)	
Test Channel Bandwidths as specified in ETSI TS 138 508-1 [7], clause 4.3.1			Lowest, Mid, Highest	
Test SCS as specified in Table 1.1-2			Highest	
Test Parameters				
	Downlink Configuration		**Uplink Configuration**	
Test ID	**Mod'n**	**RB allocation**	**Mod'n**	**RB allocation**
1	CP-OFDM QPSK	note 1	DFT-s-OFDM QPSK	note 1
NOTE 1: The specific configuration of uplink and downlink are defined in Table 5.2.3.6.1.1-1.				
NOTE 2: In a band where UE supports 4Rx, the test shall be performed only with 4Rx antennas ports connected and 4Rx REFSENS requirement (Table 4.2.2.7.2-2) is used in the test requirements.				
NOTE 3: For NR band n28, 30 MHz test channel bandwidth is tested with Low range test frequencies.				
NOTE 4: Only test frequencies within the frequency range operated in Europe as defined in Table 1.1-1 apply.				

1) Connect the SS to the aerial UE antenna connectors as shown in ETSI TS 138 508-1 [7], annex A, in Figure A.3.1.4.3 for TE diagram and clause A.3.2 for UE diagram.

2) The parameter settings for the cell are set up according to ETSI TS 138 508-1 [7], clause 4.4.3.

3) Downlink signals are initially set up according to ETSI TS 138 521-1 [4] clauses C.0, C.1, C.2, C.3.1, and uplink signals according to ETSI TS 138 521-1 [4], clauses G.0, G.1, G.2, G.3.1.

4) The DL and UL Reference Measurement channels are set according to Table 5.2.3.10.1.1-1.

5) Propagation conditions are set according to ETSI TS 138 521-1 [4], clause B.0.

6) Ensure the aerial UE is in state RRC_CONNECTED with generic procedure parameters Connectivity *NR*, Connected without release *On*, Test Mode *On* and Test Loop Function *On* according to ETSI TS 138 508-1 [7], clause 4.5. Message content is defined in ETSI TS 138 521-1 [4], clause 7.5.4.3.

5.1.3.10.1.2 Test Procedure

1) SS transmits PDSCH via PDCCH DCI format 1_1 for C_RNTI to transmit the DL RMC according to Table 5.2.3.10.1.1-1. The SS sends downlink MAC padding bits on the DL RMC.

2) SS sends uplink scheduling information for each UL HARQ process via PDCCH DCI format 0_1 for C_RNTI to schedule the UL RMC according to Table 5.2.3.10.1.1-1. Since the aerial UE has no payload data to send, the aerial UE transmits uplink MAC padding bits on the UL RMC.

3) Set the Downlink signal level to the value as defined in Table 4.2.2.11.2-1. Send uplink power control commands to the aerial UE using 1 dB power step size to ensure that the aerial UE output power measured by the test system is within the Uplink power control window in Table 5.2.3.5.1.3.2-1 of the target power level in Table 4.2.2.11.2-1 for at least the duration of the Throughput measurement.

4) Set the Interfering signal levels to the values as defined in Table 4.2.2.11.2-1 and frequency below the wanted signal.

5) Measure the average throughput according to ETSI TS 138 521-1 [4], clause G.2.

6) Repeat steps from 3 to 5, using an interfering signal above the wanted signal at step 4.

NOTE: The purpose of the Uplink power control window is to ensure that the actual aerial UE output power is no greater than the target power level, and as close as possible to the target power level. The relationship between the Uplink power control window, the target power level and the corresponding possible actual aerial UE Uplink power window is illustrated in ETSI TS 138 521-1 [4], clause F.4.3.5.2.3.

5.2.3.10.2 Test requirements

The results obtained in step 5 shall be compared to the limits in Table 4.2.2.11.2-1 in order to show compliance.

5.2.3.11 Receiver Spurious Emissions

5.2.3.11.1 Method of test

5.2.3.11.1.1 Initial conditions

Initial conditions are a set of test configurations the aerial UE needs to be tested in and the steps for the SS to take with the aerial UE to reach the correct measurement state.

The initial test configurations consist of environmental conditions, test frequencies, test channel bandwidths and sub-carrier spacing based on NR operating bands specified in Table 1.1-2. All of these configurations shall be tested with applicable test parameters for each combination of channel bandwidth and sub-carrier spacing, are shown in Table 5.2.3.11.1.1-1. The details of the uplink and downlink RMCs are specified in ETSI TS 138 521-1 [4], clauses A.2 and A.3. Configurations of PDSCH and PDCCH before measurement are specified in ETSI TS 138 521-1 [4], clause C.2.

Table 5.2.3.11.1.1-1: Test Configuration Table

Default Conditions				
Test Environment as specified in ETSI TS 138 508-1 [7], clause 4.1			Normal	
Test Frequencies as specified in ETSI TS 138 508-1 [7], clause 4.3.1			Low range, Mid range, High range (notes 3 and 4)	
Test Channel Bandwidths as specified in ETSI TS 138 508-1 [7], clause 4.3.1			Highest	
Test SCS as specified in Table 1.1-2			Highest	
Test Parameters				
	Downlink Configuration		**Uplink Configuration**	
Test ID	**Mod'n**	**RB allocation**	**Mod'n**	**RB allocation**
1	N/A	0	N/A	0
NOTE 1: The specific configuration of uplink and downlink are defined in Table 5.2.3.6.1.1-1.				
NOTE 2: In a band where UE supports 4Rx, the test shall be performed only with 4Rx antennas ports connected and 4Rx REFSENS requirement (Table 4.2.2.7.2-2) is used in the test requirements.				
NOTE 3: For NR band n28, 30 MHz test channel bandwidth is tested with Low range and High range test frequencies.				
NOTE 4: Only test frequencies within the frequency range operated in Europe as defined in Table 1.1-1 apply.				

1) Connect the SS to the aerial UE antenna connectors as shown in ETSI TS 138 508-1 [7], annex A, in Figure A.3.1.5.1 for TE diagram and clause A.3.2 for UE diagram.

2) The parameter settings for the cell are set up according to ETSI TS 138 508-1 [7], clause 4.4.3.

3) Downlink signals are initially set up according to ETSI TS 138 521-1 [4], clauses C.0, C.1, C.2, C.3.1 and uplink signals according to ETSI TS 138 521-1 [4], clauses G.0, G.1, G.2, G.3.1.

4) The DL and UL Reference Measurement channels are set according to Table 5.2.3.11.1.1-1.

5) Propagation conditions are set according to ETSI TS 138 521-1 [4], clause B.0.

6) Ensure the aerial UE is in state RRC_CONNECTED with generic procedure parameters Connectivity *NR*, Connected without release *On*, Test Mode *On* and Test Loop Function *On* according to ETSI TS 138 508-1 [7], clause 4.5. Message content is defined in ETSI TS 138 521-1 [4], clause 7.5.4.3.

5.2.3.11.1.2 Test Procedure

1) Sweep the spectrum analyser (or equivalent equipment) over a frequency range and measure the average power of spurious emission.

2) Repeat step 1 for all NR Rx antennas of the aerial UE.

5.2.3.11.2 Test requirements

5.2.3.12 Transmit OFF Power

5.2.3.12.1 Method of test

5.2.3.12.1.1 Initial condition

Initial conditions are a set of test configurations the aerial UE needs to be tested in and the steps for the SS to take with the aerial UE to reach the correct measurement state.

The initial test configurations consist of environmental conditions, test frequencies, test channel bandwidths and sub-carrier spacing based on NR operating bands specified in Table 1.1-2. All of these configurations shall be tested with applicable test parameters for each combination of channel bandwidth and sub-carrier spacing, and are shown in Table 5.2.3.12.1.1-1. The details of the uplink RMCs are specified in ETSI TS 138 521-1 [4], clause A.2. Configurations of PDSCH and PDCCH before measurement are specified in ETSI TS 138 521-1 [4], clause C.2.

Table 5.2.3.12.1.1-1: Test Configuration Table

Initial Conditions			
Test Environment as specified in ETSI TS 138 508-1 [7], clause 4.1		Normal, TL/VL, TL/VH, TH/VL, TH/VH	
Test Frequencies as specified in ETSI TS 138 508-1 [7], clause 4.3.1		Low range, Mid range, High range (note 2)	
Test Channel Bandwidths as specified in ETSI TS 138 508-1 [7], clause 4.3.1		Lowest, Mid, Highest	
Test SCS as specified in Table 1.1-2		Lowest, Highest	
Test Parameters for Channel Bandwidths			
Test ID	**Downlink Configuration**	**Uplink Configuration**	
	N/A for minimum output power test case	**Modulation**	**RB allocation (note 1)**
1		DFT-s-OFDM QPSK	Inner Full
NOTE 1: The specific configuration of each RB allocation is defined in Table 6.1-1 of ETSI TS 138 521-1 [4]. NOTE 2: For NR band n28, 30 MHz test channel bandwidth is tested with Low range and High range test frequencies.			

1) Connect the SS to the aerial UE antenna connectors as shown in ETSI TS 138 508-1 [7] annex A, Figure A.3.1.1.1 for TE diagram and clause A.3.2 for UE diagram.

2) The parameter settings for the cell are set up according to ETSI TS 138 508-1 [7] clause 4.4.3.

3) Downlink signals are initially set up according to clauses C.0, C.1, C.2 of ETSI TS 138 521-1 [4], and uplink signals according to clauses G.0, G.1, G.2, G.3.0 of ETSI TS 138 521-1 [4].

4) The UL Reference Measurement Channel is set according to Table 5.2.3.12.1.1-1.

5) Propagation conditions are set according to ETSI TS 138 521-1 [4], clause B.0.

6) Ensure the aerial UE is in State RRC_CONNECTED with generic procedure parameters Connectivity *NR*, Connected without release *On*, Test Mode *On* and Test Loop Function *On* according to ETSI TS 138 508-1 [7], clause 4.5. Message contents are defined in clause 6.3.3.2.4.3 of ETSI TS 138 521-1 [4].

5.2.3.12.1.2 Test procedure

1) SS sends uplink scheduling information via PDCCH DCI format 0_1 with TPC command 0 dB for C_RNTI to schedule the UL RMC according to Table 5.2.3.12.1.1-1. Since the aerial UE has no payload and no loopback data to send the aerial UE sends uplink MAC padding bits on the UL RMC. The UL assignment is such that the aerial UE transmits on slots 8 for 15 kHz SCS, on slots 8 and 18 for 30 kHz SCS and on slots 17 and 37 for 60 kHz SCS.

2) Send continuously uplink power control "up" commands in every uplink scheduling information to the aerial UE; allow at least 200 ms starting from the first TPC command in this step for the aerial UE to reach P_{UMAX} level.

3) Measure the aerial UE transmission OFF power during the slot prior to the PUSCH transmission, excluding a transient period of 10 µs at the end of the slot.

4) Measure the output power of the aerial UE PUSCH transmission during one slot.

5) Measure the aerial UE transmission OFF power during the slot following the PUSCH transmission, excluding a transient period of 10 µs at the beginning of the slot.

5.2.3.12.2 Test requirements

The results obtained shall be compared to the limits in clause 4.2.2.13.2 in order to show compliance.

5.2.3.13 Aerial UE Identifier

5.2.3.13.1 Method of test

The test description and procedure given in clause 10.1.4.2.3 in ETSI TS 138 523-1 [9] applies.

5.2.3.13.2 Test requirements

If the aerial UE successfully passes the test given in clause 10.1.4.2 in ETSI TS 138 523-1 [9], then the aerial UE is successfully provisioned with CAA-level UAV ID, as the abovementioned test requires as a pre-test condition that the aerial UE is capable of UAS operation and provisioned with CAA-level UAV ID.

5.2.3.14 Aerial UE Authorization and Authentication

5.2.3.14.1 Method of test

The test description and procedure given in clause 10.1.4.2.3 in ETSI TS 138 523-1 [9] applies.

5.2.3.14.2 Test requirements

If the aerial UE successfully passes the test given in clause 10.1.4.2 in ETSI TS 138 523-1 [9], then the aerial UE has successfully performed the UUAA procedure as defined in clause 5.2 in ETSI TS 123 256 [1].

Annex A (informative): Relationship between the present document and the essential requirements of Directive 2014/53/EU

The present document has been prepared under the Commission's standardisation request C(2015) 5376 final [i.6] to provide one voluntary means of conforming to the essential requirements of Directive 2014/53/EU on the harmonisation of the laws of the Member States relating to the making available on the market of radio equipment and repealing Directive 1999/5/EC [i.1].

Once the present document is cited in the Official Journal of the European Union under that Directive, compliance with the normative clauses of the present document given in Tables A-1 and A-2 confers, within the limits of the scope of the present document, a presumption of conformity with the corresponding essential requirements of that Directive and associated EFTA regulations.

Table A-1: Relationship between the present document and the essential requirements of Directive 2014/53/EU for devices supporting E-UTRA

Harmonised Standard ETSI EN 301 908-26					
Requirement				Requirement Conditionality	
No	Description	Essential requirements of Directive	Clause(s) of the present document	U/C	Condition
1	Transmitter Maximum Output Power	3.2	4.1.2.2	U	
2	Transmitter Minimum Output Power	3.2	4.1.2.3	U	
3	Transmitter Spectrum Emission Mask	3.2	4.1.2.4	U	
4	Transmitter Adjacent Channel Leakage Power Ratio		4.1.2.5		
5	Transmitter Spurious Emissions	3.2	4.1.2.6	U	
6	Receiver Reference Sensitivity Level	3.2	4.1.2.7	U	
7	Receiver Adjacent Channel Selectivity (ACS)	3.2	4.1.2.8	U	
8	Receiver Blocking Characteristics	3.2	4.1.2.9	U	
9	Receiver Spurious Response	3.2	4.1.2.10	U	
10	Receiver Intermodulation Characteristic	3.2	4.1.2.11	U	
11	Receiver Spurious Emissions	3.2	4.1.2.12	U	

Table A-2: Relationship between the present document and the essential requirements of Directive 2014/53/EU for devices supporting NR

Harmonised Standard ETSI EN 301 908-26					
Requirement				Requirement Conditionality	
No	Description	Essential requirements of Directive	Clause(s) of the present document	U/C	Condition
1	Transmitter maximum output power	3.2	4.2.2.2	U	
2	Transmitter minimum output power	3.2	4.2.2.3	U	
3	Transmitter spectrum emission mask	3.2	4.2.2.4	U	
4	Transmitter Adjacent Channel Leakage Power Ratio	3.2	4.2.2.5	U	
5	Transmitter spurious emissions	3.2	4.2.2.6	U	
6	Receiver Reference Sensitivity Level	3.2	4.2.2.7	U	
7	Receiver adjacent channel selectivity (ACS)	3.2	4.2.2.8		
8	Receiver blocking characteristics	3.2	4.2.2.9	U	
9	Receiver spurious response	3.2	4.2.2.10	U	
10	Receiver intermodulation characteristics	3.2	4.2.2.11	U	
11	Receiver spurious emissions	3.2	4.2.2.12	U	
12	Transmit OFF power	3.2	4.2.2.13	U	

Key to columns:

Requirement:

No A unique identifier for one row of the table which may be used to identify a requirement.

Description A textual reference to the requirement.

Essential requirements of Directive

Identification of article(s) defining the requirement in the Directive.

Clause(s) of the present document

Identification of clause(s) defining the requirement in the present document unless another document is referenced explicitly.

Requirement Conditionality:

U/C Indicates whether the requirement is unconditionally applicable (U) or is conditional upon the manufacturer's claimed functionality of the equipment (C).

Condition Explains the conditions when the requirement is or is not applicable for a requirement which is classified "conditional".

Presumption of conformity stays valid only as long as a reference to the present document is maintained in the list published in the Official Journal of the European Union. Users of the present document should consult frequently the latest list published in the Official Journal of the European Union.

Other Union legislation may be applicable to the product(s) falling within the scope of the present document.

Annex B (normative): Environmental profile

B.1 General

B.1.1 Introduction

This annex specifies the environmental profile of the aerial UE.

B.1.2 Temperature

The aerial UE shall fulfil all the requirements in the full temperature range of:

Table B.1.2-1: Temperature Test Environment

+15 °C to +35 °C	For normal conditions (with relative humidity up to 75 %)
-10 °C to +55 °C	For extreme conditions (see IEC 60068-2-1 [10] and IEC 60068-2-2 [11])

In no case shall the UE exceed the transmitted levels as defined in ETSI TS 138 101-1 [3], for extreme operation.

The normative reference for this requirement is ETSI TS 136 101 [2], clause E.1 for E-UTRA and ETSI TS 138 101-1 [3], clause E.2.1, for NR.

Some tests are performed also in extreme temperature conditions. These test conditions are denoted as Temperature Low (TL, -10 °C) and Temperature High (TH, +55 °C).

B.1.3 Voltage

The aerial UE shall fulfil all the requirements in the full voltage range, i.e. the voltage range between the extreme voltages.

The manufacturer shall declare the lower and higher extreme voltages and the approximate shutdown voltage. For the equipment that can be operated from one or more of the power sources listed below, the lower extreme voltage shall not be higher and the higher extreme voltage shall not be lower than that specified below.

Table B.1.3-1: Voltage Test Environment

Power source	Lower extreme Voltage	Higher extreme voltage	Normal conditions voltage
AC mains	0,9 × nominal	1,1 × nominal	nominal
Regulated lead acid battery	0,9 × nominal	1,3 × nominal	1,1 × nominal
Non regulated batteries: Leclanché Lithium Mercury/nickel and cadmium	 0,85 × nominal 0,95 × nominal 0,90 × nominal	 nominal 1,1 × nominal	 nominal 1,1 × nominal nominal
NOTE: The normative reference for this requirement is ETSI TS 136 101 [2], clause E.2 for E-UTRA and ETSI TS 138 101-1 [3], clause E.2.2, for NR.			

In no case shall the aerial UE exceed the transmitted levels as defined in ETSI TS 136 101 [2] for E-UTRA and ETSI TS 138 101-1 [3] for NR, for extreme operation.

Some tests, given in clauses 5.1.3 and 5.2.3, are performed also in extreme voltage conditions. These test conditions are denoted as Lower extreme Voltage (VL) and Higher extreme Voltage (VH).

B.1.4 Test environment

Where a normal environment is required then the normal conditions shown in clauses B.1.2 and B.1.3 shall be applied.

Where an extreme environment is required then the various combinations of extreme temperatures together with the extreme voltages shown in clauses B.1.2 and B.1.3 shall be applied. The combinations are:

- Low extreme Temperature/Low extreme Voltage (TL/VL);
- Low extreme Temperature/High extreme Voltage (TL/VH);
- High extreme Temperature/Low extreme Voltage (TH/VL);
- High extreme Temperature/High extreme Voltage (TH/VH).

Annex C (informative): Selection of receiver parameters

C.0 Introduction

Receiver parameters under article 3.2 of Directive 2014/53/EU [i.1] listed in ETSI EG 203 336 [i.2] V1.2.1 are analysed and the parameters which are applicable to the present document are specified in respective clauses. In each clause in Annex C the definition of the requirement is taken from the ETSI EG 203 336 [i.2].

C.1 Receiver sensitivity

Receiver sensitivity is the ability to receive a wanted signal at low input signal levels while providing a pre-determined level of performance. Receiver sensitivity is specified in the present document in clauses 4.1.2.7 and 4.2.2.7.

C.2 Receiver co-channel rejection

Receiver co-channel rejection is a measure of the capability of a receiver to receive a wanted signal, without exceeding a given degradation, due to the presence of an unwanted signal, both signals being at the nominal frequency of the receiver.

In frequency bands covered by the present document, any modulation, channel width, centre frequency and data rate may be used. A combination of many signals of widely different characteristics manifests itself primarily as an increase in the noise floor.

Thus, a specific test for co-channel rejection is not included because the co-channel rejection performance of the receiver combined with the receiver noise figure directly affects the sensitivity performance, which is tested. The required limits for sensitivity ensure that products have the required co-channel rejection.

C.3 Receiver adjacent channel selectivity

Adjacent channel selectivity is a measure of the receiver capability to receive a wanted modulated signal without exceeding the general performance criteria stated in the present document due to the presence of an unwanted input signal in the adjacent channels. Adjacent channel selectivity is specified in the present document in clauses 4.1.2.8 and 4.2.2.8.

C.4 Receiver spurious response rejection

The spurious response rejection is a measure of the capability of the receiver to receive a wanted signal without exceeding a given degradation due to the presence of an unwanted signal at any frequency at which a response is obtained. The frequencies of the adjacent signals (channels) are excluded.

Receiver spurious response rejection is specified in the present document in clauses 4.1.2.10 and 4.2.2.10.

C.5 Receiver blocking

Blocking is a measure of the receiver capability to receive a wanted modulated signal without exceeding a given degradation due to the presence of an unwanted input signal at any frequencies other than those of the spurious responses or the adjacent channels or bands. Receiver Blocking is specified in the present document in clauses 4.1.2.9 and 4.2.2.9.

C.6 Receiver radio-frequency intermodulation

Intermodulation rejection is a measure of the ability of a receiver to operate in the presence of two or more unwanted signals the frequencies of which have a specific frequency relationship to the wanted signal. Receiver Intermodulation is specified in the present document in clauses 4.1.2.11 and 4.2.2.11.

C.7 Receiver dynamic range

Receiver dynamic range is defined as the range of the wanted input signal level over which a receiver functions at a specified performance level. The lower end of this range is normally the sensitivity of the receiver. The upper end of a receiver's dynamic range determines how strong a received signal can be before producing degradation due to overloading.

The UEs are normally deployed far away from BS and for a few UEs which are close to BS, sufficient coupling loss is also guaranteed due to the antenna high difference between BS and UE. Such that UE will not receive a wanted signal that is high enough to provide overloading effect. Thus, receiver dynamic range is not included in the present document.

C.8 Receiver unwanted emissions in the spurious domain

As a default, the limit for unwanted emissions in the spurious domain referenced at the antenna port should respect those in ERC/REC 74-01 [i.12]. Receiver spurious emission is specified in the present document in clauses 4.1.2.12 and 4.2.2.12.

Annex D (informative): Maximum measurement uncertainty

D.1 E-UTRA

The measurements described in the present document are based on the following assumptions:

- the measured value related to the corresponding limit is used to decide whether an equipment meets the requirements of the present document;
- the value of the measurement uncertainty for the measurement of each parameter is included in the test report.

Table D.1-1 shows the recommended values for the maximum measurement uncertainty figures.

Table D.1-1: Maximum measurement uncertainty

Parameter	Uncertainty
Transmitter maximum output power	±0,7 dB
Transmitter spectrum emissions mask	±1,5 dB
Transmitter spurious emissions 9 kHz < f ≤ 4 GHz 4 GHz < f ≤ 12,75 GHz	±2,0 dB ±4,0 dB
Transmitter Minimum output power	±1,0 dB
Receiver Adjacent Channel Selectivity (ACS)	±1,1 dB
Receiver Blocking characteristics 1 MHz < $f_{interferer}$ ≤ 3 GHz 3 GHz < $f_{interferer}$ ≤ 12,75 GHz	±1,3 dB ±3,2 dB
Receiver spurious response 1 MHz < $f_{interferer}$ ≤ 3 GHz 3 GHz < $f_{interferer}$ ≤ 12,75 GHz	±1,3 dB ±3,2 dB
Receiver intermodulation characteristics	±1,4 dB
30 MHz ≤ f ≤ 4,0 GHz 4 GHz < f ≤ 12,75 GHz	±2,0 dB ±4,0 dB
Transmitter adjacent channel leakage power ratio	±0,8 dB
Receiver Reference Sensitivity Level f ≤ 4,0 GHz 4 GHz < f ≤ 12,75 GHz	±0,7 dB ±1,0 dB
NOTE 1: For RF tests it should be noted that the uncertainties in Table D.1-1 apply to the test system operating into a nominal 50 Ω load and do not include system effects due to mismatch between the EUT and the test system. NOTE 2: If the test system for a test is known to have a measurement uncertainty greater than that specified in Table D.1-1, this equipment can still be used provided that an adjustment is made as follows: any additional uncertainty in the test system over and above that specified in Table D.1-1 should be used to tighten the test requirements - making the test harder to pass (for some tests, e.g. receiver tests, this may require modification of stimulus signals). This procedure will ensure that a test system not compliant with Table D.1-1 does not increase the probability of passing an EUT that would otherwise have failed a test if a test system compliant with Table D.1-1 had been used.	

D.2 NR

The maximum measurement uncertainty values can be found in annex F in ETSI 138 521-1 [4].

Annex E (informative): Bibliography

- Directive 2004/108/EC of the European Parliament and of the Council of 15 December 2004 on the approximation of the laws of the Member States relating to electromagnetic compatibility and repealing Directive 89/336/EEC (EMC Directive).
- Directive 2006/95/EC of the European Parliament and of the Council of 12 December 2006 on the harmonisation of the laws of Member States relating to electrical equipment designed for use within certain voltage limits (LV Directive).
- Commission Decision 2008/477/EC of 13 June 2008 on the harmonisation of the 2 500-2 690 MHz frequency band for terrestrial systems capable of providing electronic communications services in the Community.
- Commission Decision (EU) 2015/750 of 8 May 2015 on the harmonisation of the 1 452-1 492 MHz frequency band for terrestrial systems capable of providing electronic communications services in the Union.
- Directive 98/34/EC of the European Parliament and of the Council of 22 June 1998 laying down a procedure for the provision of information in the field of technical standards and regulations.
- Directive 98/48/EC of the European Parliament and of the Council of 20 July 1998 amending Directive 98/34/EC laying down a procedure for the provision of information in the field of technical standards and regulations.
- Commission Decision 2005/513/EC of 11 July 2005 on the harmonised use of radio spectrum in the 5 GHz frequency band for the implementation of wireless access systems including radio local area networks (WAS/RLANs).
- Commission Decision 2007/90/EC of 12 February 2007 amending Decision 2005/513/EC on the harmonised use of radio spectrum in the 5 GHz frequency band for the implementation of Wireless Access Systems including Radio Local Area Networks (WAS/RLANs).
- Commission Implementing Decision 2012/688/EU on the harmonisation of the frequency bands 1920-1980 MHz and 2110-2170 MHz for terrestrial systems capable of providing electronic communications services in the Union - Press Release.

Annex F (informative): Change history

Version	Information about changes
1.0.0	Draft for the combined Public Enquiry and Vote phase

History

Version	Date	Status
V1.0.0	August 2026	SRdAP process EV 20261119: 2026-08-27 to 2026-11-19